Supplementary Information for: Labelling where models disagree: label-efficient comparison of classifiers on group fairness

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Abstract

Proofs, additional theoretical results, numerical checks of the theory, baseline definitions, ablation studies and data details for the main paper. Sections, tables, figures, equations and propositions are numbered with the prefix S; numbers without a prefix refer to the main paper.

Keywords: supplementary information

S1 Proofs

Throughout, $x_i = a^\top \xi_i$ with $\xi_i = t_i(Y_i) - \psi_i$ for a fixed or sequence-dependent vector a and a label-free vector ψ_i with $\|\psi_i\|_\infty \le 1$ (the plain estimator is $\psi \equiv 0$, so $\xi_i = t_i(Y_i)$), and $\bar{x}_h$, $S^2_{x,h}$, $\bar{x}_{s_h}$, $s^2_{x,h}$ are the stratum population and sample means and variances. We use repeatedly two classical facts on simple random sampling without replacement of n units from N [1, Ch. 2]: $E(\bar{x}_s) = \bar{x}_U$ and $\mathrm{Var}(\bar{x}_s) = (1 - n/N)S^2_x/n$; the same formulas hold coordinatewise and for covariances of vector-valued variables. Because all coordinates of ξ_i lie in $[-1, 1]$, any population variance of a coordinate or of a product of two coordinates satisfies $S^2 \le N/(N-1) \le 2$ for $N \ge 2$ (for $\psi \equiv 0$ the sharper bound $N/\{4(N-1)\} \le 1/2$ holds).

S1.1 Proof of Proposition 1

(a) Since $\sum_{i\in U} \psi_i = \sum_h N_h \bar{\psi}_h$ and $T = \sum_h N_h \bar{t}_h$, definition (14) gives $\hat{T}_\psi - T = \sum_h N_h(\bar{\xi}_{s_h} - \bar{\xi}_h)$: the known proxy total cancels exactly, which is where the fact that ψ_i is known for every unit of the pool is used. Within stratum h, $E(\bar{\xi}_{s_h}) = \bar{\xi}_h$ and $\mathrm{Cov}(\bar{\xi}_{s_h}) = (1 - f_h)\Sigma^\xi_h/n_h$. The strata are sampled independently and N_h, n_h and ψ_i are constants because, by (A1), they are functions of the predictions, probabilities and groups only. Hence $E(\hat{T}_\psi) = T$ and $\mathrm{Cov}(\hat{T}_\psi) = \sum_h N^2_h(1 - f_h)\Sigma^\xi_h/n_h$. For $n_h \ge 2$ the sample covariance matrix of ξ_i in stratum h is unbiased for Σ^ξ_h [1, Ch. 2], which gives the unbiased estimator of $\mathrm{Cov}(\hat{T}_\psi)$. Only $n_h \ge 1$ is used for unbiasedness, and nothing is assumed about the proxy beyond its being label-free. The case $\psi \equiv 0$ is the plain estimator.

(b) Let $\hat{\tau} = \hat{T}_\psi/N$. By (a), each coordinate satisfies $\mathrm{Var}(\hat{\tau}_c) = \sum_h W^2_h(1 - f_h)S^2_{\xi_c,h}/n_h \le 2r^2$, so $\|\hat{\tau} - \tau\| = O_p(r)$ by Chebyshev's inequality, and $r^2 \le \sum_h W^2_h / \min_k n_k \le 1/\min_k n_k \to 0$ by (A1). Let $K = \{x \in [-1, 2]^C : x_{P_g} \ge c/2,\ x_{Q_g} \ge c/2,\ g = 0, 1\}$, where x_{P_g}, x_{Q_g} denote the coordinates of the denominators (the box is larger than $[0, 1]^C$ because $\hat{\tau}$, unlike τ, can leave $[0, 1]^C$: the proxy correction $\bar{\psi}_h - \bar{\psi}_{s_h}$ is not a mean of the contributions). K is compact and convex, and

ϕ is a rational function whose denominators are bounded away from zero on K, so its first and second derivatives are bounded on K, say by B. By (A2) $\tau \in K$ with the denominators at least c; on the event $\mathcal{E} = \{\|\hat\tau - \tau\|_\infty \le c/2\}$, whose probability tends to one, the segment from τ to $\hat\tau$ lies in K, and Taylor's theorem gives

$$\left|\hat\theta_\psi - \theta - \nabla\phi(\tau)^\top(\hat\tau - \tau)\right| \le \tfrac{B}{2}\, C\, \|\hat\tau - \tau\|^2 = O_p(r^2). \tag{S1}$$

Therefore $\hat\theta_\psi - \theta = O_p(r) + O_p(r^2) = O_p(r) \to 0$.

Bias. The statement of Section 5.2 that a ratio-type estimand has bias $O(r^2)$ needs a convention on $\mathcal{E}^c$, where the ratio may be undefined or arbitrarily large: let $\hat\theta_\psi$ be truncated to the range of the estimand (an interval of length at most 4 for every comparison estimand), so that $|\hat\theta_\psi - \theta| \le 4$. (The code returns `NaN` when a denominator total is not positive and counts it as an undefined estimate; the convention is used only for this statement.) Since $E(\hat\tau - \tau) = 0$ by (a), $|E\{a^\top(\hat\tau - \tau)\mathbf{1}_{\mathcal{E}}\}| = |E\{a^\top(\hat\tau - \tau)\mathbf{1}_{\mathcal{E}^c}\}| \le 2CB\,P(\mathcal{E}^c)$, because $|\hat\tau_c - \tau_c| \le 2$ for every coordinate and $\|a\|_1 \le CB$, and Chebyshev's inequality with the variance bound above gives $P(\mathcal{E}^c) \le 8Cr^2/c^2$. On $\mathcal{E}$ the remainder is at most $(B/2)C\|\hat\tau - \tau\|^2$, with expectation $O(r^2)$, and on $\mathcal{E}^c$ the truncated error is at most $4P(\mathcal{E}^c) = O(r^2)$. Hence $|E(\hat\theta_\psi) - \theta| = O(r^2)$, with constants that depend on B, C and c only, and by (A3) this is $o(V_\psi^{1/2})$.

(c) Let $a = \nabla\phi(\tau)$ (it depends on ν and is bounded by B), $x_i = a^\top\xi_i$, and $\Lambda = a^\top(\hat\tau - \tau) = \sum_h W_h(\bar x_{s_h} - \bar x_h)$. Since $e_i = x_i/N$, $\mathrm{Var}(\Lambda) = \sum_h W_h^2(1 - f_h)S_{x,h}^2/n_h = V_\psi(\theta; n)$, abbreviated V. By (S1) and (A3), $(\hat\theta_\psi - \theta - \Lambda)/V^{1/2} = O_p(r^2)/(c^{1/2}r) = O_p(r) \to 0$, so by Slutsky's lemma it suffices to show $\Lambda/V^{1/2} \to N(0,1)$.

Since the limit does not depend on the subsequence, it suffices to show that every subsequence has a further subsequence along which the convergence holds. Passing to a subsequence, we may therefore assume that the set $\mathcal{H}_1$ of strata satisfying the first alternative of (A4), and its complement $\mathcal{H}_2$ (second alternative), do not depend on ν, and that $\lambda_h^2 = W_h^2(1 - f_h)S_{x,h}^2/(n_h V)$ converges to some $\lambda_{h,\infty}^2$ for every h (the λ_h^2 lie in $[0,1]$ and sum to one). For $h \in \mathcal{H}_2$, $\lambda_{h,\infty} = 0$, so $\sum_{h\in\mathcal{H}_2} W_h(\bar x_{s_h} - \bar x_h)/V^{1/2}$ has variance $\sum_{h\in\mathcal{H}_2}\lambda_h^2 \to 0$ and is $o_p(1)$. For $h \in \mathcal{H}_1$, put $\chi_h = (\bar x_{s_h} - \bar x_h)/\{(1 - f_h)S_{x,h}^2/n_h\}^{1/2}$. Hájek's condition for asymptotic normality of a simple random sample mean [2] requires $n_h \to \infty$, $N_h - n_h \to \infty$ and, for every $\eta > 0$,

$$\frac{1}{(N_h - 1)S_{x,h}^2} \sum_{\substack{i\in U_h:\\ |x_i - \bar x_h| > \eta m_h S_{x,h}}} (x_i - \bar x_h)^2 \to 0,$$

with $m_h = \{n_h(1 - f_h)\}^{1/2}$.

The first two hold by (A1) ($N_h - n_h \ge \varepsilon N_h \ge \varepsilon n_h$). Because $|x_i - \bar x_h| \le 2\|a\|_1 \le 2CB$ and $n_h(1 - f_h)S_{x,h}^2 \to \infty$ by (A4), the index set of the sum is empty for all large ν, so the third holds as well, and $\chi_h \to N(0,1)$. The χ_h are independent, and $\Lambda/V^{1/2} = \sum_{h\in\mathcal{H}_1}\lambda_h\chi_h + o_p(1)$. By Lévy's continuity theorem the characteristic functions φ_h of χ_h converge to $e^{-u^2/2}$ uniformly on compact sets, hence $E\exp(iu\sum_{\mathcal{H}_1}\lambda_h\chi_h) = \prod_{\mathcal{H}_1}\varphi_h(\lambda_h u) \to \prod_{\mathcal{H}_1} e^{-\lambda_{h,\infty}^2 u^2/2} = e^{-u^2/2}$, using $\sum_{\mathcal{H}_1}\lambda_{h,\infty}^2 = 1$. This proves (c).

(d) Let $\hat a = \nabla\phi(\hat\tau)$ and let $\hat\Sigma_h$ be the sample covariance matrix of ξ_i in the sample from stratum h. Since $\hat e_i = \hat a^\top\xi_i/N$,

$$\hat V_\psi = \sum_h W_h^2 \frac{1 - f_h}{n_h}\hat a^\top\hat\Sigma_h\hat a,$$

$$V_\psi = \sum_h W_h^2 \frac{1 - f_h}{n_h} a^\top\Sigma_h^\xi a,$$

and strata with $f_h = 1$ contribute zero to both. Hence $|\hat V_\psi - V_\psi| \le r^2 \max_h |\hat a^\top\hat\Sigma_h\hat a - a^\top\Sigma_h^\xi a|$.

Lemma S1 *Under (A1), $\hat\Sigma_h - \Sigma_h^\xi \to 0$ in probability for every h.*

Proof For coordinates c, d, $\hat\Sigma_{h,cd} = \frac{n_h}{n_h - 1}\{\overline{(\xi_c\xi_d)}_{s_h} - \bar\xi_{c,s_h}\bar\xi_{d,s_h}\}$ and $\Sigma_{h,cd}^\xi = \frac{N_h}{N_h - 1}\{\overline{(\xi_c\xi_d)}_h - \bar\xi_{c,h}\bar\xi_{d,h}\}$. The variables ξ_c, ξ_d and $\xi_c\xi_d$ take values in $[-1,1]$, so each sample mean differs from its population mean by $O_p(n_h^{-1/2})$. The factors $n_h/(n_h - 1)$ and $N_h/(N_h - 1)$ tend to one, and the products are of bounded variables, so the difference tends to zero in probability. □

By part (b), $\hat\tau - \tau \to 0$ in probability, and $\nabla\phi$ is uniformly continuous on K, so $\hat a - a \to 0$ in probability. All entries of $\hat a$, a, $\hat\Sigma_h$, Σ_h are bounded on $\mathcal{E}$, and with the lemma,

$\max_h |\hat{a}^\top \hat{\Sigma}_h \hat{a} - a^\top \Sigma_h^\xi a| = o_p(1)$. With (A3), $|\hat{V}_\psi / V_\psi - 1| \le r^2 o_p(1)/(cr^2) = o_p(1)$. Finally $(\hat{\theta}_\psi - \theta)/\hat{V}_\psi^{1/2} = \{(\hat{\theta}_\psi - \theta)/V_\psi^{1/2}\}(V_\psi/\hat{V}_\psi)^{1/2} \to N(0,1)$ by (c) and Slutsky's lemma, which gives the coverage statement. □

S1.2 Proof of Proposition 2

Fix g and the pair (m, j). The estimand $\theta_{mj}^{TPR} = \sum_{g'} s_{g'}(TP_{mg'} - TP_{jg'})/P_{g'}$ depends on T through $TP_{mg'}$, $TP_{jg'}$ and $P_{g'}$, with partial derivatives $s_{g'}/P_{g'}$, $-s_{g'}/P_{g'}$ and $-s_{g'}(TP_{mg'} - TP_{jg'})/P_{g'}^2 = -s_{g'}\delta_{g'}/P_{g'}$. Instance i with $G_i = g$ contributes $Y_i\hat{y}_{mi}$, $Y_i\hat{y}_{ji}$ and Y_i to the totals of its own group and nothing to the totals of the other group. By (10), $z_i^{TPR} = s_g\{Y_i\hat{y}_{mi} - Y_i\hat{y}_{ji} - \delta_g Y_i\}/P_g = s_g Y_i(D_i - \delta_g)/P_g$. Summing over the group gives $s_g\{(TP_{mg} - TP_{jg}) - \delta_g P_g\}/P_g = 0$. The FPR case is identical with $1 - Y_i$, Q_g and γ_g. For the error, $|\hat{y} - Y| = \hat{y} + Y - 2\hat{y}Y$ for binary values, so $|\hat{y}_{mi} - Y_i| - |\hat{y}_{ji} - Y_i| = (\hat{y}_{mi} - \hat{y}_{ji})(1 - 2Y_i)$ and $z_i^{ERR} = D_i(1 - 2Y_i)/N$.

For the stratum variances let $u_i = Y_i(D_i - \delta_g)$, so $z_i^{TPR} = s_g u_i/P_g$. Because $Y_i^2 = Y_i$, $\bar{u}_h = c_h - \delta_g\pi_h$ and $\overline{u^2}_h = r_h - 2\delta_g c_h + \delta_g^2\pi_h$, using $D_i^2 = |D_i|$. Therefore

$$\begin{aligned}\overline{u^2}_h - \bar{u}_h^2 &= r_h - 2\delta_g c_h + \delta_g^2\pi_h \\ &\quad - c_h^2 + 2\delta_g c_h\pi_h - \delta_g^2\pi_h^2 \\ &= (r_h - c_h^2) + \delta_g^2\pi_h(1 - \pi_h) \\ &\quad - 2\delta_g c_h(1 - \pi_h),\end{aligned}$$

and multiplying by κ_h/P_g^2 gives (17). For the error, $\{D_i(1 - 2Y_i)\}^2 = |D_i|$, so the variance with divisor N_h is $\rho_h - e_h^2$, which gives (18). If $D_i = 0$ on U_h then $r_h = c_h = \rho_h = e_h = 0$, and (19) follows; for FPR the label variance of $1 - Y$ is again $\pi_h(1 - \pi_h)$. When $K_i \in \{0, M\}$ all models predict the same label, so every pair agrees. □

S1.3 Proof of Proposition S2

By (13), $\xi_i = Y_i u_i + (1 - Y_i)v_i - \pi_i u_i - (1 - \pi_i)v_i = (Y_i - \pi_i)(u_i - v_i)$, so $e_i = \nabla f(T)^\top \xi_i = (Y_i - \pi_i)d_i$. The partial derivatives of θ_{mj}^{TPR} for an instance of group g are those listed in the proof of Proposition 2: s_g/P_g and $-s_g/P_g$ for the coordinates TP_{mg} and TP_{jg}, and $-s_g\delta_g/P_g$ for P_g. For these coordinates $u_i - v_i$ equals $\hat{y}_{mi}$, $\hat{y}_{ji}$ and 1 (they have $v_i = 0$), so $d_i = s_g(\hat{y}_{mi} - \hat{y}_{ji} - \delta_g)/P_g = s_g(D_i - \delta_g)/P_g$. For θ_{mj}^{FPR} the coordinates FP_{mg}, FP_{jg}, Q_g have $u_i = 0$, $v_i = \hat{y}_{mi}, \hat{y}_{ji}, 1$, so $u_i - v_i$ is the negative of these values and $d_i = -s_g(D_i - \gamma_g)/Q_g$, which gives $e_i^{FPR} = (Y_i - \pi_i)\{-s_g(D_i - \gamma_g)/Q_g\} = s_g\{(1 - Y_i) - (1 - \pi_i)\}(D_i - \gamma_g)/Q_g$. For the error, E_m has $u_i - v_i = (1 - \hat{y}_{mi}) - \hat{y}_{mi} = 1 - 2\hat{y}_{mi}$, the gradient is $1/N$ and $-1/N$ on E_m and E_j, so $d_i = \{(1 - 2\hat{y}_{mi}) - (1 - 2\hat{y}_{ji})\}/N = -2D_i/N$. The influence value $z_i = B_i + Y_i d_i$ with $B_i = \nabla f(T)^\top v_i$ is affine in Y_i, so $\tilde{z}_i = z_i - e_i = B_i + \pi_i d_i$ is z_i with Y_i replaced by π_i. If $D_i = 0$ for all $i \in U_h$ then $d_i^{ERR} = 0$ and $e_i^{ERR} = 0$.

For (S3) treat $d_i, \pi_i, \tilde{z}_i$ as constants (gradient at T_π). The $e_i = (Y_i - \pi_i)d_i$ are independent with mean zero and variances $\sigma_i^2 = \pi_i(1 - \pi_i)d_i^2$. Then $E_\pi \sum_{U_h}(e_i - \bar{e}_h)^2 = (1 - N_h^{-1})\sum_{U_h}\sigma_i^2$, so $E_\pi S_{e,h}^2 = N_h^{-1}\sum_{U_h}\sigma_i^2 = \bar{\sigma}_h^2$. Because $z_i = \tilde{z}_i + e_i$ with $\tilde{z}_i$ constant, $S_{z,h}^2 = S_{\tilde{z},h}^2 + 2\,\mathrm{Cov}_h(\tilde{z}, e) + S_{e,h}^2$ and $C_h = S_{\tilde{z},h}^2 + \mathrm{Cov}_h(\tilde{z}, e)$; the covariance with the constants $\tilde{z}_i$ is linear in e and has expectation zero. □

S1.4 Proof of Proposition 5

(a) The term $\sum_h N_h A_h^2$ does not depend on n. For any $n_h > 0$ with $\sum_h n_h = L$, the Cauchy–Schwarz inequality gives

$$\begin{aligned}\Big(\sum_h N_h A_h\Big)^2 &= \Big(\sum_h \frac{N_h A_h}{\sqrt{n_h}}\sqrt{n_h}\Big)^2 \\ &\le \Big(\sum_h \frac{N_h^2 A_h^2}{n_h}\Big)\Big(\sum_h n_h\Big) = \Phi_0(n)\,L,\end{aligned}$$

with equality if and only if $N_h A_h/\sqrt{n_h}$ is proportional to $\sqrt{n_h}$, that is $n_h \propto N_h A_h$. With the budget constraint this gives n^* and the value of $\Phi(n^*)$. Equivalently, a Lagrange multiplier μ for $\sum_h n_h = L$ gives the stationarity condition $-N_h^2 A_h^2/n_h^2 + \mu = 0$, that is $n_h = N_h A_h/\sqrt{\mu}$.

(b) Φ_0 is convex on $(0, \infty)^H$ (strictly convex in the coordinates with $A_h > 0$) and the constraints are linear, so the Karush–Kuhn–Tucker conditions are necessary and sufficient. With multipliers μ for the budget and $\alpha_h, \beta_h \ge 0$ for $n_h \ge \ell_h$ and $n_h \le N_h$, they read $-N_h^2 A_h^2/n_h^2 + \mu - \alpha_h + \beta_h = 0$ with complementary slackness. If neither bound is active, $n_h = N_h A_h/\lambda$ with $\lambda = \sqrt{\mu}$; if the upper bound is active, $\beta_h \ge 0$ forces $N_h A_h/\lambda \ge N_h$; if the lower bound is active, $\alpha_h \ge 0$ forces $N_h A_h/\lambda \le \ell_h$. In all

cases $n_h = \min\{N_h, \max(\ell_h, N_h A_h/\lambda)\}$. The map $\lambda \mapsto \sum_h n_h(\lambda)$ is continuous and non-increasing, from $\sum_h N_h \geq L$ as $\lambda \to 0$ to $\sum_h \ell_h \leq L$ as $\lambda \to \infty$, so a solution exists (we assume $\ell_h \leq N_h$, $\sum_h \ell_h \leq L \leq \sum_h N_h$ and $A_h > 0$; a stratum with $A_h = 0$ receives ℓ_h and the argument applies to the remaining strata). The implemented allocation (`_round_alloc` in `tcs/labeling.py`) is a close heuristic to this solution and not exactly it: it gives every stratum its floor $\min(N_h, 10)$ first, distributes the remaining labels in proportion to $N_h A_h$ with iterative capping at N_h, and then rounds. It coincides with (b) when no lower bound is active and in general differs from it by the shift of the floor. Every validity statement of the paper holds for any label-free allocation, so this affects efficiency only.

(c) If $\tilde{n}_h \geq n_h - 1$ and $n_h \geq n_0 > 1$, then $1/\tilde{n}_h \leq 1/(n_h - 1) = (1/n_h)\, n_h/(n_h - 1) \leq (1/n_h)\, n_0/(n_0 - 1)$ for every h; multiply by $N_h^2 A_h^2$ and sum. The implementation first rounds down and then distributes the remaining labels, so $\tilde{n}_h \geq \lfloor n_h \rfloor \geq n_h - 1$.

(d) Without finite-population corrections, $V_t^{SRS} = N^2 \sigma_t^2/L$, so $w_t V_t = (N^2/L) V_t / V_t^{SRS} = (N^2/L)/\mathrm{Eff}_t$. Summing over t gives $\Phi_0(n) L/N^2 = \sum_t 1/\mathrm{Eff}_t$, and at n^*

$$\sum_t \frac{1}{\mathrm{Eff}_t} = \frac{(\sum_h N_h A_h)^2}{N^2} = \Big(\sum_h W_h A_h\Big)^2.$$

For the lower bound, the within-stratum variances satisfy $\sum_h W_h \sigma_{ht}^2 \leq \sigma_t^2$ (the between-strata part of the variance decomposition is non-negative), so by Jensen's inequality

$$\Big(\sum_h W_h A_h\Big)^2 \leq \sum_h W_h A_h^2 = \sum_t \frac{\sum_h W_h \sigma_{ht}^2}{\sigma_t^2} \leq |\mathcal{T}|. \qquad \square$$

S1.5 Proof of Proposition S7

Proposition 5(a)–(c) use only that the A_h are non-negative numbers, so they apply with A_h^ψ. Proposition S9(b) applied with $A_h = A_h^\psi$ and the anticipated value $\tilde{A}_h = \tilde{A}_h^0$ gives (S7) with $x_h = \tilde{A}_h^0/A_h^\psi$. By (23) and the assumption $\tilde{A}_h^\psi = A_h^\psi$, $x_h^2 = \sum_t w_t(\bar{\sigma}_{ht}^2 + S_{\tilde{z},ht}^2)/\sum_t w_t \bar{\sigma}_{ht}^2 = 1 + \tilde{\nu}_h$. The bound is Proposition S9(c) with $x_{\min} \geq 1$ and $x_{\max} \leq (1+\nu_{\max})^{1/2}$. For the plain estimator allocated with the residual rule, $A_h = A_h^0$, $\tilde{A}_h = \tilde{A}_h^\psi$, so x_h is replaced by $1/x_h \in [(1+\nu_{\max})^{-1/2}, 1]$ and $\omega_h \propto N_h A_h^0$; the ratio $x_{\max}/x_{\min}$ is the same, hence so is the Kantorovich bound $(s+1)^2/(4s)$ with $s = (1+\nu_{\max})^{1/2}$. $\square$

S1.6 Proof of Corollary S8

Equation (24). Without finite-population corrections, $V_t^{SRS} = N^2\sigma_t^2/L$ and, by Proposition 5(a) applied to the single target, $V_t(n^*) = (\sum_h N_h \sigma_{ht})^2/L$; the ratio is (24). The variance decomposition $\sigma_t^2 = \sum_h W_h \sigma_{ht}^2 + \sum_h W_h(\bar{z}_h - \bar{z})^2$ and Jensen's inequality $\sum_h W_h \sigma_{ht}^2 \geq (\sum_h W_h \sigma_{ht})^2$ give $\mathrm{Eff}_t \geq 1$, with equality if and only if both inequalities are equalities.

Refinement. If a stratum h is split into h' and h'', the same two inequalities applied within h give $W_h \sigma_h \geq W_{h'} \sigma_{h'} + W_{h''} \sigma_{h''}$, so the denominator of (24) cannot increase.

(a) By Proposition 2 the influence values of θ_{mj}^{TPR} have mean zero in each group, hence overall, and $N^{-1}\sum_i z_i^2 = N^{-1}\sum_g N_g \overline{u^2}_g / P_g^2$, where $\overline{u^2}_g = r_g - 2\delta_g \cdot \delta_g p_g + \delta_g^2 p_g = r_g - \delta_g^2 p_g$ is the group mean of u_i^2 (we used $N_g^{-1}\sum_{G_i=g} Y_i D_i = \delta_g p_g$). With $P_g = N\zeta_g p_g$, $\sigma_t^2 = N^{-2}\sum_g (r_g - \delta_g^2 p_g)/(\zeta_g p_g^2)$. In the stratum $\mathcal{D}_g$, $|D_i| = 1$, so $r_h = \pi_h = \pi_{\mathcal{D},g}$ and $c_h = c_{\mathcal{D},g} = \delta_g p_g N_g/(N_g \rho_g)$; (17) with divisor N_h gives $\sigma_{\mathcal{D}_g} = \sigma_{\mathcal{D},g}/P_g$, and $W_{\mathcal{D}_g}\sigma_{\mathcal{D}_g} = \zeta_g \rho_g \sigma_{\mathcal{D},g}/(N\zeta_g p_g) = \rho_g \sigma_{\mathcal{D},g}/(N p_g)$. In $\mathcal{A}_g$, the number of positives is $N_g(p_g - r_g)$, so its positive rate is $\pi_{\mathcal{A},g}$, and (19) gives $W_{\mathcal{A}_g}\sigma_{\mathcal{A}_g} = (1-\rho_g)|\delta_g|\{\pi_{\mathcal{A},g}(1-\pi_{\mathcal{A},g})\}^{1/2}/(N p_g)$. Inserting into (24), the factors N^{-2} cancel and (S8) follows. For the error difference, $z_i = D_i(1-2Y_i)/N$ vanishes on $\mathcal{A}_g$, takes values $\pm 1/N$ on $\mathcal{D}_g$ with mean μ_g/N, so $\sigma_{\mathcal{D}_g} = (1-\mu_g^2)^{1/2}/N$, and $\sigma_t^2 = (\rho - \theta^2)/N^2$ because $z_i^2 = |D_i|/N^2$; (S9) follows. The last statement of (a) is the refinement property.

(b) With $\delta_g = 0$, $c_{\mathcal{D},g} = 0$ and $\sigma_{\mathcal{D},g}^2 = \pi_{\mathcal{D},g} = r_g/\rho_g$, so $\rho_g \sigma_{\mathcal{D},g} = (\rho_g r_g)^{1/2}$ and the agreement terms vanish. With $\rho_g = \rho$, $\mathrm{Eff}^{TPR} = \sum_g r_g/(\zeta_g p_g^2)/\{\rho(\sum_g \sqrt{r_g}/p_g)^2\} = \Gamma/\rho$. By Cauchy–Schwarz, $(\sum_g \sqrt{\zeta_g}\cdot\sqrt{r_g}/(\sqrt{\zeta_g}p_g))^2 \leq \sum_g \zeta_g \cdot \sum_g r_g/(\zeta_g p_g^2)$, so $\Gamma \geq 1$. Applying (24) to the two strata $\{G=0\}$, $\{G=1\}$, whose influence values have mean zero and variances

$r_g/(N\zeta_g p_g)^2$, gives exactly Γ. For the error difference with $\mu_g = 0$ (hence $\theta = \sum_g \zeta_g \rho_g \mu_g = 0$), (S9) equals $\rho/\rho^2 = 1/\rho$.

(c) With one group, $\zeta = 1$, $\delta = \rho\pi_{\mathcal{D}} a/p$ and $c_{\mathcal{D}} = a\pi_{\mathcal{D}}$. The numerator of (S8) is $(\rho\pi_{\mathcal{D}} - \delta^2 p)/p^2 = \rho\pi_{\mathcal{D}}\{1 + O(\rho)\}/p^2$. Since $\delta = O(\rho)$, $\sigma^2_{\mathcal{D}} = \pi_{\mathcal{D}} - a^2\pi^2_{\mathcal{D}} + O(\rho) = \pi_{\mathcal{D}}\beta^2 + O(\rho)$, and $(1-\rho)|\delta|\{\pi_{\mathcal{A}}(1-\pi_{\mathcal{A}})\}^{1/2} = \rho\sqrt{\pi_{\mathcal{D}}}\,\lambda + O(\rho^2)$. The denominator is therefore $\rho^2\pi_{\mathcal{D}}(\beta+\lambda)^2\{1 + O(\rho)\}/p^2$ (assuming $\beta + \lambda > 0$, which fails only in the degenerate case $|a| = \pi_{\mathcal{D}} = 1$), and the ratio is (S11). The value of $p = \rho\pi_{\mathcal{D}} + (1-\rho)\pi_{\mathcal{A}}$ changes by $O(\rho)$, which is absorbed in the $O(\rho)$ term.

(d) When $\rho_g = 1$ the strata $\mathcal{A}_g$ are empty and $\mathcal{D}_g$ is the whole group, so the design is stratification by group, and (S8), (S9) are continuous in ρ_g. For $\delta_g = 0$ the limit of (S8) is Γ by (b). For the error difference with $\rho_g = 1$, μ_g is the group-wise error difference and the limit is $(1-\theta^2)/(\sum_g \zeta_g\sqrt{1-\mu_g^2})^2$, which equals one when $\mu_0 = \mu_1 = \theta$.

Equation (S12). If $e_h = 0$ for all h, (18) gives $\sigma^2_{ht} = \rho_h/N^2$, and $\sigma^2_t = \rho/N^2$ with $\theta = 0$. Since $0 \le \rho_h \le 1$, $\sqrt{\rho_h} \ge \rho_h$, so $(\sum_h W_h\sqrt{\rho_h})^2 \ge \rho^2$ and Eff $\le 1/\rho$; Jensen's inequality gives $(\sum_h W_h\sqrt{\rho_h})^2 \le \rho$ and Eff ≥ 1. Equality in the upper bound requires $\rho_h \in \{0,1\}$ for every h. For (S13), let $\mathcal{H}_D$ be the strata with $0 < K < M$, so that $\sum_{h\in\mathcal{H}_D} W_h = \rho_{\text{any}}$ and $\rho_h = 0$ outside $\mathcal{H}_D$. Cauchy–Schwarz gives

$$\Big(\sum_{h\in\mathcal{H}_D} W_h^{1/2}\cdot W_h^{1/2}\rho_h^{1/2}\Big)^2 \le \rho_{\text{any}}\sum_h W_h\rho_h = \rho_{\text{any}}\rho,$$

so Eff $\ge \rho/(\rho_{\text{any}}\rho) = 1/\rho_{\text{any}}$, with equality if and only if ρ_h is constant on $\mathcal{H}_D$. (When strata are merged across $K = 0$ or $K = M$ and $0 < K < M$ because of the minimum stratum size, ρ_{any} is replaced by the total share of the strata that contain any instance with $0 < K < M$.) □

S1.7 Proof of Proposition S9

(a) The proxy is a fixed function of the pool, so $\tilde{n}$ satisfies (A1) whenever the minimum per stratum is at least two; Proposition 1 applies without change.

(b) With $\tilde{n}_h = LN_h\tilde{A}_h/\sum_k N_k\tilde{A}_k$,

$$\Phi_0(\tilde{n}) = \sum_h \frac{N_h^2A_h^2}{\tilde{n}_h} = \frac{1}{L}\Big(\sum_k N_k\tilde{A}_k\Big)\Big(\sum_h \frac{N_hA_h^2}{\tilde{A}_h}\Big),$$

and $\Phi_0(n^*) = L^{-1}(\sum_h N_hA_h)^2$ by Proposition 5(a). Writing $\tilde{A}_h = x_hA_h$,

$$\sum_h N_h\tilde{A}_h = \Big(\sum_k N_kA_k\Big)\sum_h \omega_h x_h,$$
$$\sum_h \frac{N_hA_h^2}{\tilde{A}_h} = \Big(\sum_k N_kA_k\Big)\sum_h \frac{\omega_h}{x_h},$$

which gives (S14). By Cauchy–Schwarz,

$$1 = \Big(\sum_h \sqrt{\omega_h x_h}\sqrt{\omega_h/x_h}\Big)^2 \le \Big(\sum_h \omega_h x_h\Big)\Big(\sum_h \omega_h/x_h\Big),$$

with equality if and only if x_h is constant on the strata with $\omega_h > 0$.

(c) For $x \in [x_{\min}, x_{\max}]$, $(x - x_{\min})(x_{\max} - x) \ge 0$, that is $x + x_{\min}x_{\max}/x \le x_{\min} + x_{\max}$. Averaging with weights ω_h, $\sum\omega_h x_h + x_{\min}x_{\max}\sum\omega_h/x_h \le x_{\min} + x_{\max}$. By the arithmetic–geometric mean inequality the left side is at least $2\{x_{\min}x_{\max}(\sum\omega_h x_h)(\sum\omega_h/x_h)\}^{1/2}$; squaring gives the bound.

(d) Proportional allocation corresponds to $\tilde{A}_h \equiv 1$, i.e. $x_h = 1/A_h$, which gives the stated comparison. For the negative statement consider two strata of equal size $N/2$ with equal true values $A_1 = A_2$ and equal means, so that SRS, proportional allocation and the optimal allocation coincide. If the proxy gives $\tilde{A}_1 \approx 0$, stratum 1 receives the minimum ℓ labels and stratum 2 receives $L-\ell$, and $\Phi_0(\tilde{n})/\Phi_0(n^*) = \frac{L}{4}\{1/\ell + 1/(L-\ell)\}$, which is about $L/(4\ell)$; with $L = 1000$ and $\ell = 10$ the weighted variance is 25 times that of SRS.

(e) Since $n_h \ge \eta n_h^{\text{prop}}$ and $n_h \ge (1-\eta)\tilde{n}_h$, $1/n_h \le 1/(\eta n_h^{\text{prop}})$ and $1/n_h \le 1/\{(1-\eta)\tilde{n}_h\}$; multiply by $N_h^2A_h^2$ and sum; this gives the two bounds on Φ_0. For the comparison of proportional allocation with SRS, fix a target and write S^2, S_h^2, $\bar{z}$, $\bar{z}_h$ for its variances and means. The analysis-of-variance identity $(N-1)S^2 = \sum_h (N_h - 1)S_h^2 +$

$\sum_h N_h(\bar{z}_h - \bar{z})^2$ gives

$$\begin{aligned} NS^2 - \sum_h N_h S_h^2 &= \frac{N}{N-1}\sum_h N_h(\bar{z}_h - \bar{z})^2 \\ &\quad - \sum_h \frac{N - N_h}{N-1} S_h^2 \\ &\geq -\sum_h S_h^2 \\ &\geq -\frac{1}{\min_h N_h}\sum_h N_h S_h^2. \end{aligned}$$

With $f = L/N$, proportional allocation has variance $V^{\text{prop}} = (1-f)(N/L)\sum_h N_h S_h^2$ and SRS has $V^{SRS} = (1-f)(N/L)NS^2$, so $V^{\text{prop}}(1 - 1/\min_h N_h) \leq V^{SRS}$. Summing over targets with weights w_t gives the same inequality for the weighted variance Φ^{SRS}. To pass from Φ_0 to the weighted variance $\Phi = \Phi_0 - \sum_h N_h A_h^2$ with finite-population corrections, note that $\Phi_0(n^{\text{prop}}) = (N/L)\sum_h N_h A_h^2$, so $\sum_h N_h A_h^2 = \{f/(1-f)\}\,\Phi(n^{\text{prop}})$ with $f = L/N$, and $\Phi(n) \leq \Phi_0(n^{\text{prop}})/\eta - \sum_h N_h A_h^2 = \Phi(n^{\text{prop}})\{1/\eta + (1/\eta - 1)f/(1-f)\} = \Phi(n^{\text{prop}})\,(1-\eta f)/\{\eta(1-f)\}$. Combining with $\Phi(n^{\text{prop}}) \leq \Phi^{\text{SRS}}/(1 - 1/\min_h N_h)$ proves (e): the bound $1/\eta$ holds as stated for Φ_0 (finite-population corrections ignored) and, with them, with $1/\eta$ replaced by $(1-\eta f)/\{\eta(1-f)\} = \eta^{-1}\{1 + O(f)\}$. □

S1.8 Proof of Propositions 3 and S3

(a) Within stratum h, $e_i = z_i - \tilde{z}_i$, so $S_{e,h}^2 = S_{z,h}^2 - 2C_h + S_{\tilde{z},h}^2$ with $C_h = \text{Cov}_h(z, \tilde{z}) = \beta_h S_{\tilde{z},h}^2 = R_h S_{z,h} S_{\tilde{z},h}$. Multiplying by c_h and summing, $V_0 - V_\psi = \sum_h c_h(2C_h - S_{\tilde{z},h}^2) = \sum_h c_h S_{\tilde{z},h}^2(2\beta_h - 1)$. The sufficient condition is $\beta_h > 1/2 \iff R_h S_{z,h}/S_{\tilde{z},h} > 1/2$.

(b) For the member λ, the residual influence value in stratum h is $z_i - \lambda_h \tilde{z}_i$ (the proxy $\lambda_h \psi_i$ is label-free, so Proposition 1 applies and the linearised variance has this form). Hence $S_{z-\lambda\tilde{z},h}^2 = S_{z,h}^2 - 2\lambda_h C_h + \lambda_h^2 S_{\tilde{z},h}^2 = S_{z,h}^2 - \beta_h^2 S_{\tilde{z},h}^2 + (\lambda_h - \beta_h)^2 S_{\tilde{z},h}^2$, and $\beta_h^2 S_{\tilde{z},h}^2 = C_h^2/S_{\tilde{z},h}^2 = R_h^2 S_{z,h}^2$, which is the displayed formula. Setting $\lambda \equiv 0$ and $\lambda \equiv 1$ gives the two decompositions. The function $\lambda \mapsto (\lambda - \beta_h)^2$ is a parabola centred at β_h, and the clipped value is the point of $[0,1]$ closest to β_h, so it is no larger than at $\lambda = 0$ or $\lambda = 1$.

(c) If $\text{Cov}_h(e, \tilde{z}) = 0$ then $C_h = \text{Cov}_h(\tilde{z} + e, \tilde{z}) = S_{\tilde{z},h}^2$, so $\beta_h = 1$ and (a) gives $V_0 - V_\psi = \sum_h c_h S_{\tilde{z},h}^2$. □

S1.9 Proof of Propositions 4 and S4

(a) The c_h are constants (n_h is label-free), so by (S3) $E_\pi V_0 - E_\pi V_\psi = \sum_h c_h\{(S_{\tilde{z},h}^2 + \bar{\sigma}_h^2) - \bar{\sigma}_h^2\} = \sum_h c_h S_{\tilde{z},h}^2$, which vanishes iff $S_{\tilde{z},h}^2 = 0$ in every stratum with $c_h > 0$, that is every stratum that is not fully enumerated.

(b) $E_\pi V_\psi = \sum_h a_h$ and $E_\pi V_0 = \sum_h c_h(\bar{\sigma}_h^2 + S_{\tilde{z},h}^2) = \sum_h a_h(1 + \nu_h)$, which gives the ratio and $G \geq 1$. By (S3) the expected within-stratum regression slope is one and $E_\pi S_{z,h}^2 = \bar{\sigma}_h^2(1 + \nu_h)$, so the (expected) squared correlation is $S_{\tilde{z},h}^4/(S_{\tilde{z},h}^2 S_{z,h}^2) = \nu_h/(1 + \nu_h)$, hence $1 + \nu_h = 1/(1 - R_h^2)$.

(c) Consider a stratum in which $|D_i| = 1$ for all i, with δ_g, γ_g negligible, so that for TPR $z_i = s_g Y_i D_i/P_g$, $\tilde{z}_i = s_g \pi_i D_i/P_g$, $e_i = s_g(Y_i - \pi_i)D_i/P_g$ and $d_i = s_g D_i/P_g$. Ignoring the common factor $\kappa_h = N_h/(N_h - 1)$, which cancels in ν_h, and using $D_i^2 = 1$: $\bar{\sigma}_h^2 = P_g^{-2}\overline{\pi(1-\pi)}$; $S_{\tilde{z},h}^2 = P_g^{-2}\{\overline{\pi^2} - (\overline{\pi D})^2\} = P_g^{-2}\{\overline{\pi^2} - \bar{\pi}^2\bar{a}^2\}$, using that D and π are uncorrelated within the stratum. Hence $1 + \nu = \{\overline{\pi(1-\pi)} + \overline{\pi^2} - \bar{\pi}^2\bar{a}^2\}/\overline{\pi(1-\pi)} = \bar{\pi}(1 - \bar{\pi}\bar{a}^2)/\overline{\pi(1-\pi)}$. For FPR, (Y, π) are replaced by $(1-Y, 1-\pi)$, which leaves $\overline{\pi(1-\pi)}$ unchanged, and $\bar{\pi}$ by $1 - \bar{\pi}$. For ERR, $d_i = -2D_i/N$, $\tilde{z}_i = D_i(1 - 2\pi_i)/N$, so $\bar{\sigma}_h^2 = 4N^{-2}\overline{\pi(1-\pi)}$ and $S_{\tilde{z},h}^2 = N^{-2}\{\overline{(1-2\pi)^2} - \bar{a}^2(1 - 2\bar{\pi})^2\}$; since $(1-2\pi)^2 + 4\pi(1-\pi) = 1$, $\bar{\sigma}_h^2 + S_{\tilde{z},h}^2 = N^{-2}\{1 - \bar{a}^2(1-2\bar{\pi})^2\}$, which gives the third formula. In an agreement stratum $D_i = 0$: $e_i^{ERR} = \tilde{z}_i^{ERR} = 0$; for TPR $d_i = -s_g\delta_g/P_g$ is constant, $\bar{\sigma}_h^2 = (\delta_g/P_g)^2\overline{\pi(1-\pi)}$ and $S_{\tilde{z},h}^2 = (\delta_g/P_g)^2 S_{\pi,h}^2$, which gives the formula; FPR is the same. The numerical examples are the case $\pi_i \equiv \bar{\pi}$, $\bar{a} = 0$: $\overline{\pi(1-\pi)} = \bar{\pi}(1-\bar{\pi})$, so the three gains are $\bar{\pi}/\{\bar{\pi}(1-\bar{\pi})\}$, $(1-\bar{\pi})/\{\bar{\pi}(1-\bar{\pi})\}$ and $1/\{4\bar{\pi}(1-\bar{\pi})\}$. □

S1.10 Proof of Proposition S5

(a) is Proposition 1 (only label-freeness of the proxy is used).

(b) Under independent labels with $P(Y_i = 1) = q_i$, write $z_i = \tilde{z}_i^q + (Y_i - q_i)d_i$ and $e_i = z_i - \tilde{z}_i^\pi = (Y_i - q_i)d_i + (\tilde{z}_i^q - \tilde{z}_i^\pi)$, using $\tilde{z}_i^q - \tilde{z}_i^\pi = (q_i - \pi_i)d_i$. The first term has mean zero and independent coordinates, the second is constant. As in

the proof of Proposition S2, $E_q S^2_{z,h} = S^2_{\tilde{z}^q,h} + \bar{\sigma}^2_h(q)$ and $E_q S^2_{e,h} = \bar{\sigma}^2_h(q) + S^2_{\tilde{z}^q - \tilde{z}^\pi,h}$. Subtracting and multiplying by c_h gives (S5). If $\tilde{z}^\pi_i - \bar{\tilde{z}}^\pi_h = \kappa(\tilde{z}^q_i - \bar{\tilde{z}}^q_h)$, then $S^2_{\tilde{z}^q - \tilde{z}^\pi,h} = (1-\kappa)^2 S^2_{\tilde{z}^q,h}$, which is smaller than $S^2_{\tilde{z}^q,h}$ iff $0 < \kappa < 2$.

(c) By Minkowski's inequality for standard deviations, $S_{e,h} \le S_{z,h} + S_{\tilde{z},h} \le S_{z,h}(1 + \max_k S_{\tilde{z},k}/S_{z,k})$, with equality in the first step iff $z_i - \bar{z}_h = -s_h(\tilde{z}_i - \bar{\tilde{z}}_h)$ for a constant $s_h > 0$ (perfect anti-correlation, $S_{\tilde{z},h}/S_{z,h} = 1/s_h$), and in the second iff this ratio is the same in every stratum. Multiply by c_h and sum.

(d) The first statement is Proposition S3(a). With $\beta_h = 0$, (a) of that proposition gives $V_0 - V_\psi = -\sum_h c_h S^2_{\tilde{z},h}$. □

S1.11 Proof of Proposition S6

By (14) with coefficients $\hat{\lambda}_h$ and λ^∞_h, $\hat{T}_{\hat{\lambda}\psi} - \hat{T}_{\lambda^\infty\psi} = -\sum_h N_h(\hat{\lambda}_h - \lambda^\infty_h)(\bar{\psi}_{s_h} - \bar{\psi}_h)$, because $F_h = N_h\bar{\psi}_h$. Each $\bar{\psi}_{s_h} - \bar{\psi}_h$ is $O_p(\{(1-f_h)/n_h\}^{1/2})$ coordinatewise, so, with $W_h n_h^{-1/2} \le r$ and a fixed number of strata, $\|\hat{\tau}_{\hat{\lambda}\psi} - \hat{\tau}_{\lambda^\infty\psi}\| = o_p(r)$. The Taylor expansion (S1) applies to both estimators (it needs only (A2) and the bound on the contributions), so $\hat{\theta}_{\hat{\lambda}\psi} - \hat{\theta}_{\lambda^\infty\psi} = o_p(r) = o_p(V^{1/2})$ by (A3), and Slutsky's lemma gives the same limit. By Proposition S3(a), $\lambda^\infty_h = \min\{1, \max(0, \beta_h)\}$ gives $V(\lambda^\infty) \le \min(V_0, V_\psi)$. (The sample slope $\hat{\beta}_h$ is a ratio of sample moments of bounded variables and is consistent for β_h whenever $S^2_{\tilde{z},h}$ is bounded below by a positive constant times N^{-2}; otherwise the stratum's contribution is negligible by (A4).)

For the second-order statement, write $\hat{\beta}_h - \beta_h \approx \widehat{\mathrm{Cov}}(\varepsilon, \tilde{z})/S^2_{\tilde{z},h}$ with $\varepsilon_i = z_i - \beta_h \tilde{z}_i$, a population-regression residual with variance $S^2_{\varepsilon,h} = (1 - R^2_h)S^2_{z,h}$. Its variance is $\approx S^2_{\varepsilon,h}/(n_h S^2_{\tilde{z},h})$. The leading error term of the first part is the product $(\hat{\beta}_h - \beta_h)\, N_h(\bar{\psi}_{s_h} - \bar{\psi}_h)$ of two asymptotically uncorrelated factors with variances $S^2_{\varepsilon,h}/(n_h S^2_{\tilde{z},h})$ and $N^2_h(1-f_h)S^2_{\tilde{z},h}/n_h$ (after applying the gradient), whose product is $c_h S^2_{\varepsilon,h}/n_h = c_h(1 - R^2_h)S^2_{z,h}/n_h$, a fraction $1/n_h$ of the stratum's optimal contribution $c_h(1 - R^2_h)S^2_{z,h}$. The plug-in estimator uses the residuals $\hat{z}_i - \hat{\beta}_h \hat{\tilde{z}}_i$ in the sample, whose sample variance (divisor $n_h - 1$) is, by the usual least-squares calculation with an intercept and a slope, smaller in expectation than the variance of the fixed-β_h residuals by the factor $(n_h - 2)/(n_h - 1)$. Both statements are first-order expansions in $1/n_h$ and are meant as an explanation of the finite-sample behaviour, not as a bound. □

S1.12 Justification of Remarks 1 and S1

Let Z be the set of strata in which $s^2_{\hat{e},h} = 0$ and $n_h \ge 2$, $f_h < 1$. The floored estimator replaces $s^2_{\hat{e},h}$ by $\tilde{s}^2_h$ for $h \in Z$ and is unchanged otherwise; since $\tilde{s}^2_h \ge 0$, $\hat{V}^{\mathrm{floor}}_\psi \ge \hat{V}_\psi$ pointwise, and the point estimate is the same. If $d_i = 0$ for all $i \in U_h$, then $\hat{d}_i = 0$ and $\tilde{s}^2_h = 0$ (structural zero), so no floor is added. If $N^2 S^2_{e,h} \ge c > 0$, then $a^\top \Sigma^\xi_h a$ is bounded away from zero, and by Lemma S1 and the consistency of $\hat{a}$ the probability $P(s^2_{\hat{e},h} = 0) \to 0$, so $P(h \in Z) \to 0$. Let $\mathcal{H}_0$ be the strata with $N^2 S^2_{e,h} \to 0$. Their contribution to V_ψ is vanishing relative to V_ψ whenever (A4) holds in its second form, and their contribution to $\hat{V}^{\mathrm{floor}}_\psi$ is at most $\sum_{h \in \mathcal{H}_0} c_h \tilde{s}^2_h$. If this is $o(V_\psi)$, then $\hat{V}^{\mathrm{floor}}_\psi / V_\psi \to 1$ and Proposition 1(d) holds for the floored interval; if not, the floored interval is conservative, $\liminf P(\theta \in \hat{\theta}_\psi \pm z_{1-\alpha/2}\{\hat{V}^{\mathrm{floor}}_\psi\}^{1/2}) \ge 1 - \alpha$, because $\hat{V}^{\mathrm{floor}}_\psi \ge \hat{V}_\psi$ and $\hat{V}_\psi / V_\psi \to 1$. The same argument applies to $\max\{s^2_{\hat{e},h}, \eta\tilde{s}^2_h\}$ if $\eta\tilde{s}^2_h < S^2_{e,h}$ in the limit for strata that carry variance, which holds with $\eta \le 1$ for a calibrated proxy because then $E_\pi S^2_{e,h} = \bar{\sigma}^2_h$ (S3). □

S2 Additional theoretical results

This section collects the theoretical results that are not needed to follow the main text: the residual influence values, the optimal coefficient and calibrated form of the difference estimator, closed-form design effects, misspecification bounds, tuning of the difference coefficient, the zero-variance floor, allocation for the difference estimator, and the design effect as a function of the disagreement rate. Equations and statements of the main text are cited by their main-text numbers.

S2.1 Residual influence values

The residual influence values of the difference estimator have an equally explicit form, and a feature

that the plain values lack: the label enters only through $Y_i - \pi_i$.

Proposition S2 (Residual influence values) *Let $d_i = \nabla f(T)^\top (u_i - v_i)$, which does not depend on the labels. Then $e_i = (Y_i - \pi_i)\, d_i$, and for $G_i = g$*

$$e_i^{TPR} = s_g \frac{(Y_i - \pi_i)(D_i - \delta_g)}{P_g},$$
$$e_i^{FPR} = s_g \frac{\{(1 - Y_i) - (1 - \pi_i)\}(D_i - \gamma_g)}{Q_g},$$
$$e_i^{ERR} = -\frac{2(Y_i - \pi_i)D_i}{N}, \tag{S2}$$

and $\tilde{z}_i = z_i - e_i$ equals (16) with Y_i replaced by π_i. In a stratum where the pair agrees, $e_i^{ERR} = 0$ for every i: the stratum contributes neither variance nor error to the difference estimator of θ^{ERR}. Under the working model that $Y_1, \ldots, Y_N$ are independent with $P(Y_i = 1) = \pi_i$ (the proxy is calibrated), with expectation E_π, the gradient evaluated at the non-random model-expected totals $T_\pi = \sum_{i \in U} \psi_i$ (as in the design; this changes d_i by a relative $O(r)$ only), and with $\bar{\sigma}_h^2 = N_h^{-1} \sum_{U_h} \pi_i(1 - \pi_i) d_i^2$ and $C_h = \mathrm{Cov}_h(z, \tilde{z})$ the stratum covariance,

$$E_\pi(e_i) = 0, \quad \mathrm{Var}_\pi(e_i) = \pi_i(1 - \pi_i) d_i^2,$$
$$E_\pi S_{e,h}^2 = \bar{\sigma}_h^2, \quad E_\pi S_{z,h}^2 = S_{\tilde{z},h}^2 + \bar{\sigma}_h^2,$$
$$E_\pi C_h = S_{\tilde{z},h}^2. \tag{S3}$$

S2.2 Difference estimator: coefficient and calibrated form

Recall the family $\hat{T}_{\lambda\psi}$ obtained by replacing ψ by $\lambda_h \psi$ in stratum h ($\lambda \equiv 0$ is the plain estimator and $\lambda \equiv 1$ the difference estimator); every member is unbiased for T by Proposition 1(a), and β_h, R_h are as in Section 5.4. The sufficient condition of Proposition 3 can be restated as $R_h > S_{\tilde{z},h}/(2S_{z,h})$ in every stratum: the within-stratum correlation of z_i with its proxy mean must exceed half the ratio of their standard deviations.

Proposition S3 (Variance of the difference estimator, continued) *Under the notation of Proposition 3, for any label-free proxy and any allocation:*

(a) Coefficient. $V(\lambda) = \sum_h c_h \{S_{z,h}^2 (1 - R_h^2) + (\lambda_h - \beta_h)^2 S_{\tilde{z},h}^2\}$. *It is minimised at $\lambda_h = \beta_h$, with $V(\beta) = \sum_h c_h S_{z,h}^2 (1 - R_h^2)$; moreover $V_0 = V(\beta) + \sum_h c_h \beta_h^2 S_{\tilde{z},h}^2$ and $V_\psi = V(\beta) + \sum_h c_h (1 - \beta_h)^2 S_{\tilde{z},h}^2$. With the clipped coefficient $\lambda_h = \min\{1, \max(0, \beta_h)\}$, $V(\lambda) \le \min(V_0, V_\psi)$.*

(b) Calibrated proxy, design-based form. *If $\mathrm{Cov}_h(e, \tilde{z}) = 0$ in every stratum (the residual is uncorrelated with the proxy influence within strata), then $\beta_h = 1$ and $V_0 - V_\psi = \sum_h c_h S_{\tilde{z},h}^2 \ge 0$ exactly.*

In terms of correlations, $V_0 - V_\psi = \sum_h c_h\, S_{\tilde{z},h}\, (2R_h S_{z,h} - S_{\tilde{z},h})$.

Part (a) of Proposition 3 shows that, unlike what is often assumed for prediction-powered estimators, the sign of the gain is not automatic: $\lambda = 1$ is the right coefficient only if the proxy influence is neither over- nor under-dispersed relative to its regression on z; the harm from an over-dispersed proxy is quantified in Proposition S5.

S2.3 Design-effect ratio: closed forms and discussion

The following completes Proposition 4; its proof is in Supplementary Section S1.

Proposition S4 (Closed forms of the stratum gain) *Under the setting of Proposition 4, for $\theta^{TPR}, \theta^{FPR}$ and θ^{ERR} the stratum gains depend on the proxy as follows. Let $\overline{\pi(1-\pi)}$ denote the stratum mean of $\pi_i(1 - \pi_i)$. In a* disagreement stratum *($|D_i| = 1$ on U_h, D_i and π_i uncorrelated within the stratum, $\bar{a}$ the stratum mean of D_i, and δ_g, γ_g negligible),*

$$1 + \nu^{TPR} = \frac{\overline{\bar{\pi}(1 - \bar{\pi}\bar{a}^2)}}{\overline{\pi(1-\pi)}},$$
$$1 + \nu^{FPR} = \frac{(1 - \bar{\pi})\{1 - (1 - \bar{\pi})\bar{a}^2\}}{\overline{\pi(1-\pi)}},$$
$$1 + \nu^{ERR} = \frac{1 - \bar{a}^2(1 - 2\bar{\pi})^2}{4\,\overline{\pi(1-\pi)}}. \tag{S4}$$

In an agreement stratum *($D_i = 0$) the error difference has $e_i = \tilde{z}_i = 0$, and for the rate differences $1 + \nu = \{\overline{\pi(1-\pi)} + S_{\pi,h}^2\}/\overline{\pi(1-\pi)}$, the inverse of the share of the Bernoulli variance that the proxy does* not *explain.*

Four consequences. First, calibration alone guarantees $G \ge 1$: with a calibrated proxy the difference estimator is never worse than the plain one in expectation, for any allocation. Second, the gain is a within-stratum R^2, so it is governed by what the proxy influence $\tilde{z}_i$ explains of z_i *inside* a stratum, and the strata are already built to remove between-stratum variation. Third, $\tilde{z}_i$ contains two kinds of information: the proxy π_i and the known

structure of u_i, v_i (the signs and the pattern of D_i). The second kind is free and is the main source of the gain: in a two-sided disagreement stratum with a constant proxy $\pi_i \equiv \bar{\pi}$ and $\bar{a} = 0$, (S4) gives gains $1/(1-\bar{\pi})$ for θ^{TPR}, $1/\bar{\pi}$ for θ^{FPR} and $1/\{4\bar{\pi}(1-\bar{\pi})\}$ for θ^{ERR}, i.e. 1.4, 3.3 and 1.2 for $\bar{\pi} = 0.3$ and exactly 2, 2 and 1 for $\bar{\pi} = 1/2$. Only the error difference needs a sharp proxy: its gain is $\{1 - \overline{(1-2\pi)^2}\}^{-1}$ and equals one for $\pi \equiv 1/2$, which makes $\tilde{z} \equiv 0$ and the two estimators identical. Fourth, the gain is bounded for proxies of realistic sharpness, because the disagreement strata are the strata in which the compared models, and hence typically the averaged probabilities π_i, are least certain: a large G requires within-stratum R_h^2 close to one, that is a sharp proxy (π_i near 0 or 1) and a stratum in which the models still disagree. Where the proxy is that sharp the predictions would be nearly exact and few labels would be needed in any case. For disagreement strata with $\bar{\pi}$ between 0.2 and 0.5 the closed forms give gains of the order of one to three, on top of the stratification gain; Section S5.4 compares the prediction with the Monte Carlo ratios.

Gain and stratification multiply: for the same sample the efficiency of the difference estimator relative to SRS is, to the order of the approximations of Section 5.6, $\mathrm{Eff}_t \cdot G_t$, with Eff_t from (24) and G_t from (21). They are not independent: a stratifier that removes more of the proxy's within-stratum signal (a finer partition of the proxy range) reduces G and raises Eff_t.

S2.4 Misspecified proxy for the difference estimator

Proposition S5 (Misspecified proxy)*(a) For every proxy, $\hat{T}_\psi$ is design-unbiased and $\hat{V}_\psi$ is a consistent variance estimator (Proposition 1). Misspecification can affect efficiency only.*

(b) Let the labels be independent with $P(Y_i = 1) = q_i$, where q_i may differ from π_i, and let $\tilde{z}_i^q$, $\tilde{z}_i^\pi$ be the proxy influence values for q and π. With E_q the expectation under this model,

$$E_q(V_0 - V_\psi) = \sum_h c_h \left\{ S^2_{\tilde{z}^q,h} - S^2_{\tilde{z}^q - \tilde{z}^\pi,h} \right\}, \quad \text{(S5)}$$

so the difference estimator wins if and only if, within strata and on the c_h-weighted scale, the proxy error in influence units is smaller than the signal it should capture. If $\tilde{z}_i^\pi - \bar{\tilde{z}}_h^\pi = \kappa(\tilde{z}_i^q - \bar{\tilde{z}}_h^q)$ in every stratum (a proxy that is sharper by the factor κ than the truth), the estimator wins if and only if $0 < \kappa < 2$; for $\kappa < 0$ (anti-correlated proxy) it loses.

(c) In all cases $V_\psi \le (1 + \max_h S_{\tilde{z},h}/S_{z,h})^2 V_0$, with equality only if $\tilde{z}_i = -s\, z_i$ within each stratum for a constant $s > 0$ (perfect anti-correlation). Hence the possible loss is bounded by a function of the dispersion of the proxy influence relative to the plain one.

(d) With the clipped coefficient of Proposition S3(a) the variance never exceeds that of the plain estimator (to first order). In particular, an uninformative proxy, in the sense $\beta_h = 0$ ($\tilde{z}$ uncorrelated with z within strata), has optimal coefficient $\lambda_h = 0$ and then $V_\psi = V_0 + \sum_h c_h S^2_{\tilde{z},h}$: the fixed coefficient $\lambda = 1$ adds the proxy's own variance.

Two clarifications. The statement that the difference estimator can be worse than the plain estimator *only* if the proxy is anti-correlated is too strong: by (20) it is worse whenever $\beta_h < 1/2$, so a weakly aligned or over-dispersed proxy ($\kappa > 2$) also hurts, while a modestly noisy but correctly scaled proxy does not. And "uninformative" must be read in terms of $\tilde{z}_i$, not of π_i: a constant proxy $\pi_i \equiv \bar{\pi}$ still has $\tilde{z}_i$ varying with the known pattern of D_i, and for the rate differences it still gains, as the closed forms (S4) show; only for θ^{ERR} with $\bar{\pi} = 1/2$ is $\tilde{z} \equiv 0$, and only an inverted proxy ($\kappa < 0$) makes $\beta_h < 0$ for the error difference. Section S5.4 confirms both points numerically.

S2.5 Fixed versus tuned coefficient

Fixed versus tuned coefficient.

In prediction-powered inference [3, 4] and in the regression-estimator literature [5, 6] the coefficient is estimated from the labelled sample, here per stratum and per estimand: $\hat{\lambda}_h = \widehat{\mathrm{Cov}}(\hat{z}, \tilde{\hat{z}}) / \widehat{\mathrm{Var}}(\tilde{\hat{z}})$ clipped to $[0, 1]$.

Proposition S6 (Estimated coefficient) *Suppose $\hat{\lambda}_h$ is a function of the labelled sample of stratum h with $\hat{\lambda}_h - \lambda_h^\infty \to 0$ in probability, for deterministic λ_h^∞ (for the clipped sample slope, $\lambda_h^\infty = \min\{1, \max(0, \beta_h)\}$ whenever β_h converges). Under (A1)–(A4) for the fixed coefficient λ^∞, $\hat{\theta}_{\hat{\lambda}\psi} - \hat{\theta}_{\lambda^\infty \psi} = o_p(V^{1/2})$; hence*

the tuned estimator has the same asymptotic normal limit and linearised variance $V(\lambda^\infty)$ as the one with the optimal fixed coefficient, and is never asymptotically worse than the plain estimator. At second order, estimating $\hat\lambda_h$ adds to the variance of stratum h approximately $c_h(1-R_h^2)S^2_{z,h}/n_h$, a relative increase of order $1/n_h$ in a stratum that is already optimally tuned, and the plug-in estimator $\hat V_\psi$ computed with the fitted $\hat\lambda_h$ does not account for it (it is biased downwards by a factor of the same order).

The consequence is a trade-off that is independent of the particular data. The best that tuning can add to $\lambda = 1$ is the factor $V_\psi/V(\lambda^*) = 1+\sum_h c_h(1-\beta_h)^2 S^2_{\tilde z,h}/V(\beta)$ (20), which is small when the proxy is nearly calibrated ($\beta_h \approx 1$): then tuning gains little in variance but costs an $O(1/n_h)$ inflation in each of the H strata and a downward-biased variance estimator, which matters when n_h is small, as at the budgets of this paper (n_h of the order of 10). With a miscalibrated proxy the gain from tuning can be real (see the $\kappa \neq 1$ cases in Section S5.4), but it is paid for in coverage. The calibrated difference estimator with $\lambda = 1$ has no estimated parameters, exact unbiasedness and a plug-in variance that is unbiased for the totals (Proposition 1(a)), which is why we adopt it as the main estimator. Section S5.4 reports the predicted and the observed gain of tuning together with the coverage of each version.

S2.6 Zero-variance strata and the variance floor

The main text states the floor briefly (Remark 1); the full discussion follows.

Remark S1 (Zero-variance strata and the variance floor, detailed) The Wald interval needs $\hat V_\psi > 0$ and $\hat V_\psi \approx V_\psi$, and both fail when a stratum's sample shows no variation in $\hat e_i$ although $S^2_{e,h} > 0$. For the difference estimator this has three sources. (i) *Structural zeros*: strata with $d_i \equiv 0$ (the error difference where the pair agrees, Proposition S2), where $e_i \equiv 0$ in the whole stratum; the estimate is exact there and $\hat V_\psi$ correctly contributes zero. (ii) *Sampling zeros*: a stratum with few positives (or negatives) in which the sample has none. For the plain estimator $\hat z_i = 0$ for every sampled unit, so $s^2_{\hat z,h} = 0$ exactly; for the difference estimator $\hat e_i = -\pi_i \hat d_i$ varies with $\pi_i \hat d_i$, so exact zeros are rarer, but the sample variance is still far below the population variance $\bar\sigma^2_h = N_h^{-1}\sum_{U_h}\pi_i(1-\pi_i)d_i^2$ whenever the missed positives matter. With a calibrated proxy of level π and constant d, the probability of a sample without positives is $(1-\pi)^{n_h}$, which is the zero-count failure of the Wald interval for a binomial proportion [7]. (iii) *Exact proxies*: if $\pi_i \in \{0,1\}$ equals Y_i then $e_i \equiv 0$ and (A3) fails; the estimator is then exact, but the proxy is unlikely to be exact on the whole pool. The floor of Section 9 carries over: replace $s^2_{\hat e,h}$ by the anticipated residual variance

$$\tilde s^2_h = N_h^{-1}\sum_{i\in U_h}\pi_i(1-\pi_i)\hat d_i^2, \qquad \text{(S6)}$$
$$\hat d_i = \nabla f(\hat T_\psi)^\top (u_i - v_i),$$

whenever $s^2_{\hat e,h} = 0$ (or, more conservatively, use $\max\{s^2_{\hat e,h}, \eta\tilde s^2_h\}$ with $\eta \in (0,1]$). Because $\tilde s^2_h = 0$ exactly when $d_i \equiv 0$ on U_h, structural zeros are untouched. The floor is label-free, never decreases $\hat V_\psi$, leaves the point estimate unchanged, and (Supplementary Section S1) does not affect Proposition 1(c)–(d) under Assumption (A5) (the floor is asymptotically negligible): the floor is active with probability tending to zero in every stratum with $S^2_{e,h}$ bounded away from zero, and in strata with vanishing variance it is, if anything, conservative. It is a finite-sample safeguard, not an asymptotic correction, and Section S5.4 reports its effect on coverage.

S2.7 Allocation for the difference estimator

In the design the S^2_{ht} are unknown. They are replaced by their expectations under the working model $Y_i \sim \text{Bernoulli}(\pi_i)$ independently, with the gradient evaluated at the model-expected totals; up to terms of order $1/N_h$ this is the anticipated variance of Isaki and Fuller [8]. By (S3) the anticipated stratum variances are (23), with $\bar\sigma^2_{ht} = N_h^{-1}\sum_{U_h}\pi_i(1-\pi_i)d^2_{it}$. The plain rule is what the design uses (Section 4). The residual rule is the Neyman allocation for the prediction-powered residual [4]; it depends on the proxy only through $\pi_i(1-\pi_i)$ and the label-free d_i, so a proxy of any sharpness gives a well-defined allocation.

Proposition S7 (Allocation for the difference estimator) *Let n^ψ be the optimal allocation for the difference estimator ($A_h = A_h^\psi$, $A_h^{\psi\,2} = \sum_t w_t S^2_{e,ht}$) and $\tilde n^0$ the allocation obtained from the plain anticipated variances. Put $\tilde\nu_h = \sum_t w_t S^2_{\tilde z,ht}/\sum_t w_t \bar\sigma^2_{ht}$, $x_h = (1+\tilde\nu_h)^{1/2}$ and assume the anticipated values are exact*

for the residual variances ($\tilde{A}_h^\psi = A_h^\psi$). Ignoring caps, minima and rounding, the weighted variance of the difference estimator with the wrong *(plain) allocation satisfies*

$$\frac{\Phi_\psi(\tilde{n}^0)}{\Phi_\psi(n^\psi)} = \Big(\sum_h \omega_h x_h\Big)\Big(\sum_h \omega_h / x_h\Big) \le \frac{(x_{\max}+x_{\min})^2}{4 x_{\max} x_{\min}}, \qquad \omega_h = \frac{N_h A_h^\psi}{\sum_k N_k A_k^\psi}, \tag{S7}$$

with equality to one iff $\tilde{\nu}_h$ is constant over the strata with $\omega_h > 0$. The same expressions hold, with x_h replaced by $1/x_h$ and $\omega_h \propto N_h A_h^0$, for the plain estimator allocated with the residual variances. Consequently, if $\tilde{\nu}_h \in [0, \nu_{\max}]$, the loss from using the wrong rule is at most $\{1+(1+\nu_{\max})^{1/2}\}^2/\{4(1+\nu_{\max})^{1/2}\}$ in either direction, for instance $9/8$ for $\nu_{\max} = 3$.

So the choice between the two rules is a second-order matter wherever the proxy's signal-to-noise ratio $\tilde{\nu}_h$ is similar across strata, which is the case for strata with comparable disagreement structure, and it matters when $\tilde{\nu}_h$ is large in some strata and small in others (for example, very sharp proxies in the agreement strata of the rate differences). Neither rule is optimal for the other estimator, and the plain rule remains the sensible default for the difference estimator: it is exactly Neyman for the plain estimator, is robust to a proxy whose calibration is not trusted (the residual rule depends on $\pi_i(1-\pi_i)$ and gives zero weight to strata in which the proxy is extreme), and its efficiency loss for the difference estimator is bounded by (S7). Section S5.4 reports the predicted and the observed loss.

S2.8 Design effect and the disagreement rate

Let $\sigma_{ht}^2 = (1-1/N_h)S_{ht}^2$ and $\sigma_t^2 = (1-1/N)S_t^2$ be the variances with divisor N_h and N. For a single target allocated by its own Neyman rule, and with $L/N \to 0$, the efficiency relative to SRS is (24) of the main text, $\mathrm{Eff}_t = \sigma_t^2/(\sum_h W_h \sigma_{ht})^2 \ge 1$, because $\sigma_t^2 = \sum_h W_h \sigma_{ht}^2 + \sum_h W_h(\bar{z}_h - \bar{z})^2 \ge (\sum_h W_h \sigma_{ht})^2$; equality holds only if all within-stratum standard deviations are equal and the stratum means coincide.

Combined with Proposition 2, equation (24) gives explicit formulas for the plain estimator. We state them for an idealised design in which, within each group, the strata separate the instances on which m and j disagree (set $\mathcal{D}_g$, share ρ_g of group g) from those on which they agree (set $\mathcal{A}_g$). This is exactly the pattern design when $M = 2$; for $M > 2$ it describes a design stratified for the pair (m, j), and is discussed below. Finite-population corrections are ignored ($L/N \to 0$ and $n_h/N_h \to 0$); they are retained in the exact linearised efficiency used in Section S5.

Corollary S8 (Design effect as a function of the disagreement rate) *Let $\zeta_g = N_g/N$, $p_g = P_g/N_g$, $r_g = N_g^{-1}\sum_{G_i=g} Y_i|D_i|$ (positives on which the pair disagrees, as a share of the group), $\pi_{\mathcal{D},g} = r_g/\rho_g$, $\pi_{\mathcal{A},g} = (p_g - r_g)/(1-\rho_g)$ and $a_g = c_{\mathcal{D},g}/\pi_{\mathcal{D},g}$, where $c_{\mathcal{D},g} = \delta_g p_g/\rho_g$ is the mean of $Y_i D_i$ over $\mathcal{D}_g$; $a_g \in [-1,1]$ is the net share of disputed positives on which m is right and j wrong.*

(a) For θ_{mj}^{TPR}, with own-target Neyman allocation on the strata $\{\mathcal{A}_g, \mathcal{D}_g\}_{g=0,1}$ and $L/N \to 0$,

$$\mathrm{Eff}^{TPR} = \frac{\sum_g (r_g - \delta_g^2 p_g)/(\zeta_g p_g^2)}{\big[\sum_g \{\rho_g \sigma_{\mathcal{D},g} + (1-\rho_g)\sigma_{\mathcal{A},g}\}/p_g\big]^2}, \tag{S8}$$

$\sigma_{\mathcal{A},g} = |\delta_g|\{\pi_{\mathcal{A},g}(1-\pi_{\mathcal{A},g})\}^{1/2}$ and $\sigma_{\mathcal{D},g}^2 = \pi_{\mathcal{D},g} - c_{\mathcal{D},g}^2 + \delta_g^2 \pi_{\mathcal{D},g}(1-\pi_{\mathcal{D},g}) - 2\delta_g c_{\mathcal{D},g}(1-\pi_{\mathcal{D},g})$. The same holds for θ_{mj}^{FPR} with Y replaced by $1-Y$. For θ_{mj}^{ERR}, with $\rho = \sum_g \zeta_g \rho_g$ and μ_g the mean of $D_i(1-2Y_i)$ over $\mathcal{D}_g$,

$$\mathrm{Eff}^{ERR} = \frac{\rho - (\theta_{mj}^{ERR})^2}{\big(\sum_g \zeta_g \rho_g \sqrt{1-\mu_g^2}\big)^2}. \tag{S9}$$

Any further split of $\mathcal{A}_g$ or $\mathcal{D}_g$ (for example $\mathcal{A}_g$ into $K = 0$ and $K = M$) can only increase (S8) and (S9).

(b) Small targets. *If $\delta_g = 0$ and $\rho_g = \rho$ for both groups, then*

$$\mathrm{Eff}^{TPR} = \frac{1}{\rho}\,\Gamma, \qquad \Gamma = \frac{\sum_g r_g/(\zeta_g p_g^2)}{(\sum_g \sqrt{r_g}/p_g)^2} \ge 1, \tag{S10}$$

where Γ is the efficiency of Neyman allocation between the two groups alone; and if $\theta_{mj}^{ERR} = 0 = \mu_g$, then $\mathrm{Eff}^{ERR} = 1/\rho$.

(c) Target size. *For one group (drop g), with $\pi_{\mathcal{D}}$, $\pi_{\mathcal{A}}$ and a held fixed, as $\rho \to 0$,*

$$\begin{aligned} \mathrm{Eff}^{TPR} &= \frac{1 + O(\rho)}{\rho\,(\beta + \lambda)^2}, \\ \beta &= \sqrt{1 - a^2 \pi_{\mathcal{D}}}, \\ \lambda &= \frac{|a| \sqrt{\pi_{\mathcal{D}}\, \pi_{\mathcal{A}} (1 - \pi_{\mathcal{A}})}}{p}. \end{aligned} \tag{S11}$$

Since $\delta = \rho \pi_{\mathcal{D}} a / p$, the target is proportional to ρ and to a. A more one-sided disagreement set (larger $|a|$, hence a larger target) changes the gain by the bounded factor $(\beta + \lambda)^{-2}$, which is below one when the agreement region is noisy ($\lambda > 1 - \beta$), but it does not change the order $1/\rho$.

(d) When the gain vanishes. *As $\rho_g \to 1$ the agreement strata disappear and (S8) and (S9) tend to the efficiency of stratification by group alone (for the rate differences with $\delta_g = 0$ this is Γ; for the error difference it equals 1 when $\mu_0 = \mu_1$). The gain beyond group stratification is therefore governed by $1 - \rho$: it is large when the compared models disagree on a small share of the pool, it is reduced by a constant factor when the pair's disagreements are one-sided ($|a_g|$ near 1) and the agreement region has substantial label noise, and it disappears when the models disagree almost everywhere.*

The heuristic *efficiency* $\approx 1/\rho$ is thus exact for an error difference with a two-sided disagreement set ($\mu_g = 0$), and correct up to the group factor Γ and the target-size factor $(\beta + \lambda)^{-2}$ for the rate differences. For a pair that disagrees on 5% of the pool it promises a gain of the order of 20 relative to SRS, and of 2.5 for a pair that disagrees on 40%. These are gains of the stratification; the difference estimator multiplies them by the factor G of (21), which by (S4) is of order one to three and does not grow as $\rho \to 0$: the difference estimator removes the part of the label noise that a proxy of moderate sharpness can explain, whereas the stratification removes the unlabelled agreement region.

For $M > 2$ the pattern strata (G_i, K_i) do not separate the disagreements of a particular pair: an instance with $K_i = 4$ out of nine may or may not be disputed by m and j. The general formula (24) still applies with the stratum variances of Proposition 2. For an error difference whose disagreements are two-sided within every stratum ($e_h = 0$ for all h, hence $\theta^{ERR}_{mj} = 0$), $\sigma^2_{ht} \propto \rho_h$ and (24) becomes

$$\mathrm{Eff}^{ERR} = \frac{\rho}{(\sum_h W_h \sqrt{\rho_h})^2}, \qquad 1 \le \mathrm{Eff}^{ERR} \le \frac{1}{\rho}, \tag{S12}$$

since $\rho_h \le \sqrt{\rho_h}$ and $(\sum_h W_h \sqrt{\rho_h})^2 \le \sum_h W_h \rho_h = \rho$. The bound $1/\rho$ is attained only when every stratum is either entirely disputed or entirely agreed; the pattern strata approach it to the extent that the pair's disagreements concentrate in a few values of K. A lower bound of the same type holds for the pattern strata. Let ρ_{any} be the share of instances on which not all M models agree, i.e. with $0 < K_i < M$. The pair can disagree only there, so $\rho_h = 0$ in the strata with $K \in \{0, M\}$, and Cauchy–Schwarz over the remaining strata gives $(\sum_h W_h \sqrt{\rho_h})^2 \le \rho_{\text{any}}\, \rho$, hence

$$\frac{1}{\rho_{\text{any}}} \;\le\; \mathrm{Eff}^{ERR} \;\le\; \frac{1}{\rho}, \tag{S13}$$

with equality on the left when the pair's disagreements are spread evenly over the strata with $0 < K < M$ ($\rho_h = \rho / \rho_{\text{any}}$ there). With nine models ρ_{any} is large (between 0.18 and 0.73 in our pools) while the pairwise ρ is small, and the position of a pair within this interval measures how well the counts K locate its disagreements. Two further losses separate (S8) and (S9) from the efficiency of the implemented design: a single allocation serves all $3\binom{M}{2}$ comparison estimands (Proposition 5(d)), and it is computed from the proxy (Section 5.7). The exact linearised efficiency of the implemented design is $N^2(1/L - 1/N) S_t^2 / V_t(n)$ with V_t from (12) (or (15) for the difference estimator); Section S5 compares it, and the closed forms above, with Monte Carlo results.

S2.9 Imperfect proxy in the allocation

Let $\tilde{n}$ be the allocation computed from the anticipated values $\tilde{A}_h$, and A_h the values computed from the true labels, both with the same weights. This subsection concerns the proxy in the allocation; its role in the difference estimator is Proposition S5.

Proposition S9 (Imperfect proxy)*(a) Proposition 1 holds for $\tilde{n}$ whatever the proxy, provided $\tilde{n}_h \geq 2$ in every stratum (guaranteed by the minimum of 10 labels).*

(b) Ignoring caps, minima and rounding, the relative loss in the weighted variance is

$$\frac{\Phi_0(\tilde{n})}{\Phi_0(n^*)} = \frac{(\sum_h N_h \tilde{A}_h)(\sum_h N_h A_h^2/\tilde{A}_h)}{(\sum_h N_h A_h)^2} = \Big(\sum_h \omega_h x_h\Big)\Big(\sum_h \omega_h / x_h\Big) \geq 1, \tag{S14}$$

with $\omega_h = N_h A_h / \sum_k N_k A_k$ and $x_h = \tilde{A}_h / A_h$. The loss is zero if and only if $\tilde{A}_h \propto A_h$; a proxy that is wrong by a common factor costs nothing.

(c) If $x_{\min} \leq x_h \leq x_{\max}$, the loss is at most $(x_{\max}+x_{\min})^2/(4x_{\max}x_{\min})$ (Kantorovich inequality). For instance, if every anticipated value $\tilde{A}_h$ is within a factor 2 of A_h up to a common scale ($x_{\max}/x_{\min} \leq 4$), the weighted variance is at most $25/16$ times the optimum on the same strata, so the design keeps at least 64% of the efficiency of the optimal allocation, whatever the number of strata.

(d) Proportional allocation is the special case $\tilde{A}_h \equiv 1$. Neither allocation dominates the other for every population: (S14) is smaller for the proxy allocation if and only if

$$\Big(\sum_h \omega_h x_h\Big)\Big(\sum_h \omega_h / x_h\Big) \leq \Big(\sum_h \omega_h / A_h\Big)\Big(\sum_h \omega_h A_h\Big).$$

A proxy that is badly wrong in a high-variance stratum can make the design worse than proportional allocation, and, through the minimum of ℓ labels in that stratum, worse than SRS.

(e) A defensive mixture $n_h = \eta\, n_h^{\text{prop}} + (1-\eta)\tilde{n}_h$, $0 < \eta < 1$, satisfies $\Phi_0(n) \leq \Phi_0(n^{\text{prop}})/\eta$ and $\Phi_0(n) \leq \Phi_0(\tilde{n})/(1-\eta)$; since proportional allocation is never worse than SRS up to the factor $(1-1/\min_h N_h)^{-1}$, the mixture is never worse than $1/\eta$ times SRS in the weighted variance without finite-population corrections, and with them (sampling fraction $f = L/N$) never worse than $(1-\eta f)/\{\eta(1-f)\}$ times SRS, both up to the same factor; the second bound is $\eta^{-1}\{1+O(f)\}$.

What can be guaranteed is therefore validity (part (a)) and a loss that is controlled by how far the anticipated stratum standard deviations are from proportional to the true ones (parts (b) and (c)). What cannot be guaranteed for the implemented rule is improvement over SRS for every population; that requires a defensive component as in part (e). Two features make large losses unlikely in our setting. The dominant part of the structure, namely which strata carry the numerators of the comparison estimands, is determined by the predictions and does not depend on the proxy (Proposition 2); and the proxy enters $\tilde{A}_h$ only through $\pi_h(1-\pi_h)$ and r_h, which are smooth in the proxy probabilities. Section S5 quantifies the loss for good, uninformative and adversarial proxies.

S3 Proofs for the testing results

Conventions.

All levels α of one-sided, TOST and Holm procedures are taken below $1/2$, so that the critical constants are positive as Lemma S10 requires. All limits are along a sequence of pools indexed by N, with H and Q fixed; Φ, φ are the standard normal distribution function and density, $z_q = \Phi^{-1}(q)$, and "$\to_p$", "$\Rightarrow$" denote convergence in probability and in law. As in Supplementary Section S1, $a_k = \nabla\phi_k(\tau)$ is the gradient of the kth contrast on the scale of means, $x_i^{(k)} = a_k^\top \xi_i = N e_i^{(k)}$, so that $|x_i^{(k)}| \leq B'$ with a constant B' that depends only on the bound B on the derivatives of the ϕ_k on the compact set K of Supplementary Section S1 and on C, and $\Lambda_k = a_k^\top(\hat{\tau}-\tau) = \sum_h W_h(\bar{x}_{s_h}^{(k)} - \bar{x}_h^{(k)})$ with $\mathrm{Var}(\Lambda_k) = V_k$. Recall $r^2 = \sum_h W_h^2/n_h$ and, from Proposition 1, that under (A1)–(A3) for contrast k,

$$X_k = \frac{\hat{\theta}_k - \theta_k}{V_k^{1/2}} \Rightarrow N(0,1) \text{ (under (A4))}, \qquad \varrho_k = \Big(\frac{\hat{V}_k}{V_k}\Big)^{1/2} \to_p 1, \tag{S15}$$

and the second statement also holds for the floored estimator under (A5). These two facts are the only inputs to the results for a single contrast.

S3.1 A uniform Slutsky lemma

Lemma S10 *Let $X_N \Rightarrow N(0,1)$ and let $\varrho_N > 0$ with $\varrho_N \to_p 1$. For every $0 < z_- < z_+ < \infty$, uniformly in $z \in [z_-, z_+]$ and $\mu, a, b \in \mathbb{R}$,*

$$P(X_N + \mu > z\varrho_N) = \Phi(\mu - z) + o(1), \tag{S16}$$

$$P(z\varrho_N - a < X_N < b - z\varrho_N) = \big[\Phi(b - z) - \Phi(z - a)\big]_+ + o(1) \tag{S17}$$

Proof Let F_N be the distribution function of X_N. Since the limit is continuous, Pólya's theorem [9, Lemma 2.11] gives $\varepsilon_N = \sup_x |F_N(x) - \Phi(x)| \to 0$, and the same bound holds for the left limits $F_N(x-)$. Fix $\eta \in (0, \frac{1}{2})$, let $\Omega_\eta = \{|\varrho_N - 1| \le \eta\}$ and $\pi_N = P(\Omega_\eta^c) \to 0$. On Ω_η we have $z(1-\eta) \le z\varrho_N \le z(1+\eta)$.

For (S16): on Ω_η, $\{X_N > z(1+\eta) - \mu\} \subseteq \{X_N + \mu > z\varrho_N\} \subseteq \{X_N > z(1-\eta) - \mu\}$, so

$$\Phi(\mu - z(1+\eta)) - \varepsilon_N - \pi_N \le P(X_N + \mu > z\varrho_N) \le \Phi(\mu - z(1-\eta)) + \varepsilon_N + \pi_N,$$

using $P(X_N > x) = 1 - F_N(x)$ and $1 - \Phi(x) = \Phi(-x)$. The function Φ is Lipschitz with constant $\varphi(0)$, hence both bounds are within $\varphi(0)z_+\eta + \varepsilon_N + \pi_N$ of $\Phi(\mu - z)$, uniformly in (μ, z). Let $N \to \infty$ and then $\eta \to 0$.

For (S17): on Ω_η the interval $(z\varrho_N - a, b - z\varrho_N)$ contains $(z(1+\eta) - a,\ b - z(1+\eta))$ and is contained in $(z(1-\eta) - a,\ b - z(1-\eta))$. For an interval (l, u) with $l < u$, $P(l < X_N < u) = F_N(u-) - F_N(l)$ lies within $2\varepsilon_N$ of $\Phi(u) - \Phi(l)$, and for $u \le l$ the event is empty and $[\Phi(u) - \Phi(l)]_+ = 0$. The map $(s, t) \mapsto [\Phi(t) - \Phi(s)]_+$ is Lipschitz with constant $\varphi(0)$ in each argument, and replacing z by $z(1 \pm \eta)$ moves each endpoint by at most $z_+\eta$. The sandwich gives (S17) as before. □

S3.2 Proof of Theorems 6(a)–(c) and 7

Fix k and drop it from the notation. For any θ_0, $\hat\theta - \theta_0 = V^{1/2}(X + \mu)$ with $\mu = (\theta - \theta_0)/V^{1/2}$, hence $Z(\theta_0) = (X + \mu)/\varrho$. By (S15) and Lemma S10, uniformly in $\mu \in \mathbb{R}$ (the range of μ as θ_0 varies over $\mathbb{R}$),

$$P\big(Z(\theta_0) > z\big) = \Phi(\mu - z) + o(1), \qquad P\big(Z(\theta_0) < -z\big) = \Phi(-\mu - z) + o(1),$$

the second because $-X \Rightarrow N(0,1)$ too. Adding the two tails gives (26) of Theorem 7 for $z = z_{1-\alpha/2}$, and the two one-sided powers are the separate tails with $z = z_{1-\alpha}$. For the TOST, with $a = (\theta + \delta)/V^{1/2}$ and $b = (\delta - \theta)/V^{1/2}$ we have $Z(-\delta) = (X + a)/\varrho$ and $Z(\delta) = (X - b)/\varrho$, so the TOST rejects exactly when $z\varrho - a < X < b - z\varrho$ with $z = z_{1-\alpha}$, and (S17) gives its power uniformly in (a, b), hence in δ. Consistency: if $|\mu| \to \infty$ then $\Psi_\alpha(\mu) \to 1$, and the supremum bound transfers this to the rejection probability. This proves Theorem 7.

Theorem 6. (a) At $\theta_0 = \theta$, $\mu = 0$ and the rejection probability is $2\Phi(-z_{1-\alpha/2}) + o(1) = \alpha + o(1)$; moreover $Z = X/\varrho \Rightarrow N(0,1)$ by Slutsky's lemma, and applying the same computation to the level u gives $P(2\{1 - \Phi(|Z|)\} \le u) = P(|Z| \ge z_{1-u/2}) \to u$ for every fixed $u \in (0,1)$. (b) If $\theta \le \theta_0$ then $\mu \le 0$ and $P(Z > z_{1-\alpha}) = \Phi(\mu - z_{1-\alpha}) + o(1) \le \Phi(-z_{1-\alpha}) + o(1) = \alpha + o(1)$; if $\mu \to 0$ the first bound becomes α by continuity of Φ. (c) Non-inferiority: if $\theta \ge \delta$ then $\mu' = (\theta - \delta)/V^{1/2} \ge 0$ and $P(Z(\delta) < -z_{1-\alpha}) = \Phi(-\mu' - z_{1-\alpha}) + o(1) \le \alpha + o(1)$. TOST: if $\theta \ge \delta$ it rejects only if $Z(\delta) < -z_{1-\alpha}$, whose probability is at most $\alpha + o(1)$ by the same computation; if $\theta \le -\delta$ it rejects only if $Z(-\delta) > z_{1-\alpha}$, i.e. $X + a > z_{1-\alpha}\varrho$ with $a \le 0$, of probability $\Phi(a - z_{1-\alpha}) + o(1) \le \alpha + o(1)$. Finally let $\theta = \delta$ and $\delta/V^{1/2} \to \infty$: then $b = 0$, $a = 2\delta/V^{1/2} \to \infty$, and (S17) gives $P(\text{reject}) = [\Phi(-z_{1-\alpha}) - \Phi(z_{1-\alpha} - a)]_+ + o(1) \to \alpha$; the case $\theta = -\delta$ is symmetric. □

Corollary S11 (Uniformity) *Let $\mathcal{C}_N$ be sets of pools such that every sequence $u_N \in \mathcal{C}_N$ satisfies (A1)–(A4) (and (A5)). Then the statements of Theorems 6(a)–(c) and 7 hold with $\sup_{u \in \mathcal{C}_N}$ in front of the error terms.*

Proof If the supremum of an error term did not tend to zero, choose $u_N \in \mathcal{C}_N$ attaining it up to $1/N$ for every N. The resulting sequence satisfies the assumptions, so the error along it tends to zero, a contradiction. □

S3.3 Proof of Corollary 8

With $n_h = \lambda_h L \le N_h$ and $V = \sum_h N_h^2(1/n_h - 1/N_h)S_h^2$ we get $V^{(d)}(L) = v_d/L - w_d$ with $v_d = \sum_h N_h^2 S_h^2/\lambda_h$ and $w_d = \sum_h N_h S_h^2$ exactly. By Theorem 7 the (two-sided) power against $|\theta - \theta_0| = \Upsilon$ is $\Psi_\alpha(\Upsilon/V^{1/2}) + o(1)$. The function Ψ_α is continuous with $\Psi_\alpha(0) = \alpha$, tends to one, and is strictly increasing on $[0, \infty)$ since $\Psi'_\alpha(\mu) = \varphi(\mu - z) - \varphi(\mu + z) > 0$ for $\mu > 0$, $z > 0$ (because $|\mu - z| < \mu + z$). So for $\alpha < 1 - \beta < 1$ there is a unique root $\mu_{\alpha,\beta} > 0$ of $\Psi_\alpha = 1 - \beta$, and power $1 - \beta$ holds iff $\Upsilon/V^{1/2} \ge \mu_{\alpha,\beta}$, i.e. $V(L) \le \Delta^2/\mu_{\alpha,\beta}^2$, i.e. $v_d/L \le \Delta^2/\mu^2 + w_d$, which

is (27). Since $\Psi_\alpha(\mu) \geq \Phi(\mu - z)$, the root satisfies $\mu_{\alpha,\beta} \leq z_{1-\alpha/2} + z_{1-\beta}$; since $\Psi_\alpha(\mu) \leq \Phi(\mu - z) + \Phi(-2z)$ for $\mu \geq z$, the power at $z_{1-\alpha/2} + z_{1-\beta}$ exceeds $1-\beta$ by at most $\Phi(-2z_{1-\alpha/2})$. For one-sided tests the root is exactly $z_{1-\alpha} + z_{1-\beta}$. The ratio in (28) is algebra. For the limit, $w_d L / v_d = \sum_h N_h S_h^2 L / \sum_h N_h^2 S_h^2 / \lambda_h \leq \max_h \lambda_h L / N_h = \max_h f_h =: \bar{f}_d$, because the hth term of the numerator is at most $\bar{f}_d$ times the hth term of the denominator. Hence $V^{(d)}(L) = (v_d/L)(1 - \vartheta_d)$ with $0 \leq \vartheta_d \leq \bar{f}_d$, and $V^{(2)}(L)/V^{(1)}(L) \to v_2/v_1$ when the sampling fractions tend to zero. At the labels L_d of (27) we have $v_d/L_d = \Delta^2/\mu^2 + w_d$, so $w_d/(\Delta^2/\mu^2) = \vartheta_d/(1-\vartheta_d) \to 0$ and the second factor in (28) tends to one. For the Pitman efficiency: two tests with power curves $\Psi_\alpha(\Upsilon/V_1^{1/2})$ and $\Psi_\alpha(\Upsilon/V_2^{1/2})$ have the same power at every Υ iff $V_1 = V_2$; if $V_d = v_d/L_d$ (neglecting w_d) then equal power at the same Υ needs $L_2/L_1 = v_2/v_1$, the ratio of asymptotic variances (the label ratio at equal power for a fixed effect; the analogue of the Pitman efficiency of van der Vaart 9, Ch. 14 and Noether 10, there being no local parametric sequence). For PATTERN-DIF against SRS, $v_{\mathrm{SRS}}/v_{\mathrm{PD}}$ is the large-N limit of $V^{\mathrm{SRS}}/V^{\mathrm{PD}}$, i.e. the efficiency of the paper. For the population target the variance is $V_{\mathrm{tot}}^{(d)}(L) = v_d/L - w_d + v_{\mathrm{P}}$ with v_{P} the same for all designs (Supplementary Section S3.8); the equation $V_{\mathrm{tot}}^{(d)}(L_d) = \Delta^2/\mu^2$ gives $L_d = \mu^2 v_d / \{\Delta^2 + \mu^2(w_d - v_{\mathrm{P}})\}$, and since $v_d/L - w_d = V^{(d)}(L) \geq 0$ the total variance is at least v_{P} at every L, so power $1-\beta$ requires $\Delta^2 \geq \mu^2 v_{\mathrm{P}}$. With w_d negligible, $L_2/L_1 = v_2/v_1$ again. The mean squared error ratio at equal L is $(V^{(2)} + v_{\mathrm{P}})/(V^{(1)} + v_{\mathrm{P}})$, which is $(\mathrm{Eff} + \varpi)/(1 + \varpi)$ for $\varpi = v_{\mathrm{P}}/V^{(1)}$ and $\mathrm{Eff} = V^{(2)}/V^{(1)}$. □

S3.4 A central limit lemma for stratified means

The following lemma isolates the argument of the proof of Proposition 1(c) for an arbitrary real variable.

Lemma S12 *For each N let y_i, $i \in U$, be real numbers and put $\Lambda = \sum_h W_h(\bar{y}_{s_h} - \bar{y}_h)$, $v_h = W_h^2(1 - f_h)S_{y,h}^2/n_h$ and $q = \mathrm{Var}\,\Lambda = \sum_h v_h$. Suppose (A1) holds (so $n_h \to \infty$ and $N_h - n_h \geq \varepsilon N_h$ in strata that are not fully enumerated; fully enumerated strata have $v_h = 0$), q is bounded away from zero, and for every stratum either $v_h/q \to 0$ or Hájek's condition holds: for every $\eta > 0$,*

$$\frac{1}{(N_h - 1)S_{y,h}^2} \sum_{i \in U_h:\ |y_i - \bar{y}_h| > \eta m_h S_{y,h}} (y_i - \bar{y}_h)^2 \to 0, \qquad m_h = \{n_h(1-f_h$$

Then $\Lambda/q^{1/2} \Rightarrow N(0,1)$. The Hájek condition holds in particular if $\max_{i \in U_h} |y_i - \bar{y}_h|/(m_h S_{y,h}) \to 0$.

Proof It suffices to show that every subsequence has a further subsequence along which the claim holds. Pass to a subsequence along which the set $\mathcal{H}_1$ of strata satisfying Hájek's condition and its complement $\mathcal{H}_2$ (where $v_h/q \to 0$) are constant and $v_h/q \to \lambda_{h,\infty}^2$ for every h (the ratios lie in $[0,1]$ and sum to one); then $\lambda_{h,\infty} = 0$ on $\mathcal{H}_2$ and $\sum_{\mathcal{H}_1} \lambda_{h,\infty}^2 = 1$. The variance of $\sum_{h \in \mathcal{H}_2} W_h(\bar{y}_{s_h} - \bar{y}_h)/q^{1/2}$ is $\sum_{\mathcal{H}_2} v_h/q \to 0$, so this term is $o_p(1)$. For $h \in \mathcal{H}_1$ put $\chi_h = (\bar{y}_{s_h} - \bar{y}_h)/\{(1-f_h)S_{y,h}^2/n_h\}^{1/2}$. By Hájek's theorem for simple random sampling without replacement [2], $\chi_h \Rightarrow N(0,1)$. The χ_h are independent across strata and $\Lambda/q^{1/2} = \sum_{\mathcal{H}_1}(v_h/q)^{1/2}\chi_h + o_p(1)$. The characteristic functions satisfy $\varphi_{\chi_h}(t) \to e^{-t^2/2}$ uniformly on compacts, so $E\exp\{it\sum_{\mathcal{H}_1}(v_h/q)^{1/2}\chi_h\} = \prod_{\mathcal{H}_1}\varphi_{\chi_h}\{(v_h/q)^{1/2}t\} \to e^{-t^2/2}$. For the last claim: if $\max|y_i - \bar{y}_h| \leq \eta m_h S_{y,h}$ for large N, the index set in Hájek's sum is empty. □

S3.5 Proof of Theorem 9

Linearisation. By the argument of Supplementary Section S1, part (b), on the event $\mathcal{E} = \{\|\hat{\tau} - \tau\|_\infty \leq c/2\}$, whose probability tends to one, $|\hat{\theta}_k - \theta_k - \Lambda_k| \leq (B/2)C\|\hat{\tau} - \tau\|^2 = O_p(r^2)$ for every $k \leq Q$, and $V_k \geq c_k r^2$ by (A3), so $(\hat{\theta}_k - \theta_k - \Lambda_k)/V_k^{1/2} = O_p(r) \to_p 0$. Let $\Sigma_{kl} = \mathrm{Cov}(\Lambda_k, \Lambda_l) = \sum_h W_h^2(1 - f_h)n_h^{-1}\,\mathrm{Cov}_h(x^{(k)}, x^{(l)}) = \sum_h c_h\,\mathrm{Cov}_h(e^{(k)}, e^{(l)})$; it has $\Sigma_{kk} = V_k$, and $R = D^{-1}\Sigma D^{-1}$ with $D = \mathrm{diag}(V_k^{1/2})$. Note that $V_k \leq 2B'^2 r^2$ since $|x^{(k)}| \leq B'$ (the variance bound of Supplementary Section S1), so $V_k \asymp r^2$.

Cramér–Wold. Let $Y = D^{-1}\Lambda$ with $\Lambda = (\Lambda_1, \ldots, \Lambda_Q)^\top$. The set of $Q \times Q$ correlation matrices is compact. Take a subsequence with $R_N \to R_\infty$; we show that $Y \Rightarrow N(0, R_\infty)$ along it, which proves the first claim of the theorem by the subsequence principle (every subsequence has a further subsequence with a convergent R_N). Fix $u \in \mathbb{R}^Q$ with $\|u\| = 1$ and put $q_N = u^\top R_N u = \mathrm{Var}(u^\top Y) \to q_\infty = u^\top R_\infty u$. If $q_\infty = 0$ then $u^\top Y \to 0$ in L^2 and the limit $N(0,0)$ is correct.

If $q_\infty > 0$ write $u^\top Y = \sum_h W_h(\bar{y}_{s_h} - \bar{y}_h)$ with $y_i = b^\top x_i/r$, $x_i = (x_i^{(k)})_k$ and $b_k = u_k r/V_k^{1/2}$; the vector $b = b_N$ is bounded because $V_k \geq c_k r^2$. Its variance is $q_N \geq q_\infty/2 > 0$ for large N, so $b^\top \Lambda_x$ has variance of exact order r^2, and $(\mathrm{A4}_Q)$ applies to b_N: for each stratum either $n_h(1-f_h)b^\top \Sigma_h^x b \to \infty$ (Σ_h^x the stratum covariance matrix of x), or the stratum's share of the variance tends to zero. In the first case, $m_h S_{y,h} = \{n_h(1-f_h)b^\top \Sigma_h^x b\}^{1/2}/r$ while $\max|y_i - \bar{y}_h| \leq 2\|b\|_1 B'/r$, so $\max|y_i - \bar{y}_h|/(m_h S_{y,h}) \to 0$ and Hájek's condition holds by the last sentence of Lemma S12. Lemma S12 gives $u^\top Y/q_N^{1/2} \Rightarrow N(0,1)$, hence $u^\top Y \Rightarrow N(0, q_\infty)$. Since u was arbitrary, the Cramér–Wold device gives $Y \Rightarrow N(0, R_\infty)$. With the linearisation, $\{(\hat{\theta}_k - \theta_k)/V_k^{1/2}\}_k = Y + o_p(1)$ has the same limit.

Covariance estimation. Let $\hat{a}_k = \nabla \phi_k(\hat{\tau})$ and $\hat{S}_h$ the sample covariance matrix of ξ_i in stratum h. Then $\hat{\Sigma}_{kl} = \sum_h W_h^2(1-f_h)n_h^{-1}\hat{a}_k^\top \hat{S}_h \hat{a}_l$ and $|\hat{\Sigma}_{kl} - \Sigma_{kl}| \leq r^2 \max_h |\hat{a}_k^\top \hat{S}_h \hat{a}_l - a_k^\top \Sigma_h^\xi a_l| = r^2 o_p(1)$ by Lemma **??**, the consistency $\hat{a}_k - a_k \to_p 0$ and boundedness on $\mathcal{E}$, exactly as in the proof of Proposition 1(d). The matrix $\hat{\Sigma} = \sum_h W_h^2(1-f_h)n_h^{-1}\hat{A}\hat{S}_h\hat{A}^\top$, with $\hat{A}$ the $Q \times C$ matrix of the $\hat{a}_k^\top$, is positive semi-definite. The floored estimator is $\hat{\Sigma}^\flat = \hat{\Sigma} + \mathrm{diag}(\hat{V}_k^\flat - \hat{V}_k)$, with non-negative increments, hence positive semi-definite, and $\hat{V}_k^\flat/V_k = (\hat{V}_k^\flat/\hat{V}_k)(\hat{V}_k/V_k) \to_p 1$ by (A5) and Proposition 1(d). With $\hat{D} = \mathrm{diag}((\hat{V}_k^\flat)^{1/2})$ and $\hat{R} = \hat{D}^{-1}\hat{\Sigma}^\flat \hat{D}^{-1}$ (a correlation matrix), $\hat{R}_{kl} - R_{kl} = \{\hat{\Sigma}_{kl}/(\hat{V}_k^\flat \hat{V}_l^\flat)^{1/2}\} - \Sigma_{kl}/(V_k V_l)^{1/2} \to_p 0$ for $k \neq l$, because $\hat{\Sigma}_{kl} - \Sigma_{kl} = r^2 o_p(1)$, $V_k \geq c_k r^2$ and $\hat{V}_k^\flat/V_k \to_p 1$; and $\hat{R}_{kk} = R_{kk} = 1$. □

S3.6 The critical value of the max-$|t|$ procedure

Let $\mathcal{R}_Q$ be the set of $Q \times Q$ correlation matrices (symmetric, positive semi-definite, unit diagonal; compact). For $R \in \mathcal{R}_Q$ and $N \sim N(0, R)$ let $F_R(c) = P(\max_{k \leq Q}|N_k| \leq c)$. For a non-empty $S \subseteq \{1, \ldots, Q\}$, $R_S \in \mathcal{R}_{|S|}$ is the principal submatrix.

Lemma S13 *For every $\alpha \in (0,1)$ and $R \in \mathcal{R}_Q$:*

(i) F_R is continuous and strictly increasing on $[0, \infty)$, so $c_\alpha(R) = F_R^{-1}(1-\alpha)$ is well defined;
(ii) $z_{1-\alpha/2} \leq c_\alpha(R) \leq z_{1-\alpha/(2Q)}$;
(iii) $c_\alpha(R_S) \leq c_\alpha(R_{S'})$ for $S \subseteq S'$;
(iv) $(R, c) \mapsto F_R(c)$ is jointly continuous, and $R \mapsto c_\alpha(R)$ is continuous, hence uniformly continuous on $\mathcal{R}_Q$.

Proof (i) Each $N_k \sim N(0,1)$, so $P(|N_k| = c) = 0$ and $P(\max_k |N_k| = c) = 0$: F_R is continuous. For $0 \leq c_1 < c_2$, the law of N is Gaussian with support the subspace range$(R) \neq \{0\}$ (as $R_{kk} = 1$), which contains a vector x with $\max_k |x_k| = (c_1 + c_2)/2$ by scaling. The set $\{c_1 < \max_k |x_k| < c_2\}$ is open and meets the support, so it has positive probability, and $F_R(c_2) > F_R(c_1)$. (ii) $F_R(c) \leq P(|N_1| \leq c) = 2\Phi(c) - 1$, which equals $1-\alpha$ at $c = z_{1-\alpha/2}$; so $F_R(z_{1-\alpha/2}) \leq 1 - \alpha = F_R(c_\alpha)$ and $z_{1-\alpha/2} \leq c_\alpha$ by monotonicity. By the union bound, $1 - F_R(c) \leq \sum_k P(|N_k| > c) = 2Q\{1 - \Phi(c)\}$, which equals α at $c = z_{1-\alpha/(2Q)}$; so $F_R(z_{1-\alpha/(2Q)}) \geq 1 - \alpha$. (iii) $\{\max_{k \in S'}|N_k| \leq c\} \subseteq \{\max_{k \in S}|N_k| \leq c\}$ gives $F_{R_{S'}} \leq F_{R_S}$, so $F_{R_S}(c_\alpha(R_{S'})) \geq 1 - \alpha = F_{R_S}(c_\alpha(R_S))$. (iv) Let $R_n \to R$ and $c_n \to c$. With $\zeta \sim N(0, I_Q)$, $N^{(n)} = R_n^{1/2}\zeta \to R^{1/2}\zeta$ almost surely (the matrix square root is continuous), so $\max|N^{(n)}| - c_n \to \max|N| - c$ almost surely and, since $P(\max|N| = c) = 0$, $F_{R_n}(c_n) \to F_R(c)$. Now let $c_n = c_\alpha(R_n) \in [z_{1-\alpha/2}, z_{1-\alpha/(2Q)}]$; any limit point c^* satisfies $F_R(c^*) = \lim F_{R_n}(c_n) = 1 - \alpha$, so $c^* = c_\alpha(R)$ by (i). Uniform continuity follows from compactness of $\mathcal{R}_Q$. □

S3.7 Proof of Theorem 10

Let $\mathcal{Z} = (Z_k(\theta_k))_{k \leq Q} = (X_k/\varrho_k)_k$. By Theorem 9 and $\varrho_k \to_p 1$, along a subsequence with $R \to R_\infty$, $\mathcal{Z} \Rightarrow N(0, R_\infty)$, and $\hat{R} - R \to_p 0$. By Lemma S13(iv), $c_\alpha(\hat{R}_S) - c_\alpha(R_S) \to_p 0$ for every fixed S (a principal submatrix map is continuous), and $c_\alpha(R_S) \to c_\alpha(R_{\infty,S})$. Since limsup statements along the whole sequence are attained along subsequences, and there are finitely many sets I_0, it suffices to consider subsequences with $R \to R_\infty$ and a constant true set I_0. If $I_0 = \emptyset$ there is nothing to prove.

(a), single step. For $k \in I_0$, $\theta_k = \theta_{k,0}$ so $Z_k(\theta_{k,0}) = \mathcal{Z}_k$. The false nulls do not enter $\mathcal{Z}_{I_0}$ or $\hat{R}$ (the critical value depends on the data only through $\hat{R}$, which is consistent for R whatever the values of the θ_k). Thus $(\max_{I_0}|\mathcal{Z}_k|, c_\alpha(\hat{R})) \Rightarrow (\max_{I_0}|N_k|, c_\alpha(R_\infty))$, the second component being constant in the limit and the first having a continuous law. By the

portmanteau theorem,

$$P\big(\max_{I_0}|\mathcal{Z}_k| > c_\alpha(\hat{R})\big) \to P\big(\max_{I_0}|N_k| > c_\alpha(R_\infty)\big) \le P\big(\max_{I_0}|N_k| > c_\alpha(R_{\infty,I_0})\big) = \alpha,$$

using Lemma S13(iii) for the inequality. If $I_0 = \{1,\dots,Q\}$ the inequality is an equality. This is the subset pivotality of the Gaussian limit: the limiting law of the vector of statistics of the true nulls does not depend on which other hypotheses are false.

(a), step-down. The procedure sets $S_1 = \{1,\dots,Q\}$, rejects $\{k \in S_j : |Z_k(\theta_{k,0})| > c_\alpha(\hat{R}_{S_j})\}$, sets $S_{j+1} = S_j$ minus the rejected, and stops when nothing is rejected. Let j be the first step at which a true null is rejected. No true null was rejected before, so $I_0 \subseteq S_j$, and $\max_{I_0}|\mathcal{Z}_k| > c_\alpha(\hat{R}_{S_j}) \ge c_\alpha(\hat{R}_{I_0})$ by Lemma S13(iii) applied to $\hat{R}$. Hence the familywise error event is contained in $\{\max_{I_0}|\mathcal{Z}_k| > c_\alpha(\hat{R}_{I_0})\}$, whose probability tends to $P(\max_{I_0}|N_k| > c_\alpha(R_{\infty,I_0})) = \alpha$ by the same portmanteau argument.

Bonferroni and Holm. By Theorem 6(a), for $k \in I_0$ and any fixed level u, $P(|\mathcal{Z}_k| > z_{1-u/2}) \to u$. For Bonferroni, $P(\exists k \in I_0 : |\mathcal{Z}_k| > z_{1-\alpha/(2Q)}) \le \sum_{I_0} P(|\mathcal{Z}_k| > z_{1-\alpha/(2Q)}) \to |I_0|\alpha/Q \le \alpha$. Holm's procedure in step-down form is that above with the critical value $z_{1-\alpha/(2|S_j|)}$ replacing $c_\alpha(\hat{R}_{S_j})$ (this rejects the same set as the one-at-a-time description with ordered p-values and thresholds $\alpha/(Q-j+1)$, because the thresholds are non-decreasing in j). If a true null is first rejected at step j then $|S_j| \ge m_0 = |I_0|$, so the critical value is at least $z_{1-\alpha/(2m_0)}$, and the probability of this event is at most $\sum_{I_0} P(|\mathcal{Z}_k| > z_{1-\alpha/(2m_0)}) \to m_0 \cdot \alpha/m_0 = \alpha$.

(b). The band covers all θ_k iff $\max_k|\mathcal{Z}_k| \le c_\alpha(\hat{R})$, and

$$P\big(\max_k|\mathcal{Z}_k| \le c_\alpha(\hat{R})\big) \to F_{R_\infty}\big(c_\alpha(R_\infty)\big) = 1-\alpha.$$

If every θ_k is covered then so is $\theta_{\hat{k}}$ for any data-dependent index $\hat{k}$.

(c). The bounds are Lemma S13(ii),(iii), valid for every $\hat{R} \in \mathcal{R}_Q$ and hence for every data set. Single-step max-$|t|$: $|\mathcal{Z}_k| > z_{1-\alpha/(2Q)} \ge c_\alpha(\hat{R})$ implies rejection. For the step-down procedures let S_j^M, S_j^H be the active sets of max-$|t|$ and Holm. We show $S_j^M \subseteq S_j^H$ by induction; $S_1^M = S_1^H$. If it holds at j, then $c_\alpha(\hat{R}_{S_j^M}) \le z_{1-\alpha/(2|S_j^M|)} \le z_{1-\alpha/(2|S_j^H|)}$ (using (ii) for the submatrix and $|S_j^M| \le |S_j^H|$). Take $k \in S_{j+1}^M$: then $k \in S_j^M \subseteq S_j^H$ and $|Z_k| \le c_\alpha(\hat{R}_{S_j^M}) \le z_{1-\alpha/(2|S_j^H|)}$, so Holm does not reject k at step j and $k \in S_{j+1}^H$. Hence every hypothesis rejected by Holm is rejected by step-down max-$|t|$. Nested rejection regions give the power statement for every sample size. □

S3.8 The population target: proof of Theorem 6(d)

Fix one contrast. Write $\theta_U = f(T_U) = \phi(\bar{t})$ for the pool contrast, with $\bar{t} = N^{-1}\sum_{i\le N} t_i$ the mean of the pool contributions, and $\theta_{\text{pop}} = \phi(\tau)$, $\tau = E\,t_i$ (fixed, by (P1)). Put $\text{er}_\text{D} = \hat{\theta} - \theta_U$ (design error), $\text{er}_\text{P} = \theta_U - \theta_{\text{pop}}$ (pool error), $V_D(u) = V_\psi(\theta; n)$ computed from the pool u, $\mathbf{b} = \nabla\phi(\tau)$, $\zeta_i = \mathbf{b}^\top t_i$ with $\operatorname{Var}\zeta_i = \sigma_\zeta^2 > 0$ by (P2), $v_\text{P} = \sigma_\zeta^2/N$, v_D as in (P3) and $s^2 = v_\text{D} + v_\text{P}$; $\varpi_N = v_\text{P}/v_\text{D}$.

Step 1: design-based statements, uniformly over $\mathcal{E}_N$. Write $\mathcal{E}_N = \mathcal{P}_N$ for the class of (P2). For any sequence $u_N \in \mathcal{P}_N$, (A1)–(A4) hold (the constants c, ε are common, $m_N \to \infty$ gives $\min n_h \to \infty$, $\eta_N \to 0$ gives the dichotomy of (A4), and τ is handled along subsequences), so by Proposition 1(c),(d) (and (A5)), $\text{er}_\text{D}/V_D(u_N)^{1/2} \Rightarrow N(0,1)$ and $\hat{V}_D/V_D(u_N) \to_p 1$ conditionally on the pool. By the argument of Corollary S11 (this is where the quantitative class $\mathcal{P}_N$ replaces "(A1)–(A4) hold", which is a statement about sequences), for every $\eta > 0$ and $t \in \mathbb{R}$,

$$\sup_{u\in\mathcal{E}_N}\Big\{\sup_x\Big|P\Big(\tfrac{\text{er}_\text{D}}{V_D(u)^{1/2}} \le x \,\Big|\, u\Big) - \Phi(x)\Big| + P\Big(\Big|\tfrac{\hat{V}_D}{V_D(u)} - 1\Big| > \eta \,\Big|\, u\Big)\Big\} \to$$

and, by Lévy's continuity theorem applied along nearly-worst sequences, for each T

$$\sup_{u\in\mathcal{E}_N}\sup_{|\tau'|\le T}\Big|E\big(e^{i\tau'\text{er}_\text{D}/V_D(u)^{1/2}} \mid u\big) - e^{-\tau'^2/2}\Big| \to 0.$$

Step 2: the pool error. By (P2) ϕ is C^2 with bounded derivatives on a neighbourhood of τ and $\|\bar{t} - \tau\| = O_p(N^{-1/2})$; Taylor's theorem gives $\text{er}_\text{P} = \mathbf{b}^\top(\bar{t} - \tau) + \Xi_N$ with $\Xi_N = O_p(N^{-1})$. By the classical central limit theorem for the i.i.d. bounded variables ζ_i, $N^{1/2}\mathbf{b}^\top(\bar{t} - \tau) \Rightarrow N(0, \sigma_\zeta^2)$. Since $s^2 \ge v_\text{P} = \sigma_\zeta^2/N$, $\Xi_N/s = O_p(N^{-1/2}) = o_p(1)$.

Step 3: the sum. Fix $t \in \mathbb{R}$ and write $\varphi_N = E\exp\{it(\mathrm{er_D} + \mathrm{er_P})/s\}$. Condition on the pool:

$$\varphi_N = E\Big[e^{it\,\mathrm{er_P}/s}\,E\big(e^{it\,\mathrm{er_D}/s} \mid u\big)\Big].$$

Let $\Omega_N = \{u \in \mathcal{E}_N,\ |V_D(u)/v_\mathrm{D} - 1| \le \frac{1}{2}\}$; $P(\Omega_N) \to 1$ by (P2) and (P3). On Ω_N, $t\,V_D(u)^{1/2}/s \in [-2|t|, 2|t|]$ (since $v_\mathrm{D} \le s^2$), so by Step 1, $E(e^{it\,\mathrm{er_D}/s} \mid u) = \exp\{-t^2 V_D(u)/(2s^2)\} + \epsilon_N(u)$ with $\sup_{u\in\Omega_N}|\epsilon_N(u)| \to 0$. Hence

$$\varphi_N = E\big[e^{it\,\mathrm{er_P}/s}e^{-t^2V_D(u)/(2s^2)}\big] + o(1) = e^{-t^2 v_\mathrm{D}/(2s^2)}\,E\,e^{it\,\mathrm{er_P}/s} + o(1),$$

where the last step uses $|e^{-t^2V_D/(2s^2)} - e^{-t^2 v_\mathrm{D}/(2s^2)}| \le \{t^2/(2s^2)\}|V_D - v_\mathrm{D}| \wedge 2 \to_p 0$ and bounded convergence (here (P3) is used: it is the only place where $V_D(u)$ is treated as non-random). By Step 2 and the uniform convergence of characteristic functions on compacts, with $m_N = 1/(sN^{1/2}) \le 1/\sigma_\zeta$, $E\,e^{it\,\mathrm{er_P}/s} = \exp\{-t^2 m_N^2\sigma_\zeta^2/2\} + o(1) = \exp\{-t^2 v_\mathrm{P}/(2s^2)\} + o(1)$. Therefore $\varphi_N = \exp\{-t^2(v_\mathrm{D} + v_\mathrm{P})/(2s^2)\} + o(1) = e^{-t^2/2} + o(1)$, that is, $(\hat\theta - \theta_\mathrm{pop})/s \Rightarrow N(0,1)$, with the remainder Ξ_N/s already accounted for. No covariance between the two errors was computed; the conditional argument shows that the design error is asymptotically independent of the pool.

Step 4: variance estimators. By Step 1 and (P3), $\hat V_D/v_\mathrm{D} \to_p 1$. For the pool term, per-unit scale: $N\hat V_\mathrm{pool} = \hat g^\top \hat M \hat g$ where $\hat g = \nabla\phi(\hat T_\psi/N)$ and $\hat M$ is the weighted covariance matrix $N^{-1}\sum_{i\in s} w_i t_i t_i^\top - (N^{-1}\sum_{i\in s} w_i t_i)(N^{-1}\sum_{i\in s} w_i t_i)^\top$ of the labelled sample. Each entry of $N^{-1}\sum_{i\in s} w_i t_{ic} t_{id}$ and of $N^{-1}\sum_{i\in s} w_i t_{ic}$ is a stratified expansion estimator of the corresponding pool mean of a bounded variable, with conditional variance at most $Cr^2 \to 0$; the pool means of $t_{ic}t_{id}$ and t_{ic} converge in probability to their population means by the law of large numbers. Hence $\hat M \to_p \mathrm{Cov}(t_i)$. Also $\hat T_\psi/N \to_p \tau$ (Proposition 1(b) and the law of large numbers), so $\hat g \to_p g$ and $N\hat V_\mathrm{pool} \to_p g^\top \mathrm{Cov}(t_i) g = \sigma_\zeta^2$, i.e. $\hat V_\mathrm{pool}/v_\mathrm{P} \to_p 1$. Consequently $\varrho^2 = (\hat V_D + \hat V_\mathrm{pool})/s^2$ lies between $\hat V_D/v_\mathrm{D}$ and $\hat V_\mathrm{pool}/v_\mathrm{P}$ and tends to one in probability.

Step 5. Under $\theta_\mathrm{pop} = \theta_0$, $Z^\mathrm{pop} = \{(\hat\theta - \theta_0)/s\}/\varrho$ with $(\hat\theta - \theta_0)/s \Rightarrow N(0,1)$ and $\varrho \to_p 1$, which are the two inputs of Lemma S10; the proofs of Theorem 6(a)–(c) and Theorem 7 then apply verbatim to Z^pop (with $\mu = (\theta_\mathrm{pop} - \theta_0)/s$), which gives in particular the population power statements of Theorem 7; $\alpha < 1/2$ is needed for the one-sided and TOST statements because Lemma S10 requires $z > 0$. For the pool-conditional statistic, $Z = \{(\hat\theta - \theta_0)/s\}\cdot s/\hat V_D^{1/2}$ and $s^2/\hat V_D \to_p 1 + \varpi_N$. Along a subsequence with $\varpi_N \to \varpi_\infty < \infty$, $Z \Rightarrow N(0, 1+\varpi_\infty)$ and the rejection probability tends to $P(|N(0,1)| > z_{1-\alpha/2}/\sqrt{1+\varpi_\infty}) = 2\{1 - \Phi(z_{1-\alpha/2}/\sqrt{1+\varpi_\infty})\}$. If $\varpi_N \to \infty$ then $|Z| \ge |(\hat\theta - \theta_0)/s|\cdot s/\hat V_D^{1/2}$ with the first factor converging to $|N(0,1)|$, which has no atom at zero, and the second tending to infinity in probability, so $P(|Z| > z) \to 1$, consistent with the formula at $\varpi_\infty = \infty$. □

S3.9 Proof of Theorem 11

Write $F(n,w) = \sum_k w_k V_k(n)/\delta_k^2$, $\mathcal{N} = \{n : \sum_h n_h = L,\ \ell_h \le n_h \le N_h\}$ (compact, convex, non-empty by $\sum \ell_h \le L \le \sum N_h$), and note $V_k(n) = \sum_h a_{hk}/n_h - b_k$ with $a_{hk} = N_h^2 S_{hk}^2 \ge 0$, $b_k = \sum_h N_h S_{hk}^2$, so V_k is convex, and $F(n,w)$ equals the objective $\Phi(n)$ of Proposition 5 with weights $\omega = w/\delta^2$ (we write $F(\cdot,w)$ for it; the symbol Φ without argument n is the normal distribution function).

(a) By Theorem 7 the asymptotic power depends on the design only through $\mu = \Upsilon/V^{1/2}$ and Ψ_α is strictly increasing in $|\mu|$ (Supplementary Section S3.3); the one-sided power $\Phi(\mu - z)$ is increasing in μ. So any n minimising V maximises power at every Υ. With $Q = 1$, Proposition 5(b) gives the minimiser $n(\omega)$ (weights $\omega = 1/\delta^2$ do not change the allocation).

(b) Saddle point. F is continuous, convex in n and linear in w, and both $\mathcal{N}$ and the simplex $\mathcal{S}_Q$ are compact and convex. Sion's minimax theorem [11] gives $\min_n \max_w F = \max_w \min_n F$. Now $\max_{w\in\mathcal{S}_Q} F(n,w) = \max_k V_k(n)/\delta_k^2$, a continuous function with a minimiser n°, and $\Gamma(w) = \min_n F(n,w)$ is a minimum of affine functions, hence concave and upper semi-continuous, with a maximiser w° on $\mathcal{S}_Q$. Let $t^* = \Gamma(w^\circ) = \max_k V_k(n^\circ)/\delta_k^2$. Then $t^* = \Gamma(w^\circ) \le F(n^\circ, w^\circ) \le \max_k V_k(n^\circ)/\delta_k^2 = t^*$, so equality holds throughout: $n^\circ \in \arg\min_n F(\cdot, w^\circ)$, and $F(n^\circ, w^\circ) = \max_w F(n^\circ, w)$, which means that w° is supported on $\{k : V_k(n^\circ)/\delta_k^2 = t^*\}$ (complementary slackness). Conversely, any pair (n,w) with

$n \in \arg\min F(\cdot, w)$ and w supported on the argmax set of $V_k(n)/\delta_k^2$ is optimal: for every n', $\max_k V_k(n')/\delta_k^2 \geq F(n', w) \geq F(n, w) = \max_k V_k(n)/\delta_k^2$. *The primal solution from the dual one.* The argument above shows that for every maximiser w^* of Γ and every primal optimum n°, $n^\circ \in \arg\min F(\cdot, w^*)$ (the equality chain with w° replaced by w^*, which has $\Gamma(w^*) = t^*$). Now $\arg\min F(\cdot, w^*) = \arg\min \Phi_{\omega^*}$; if $A_h(\omega^*) > 0$ for all h, $\Phi_{\omega^*}(n) = \sum_h N_h^2 A_h(\omega^*)^2/n_h -$ const is strictly convex, so the minimiser is unique and equals $n(\omega^*)$ by Proposition 5(b). Thus the optimum is unique and given by (29). Without bounds, Proposition 5(a) gives $\Gamma(w) = (\sum_h N_h A_h)^2/L - \sum_h N_h A_h^2$ with $A_h^2 = \sum_k w_k S_{hk}^2/\delta_k^2$. *Gradient.* For w with $A_h(w/\delta^2) > 0$ for all h the minimiser $n(w/\delta^2)$ of $F(\cdot, w)$ over the compact $\mathcal{N}$ is unique; the superdifferential of the concave function Γ is the closed convex hull of the gradients $\nabla_w F(n, w) = (V_k(n)/\delta_k^2)_k$ over minimisers n (Danskin's theorem), here the single vector $(V_k(n(w/\delta^2))/\delta_k^2)_k$. So Γ is differentiable there with this gradient, and the first-order condition for maximising Γ on $\mathcal{S}_Q$ ($w_k > 0 \Rightarrow \partial_k g = \max_j \partial_j g$) is the complementary slackness above. *Support bound.* Let (n^*, w^*) be a saddle point and $J = \operatorname{supp} w^*$. Since $n_h^* \geq \ell_h > 0$, Φ_ω is differentiable at n^* with $\partial_h \Phi_\omega = -\sum_k (w_k/\delta_k^2) N_h^2 S_{hk}^2/(n_h^*)^2 = -\sum_k w_k G_{hk}$, $G_{hk} = N_h^2 S_{hk}^2/\{\delta_k^2 (n_h^*)^2\}$. Because Φ_{ω^*} is convex, n^* minimises it over the polytope $\mathcal{N}$ iff $\sum_h (\sum_k w_k^* G_{hk})(n_h - n_h^*) \leq 0$ for all $n \in \mathcal{N}$. Every $n - n^*$ has coordinate sum zero, so the condition depends on the vector $p = \sum_{k \in J} w_k^* G_{\cdot k} \in \mathbb{R}^H$ only modulo $\mathbb{R}\mathbf{1}$. In the quotient $\mathbb{R}^H/\mathbb{R}\mathbf{1} \cong \mathbb{R}^{H-1}$ the image of p lies in the convex hull of the images of $\{G_{\cdot k} : k \in J\}$, so by Carathéodory's theorem it is a convex combination, with weights w', of at most H of them. Then $w' \in \mathcal{S}_Q$ is supported in J, $\sum_k w_k' G_{\cdot k} - p \in \mathbb{R}\mathbf{1}$, hence n^* also minimises $F(\cdot, w')$, and w' is supported on the argmax set, so (n^*, w') is a saddle point and w' maximises Γ.

(c) Let $n^{\mathrm{def}} = n(\omega^{\mathrm{def}})$ be built from the same array S_{hk}^2 that defines the V_k. By Proposition 5(b) it minimises $\sum_k V_k(n)/\sigma_k^2$ over $\mathcal{N}$; with $V_k^{\mathrm{SRS}} = N^2 \sigma_k^2 (1 + O(1/N))/L$ (up to the finite-population correction, common to all k) this is the sum of inverse efficiencies up to a factor not depending on k or n (Proposition 5(d)). If the paper's allocation (computed from plain variances) is used with the difference estimator, V_k is the residual variance and that allocation is not $n(\omega^{\mathrm{def}})$ for the residual array; none of the statements of (c) then applies to it, and only the weighted-sum loss bound of Proposition S7 is available. *Sufficiency.* Suppose $V_k(n^{\mathrm{def}})/\delta_k^2 = t$ for all k. Put $w_k \propto \delta_k^2 \omega_k^{\mathrm{def}}$ on the simplex. The allocation (29) is invariant under a common rescaling of ω (λ rescales accordingly), so $n(w/\delta^2) = n^{\mathrm{def}}$, and every k is in the argmax set; by the converse in (b), n^{def} is optimal. *Necessity.* Assume n^{def} is optimal, no stratum is at a bound, $A_h(\omega^{\mathrm{def}}) > 0$ for all h, and the vectors $(S_{hk}^2)_{h \leq H}$, $k \leq Q$, are linearly independent. By (b) there is a maximiser w^* with $n^{\mathrm{def}} = n(\omega^*)$ (we also assume $A_h(\omega^*) > 0$ for all h, so that the minimiser is unique). No stratum being at a bound, $n_h \propto N_h A_h$, so $A_h(\omega^*)^2 = c A_h(\omega^{\mathrm{def}})^2$ for all h, i.e. $\sum_k (\omega_k^* - c\,\omega_k^{\mathrm{def}}) S_{hk}^2 = 0$ for all h; linear independence of the Q vectors forces $\omega^* = c\,\omega^{\mathrm{def}}$, so $w_k^* \propto \delta_k^2/\sigma_k^2 > 0$ for every k, and complementary slackness makes all $V_k(n^{\mathrm{def}})/\delta_k^2$ equal. (Linear independence needs $Q \leq H$. For $Q > H$ equal relative variances are not necessary: a numerical example with $H = 2$, $Q = 3$, in which n^{def} has relative variances $(1, 1, 0.01)$ and is maximin-optimal, is in `docs/testing_proof_check.md`; it is not proved here. For the full family, optimality of n^{def} has to be checked numerically from the saddle-point conditions.) *Certificate.* For any $w \in \mathcal{S}_Q$ and any n, weak duality gives $\Gamma(w) \leq \min_{n'} \max_k V_k(n')/\delta_k^2 = t^* \leq \max_k V_k(n)/\delta_k^2$, so the ratio $\max_k V_k(n^{\mathrm{def}})/\delta_k^2$ to $\Gamma(w)$ bounds the loss of the default (the factor by which the labels needed for a common noncentrality exceed the optimum, when finite-population corrections are negligible).

(d) For $\tilde{n}$ with $\tilde{n}_h \geq n_h - 1$ and $n_h \geq n_0 > 1$, $V_k(\tilde{n}) + b_k = \sum_h a_{hk}/\tilde{n}_h \leq \sum_h a_{hk}/(n_h - 1) = \sum_h \{n_h/(n_h - 1)\} a_{hk}/n_h \leq \{n_0/(n_0 - 1)\} \sum_h a_{hk}/n_h = \{n_0/(n_0 - 1)\}\{V_k(n) + b_k\}$, because $x \mapsto x/(x-1)$ is decreasing on $(1, \infty)$. An integer allocation with $\tilde{n}_h \in \{\lfloor n_h \rfloor, \lceil n_h \rceil\}$ and $\sum_h \tilde{n}_h = L$ exists, satisfies $\tilde{n}_h \geq n_h - 1$, and $\tilde{n}_h \leq \lceil n_h \rceil \leq N_h$ and $\tilde{n}_h \geq \ell_h$ if the bounds are integers. □

S3.10 Proof of Proposition 12

Let $(\hat{a}, \hat{b})$ be the plug-in gaps with variances $V_a, V_b \asymp r^2$ (assumption (A3) for the two gaps, which are smooth functions of the totals like the comparison contrasts) and suppose the

standardised vector $((\hat{a}-a)/V_a^{1/2}, (\hat{b}-b)/V_b^{1/2})$ is asymptotically $N(0, R_\infty)$ along subsequences (Theorem 9 with $Q=2$), so that $Z_a^N = (\hat{a}-a)/r$ and $Z_b^N = (\hat{b}-b)/r$ are tight with a Gaussian limit (Z_a, Z_b) along subsequences. Let $s_a = \operatorname{sgn} a$, $s_b = \operatorname{sgn} b$.

(a) Take $|a| > |b|$ (the other case is symmetric) and put $\varsigma_N = |a| - |b|$, with $\varsigma_N/r \to \infty$. On $\Omega_N = \{|\hat{a}-a| < \varsigma_N/4,\ |\hat{b}-b| < \varsigma_N/4\}$, whose probability tends to one because $|\hat{a}-a| + |\hat{b}-b| = O_p(r) = o_p(\varsigma_N)$, we have $|\hat{a}-a| < \varsigma_N \le |a|$, so $\operatorname{sgn}\hat{a} = s_a$, and $|\hat{a}| \ge |a| - \varsigma_N/4 > |b| + \varsigma_N/4 > |\hat{b}|$. Hence on Ω_N, $\widehat{\mathrm{EO}} = |\hat{a}| = s_a\hat{a}$ and $\widehat{\mathrm{EO}} - \mathrm{EO} = s_a(\hat{a}-a)$. Therefore $(\widehat{\mathrm{EO}} - \mathrm{EO})/V_a^{1/2} = s_a(\hat{a}-a)/V_a^{1/2} + o_p(1) \Rightarrow N(0,1)$ by Theorem 9 (symmetry of $N(0,1)$); in the case $|b| > |a|$ the same argument gives the limit with V_b. With $\hat{V}_a/V_a \to_p 1$ (Proposition 1(d)) the Wald interval has asymptotic coverage $1-\alpha$ by Slutsky's lemma.

(b) Let $\vartheta_N = (|a|-|b|)/r \to \vartheta \in \mathbb{R}$ (either sign) and $|a|/r, |b|/r \to \infty$. Then $P(\operatorname{sgn}\hat{a} = s_a, \operatorname{sgn}\hat{b} = s_b) \to 1$, since $|\hat{a}-a| = O_p(r) = o_p(|a|)$ and likewise for b. On this event $|\hat{a}| = |a| + s_a(\hat{a}-a)$ and $|\hat{b}| = |b| + s_b(\hat{b}-b) = |a| - r\vartheta_N + s_b(\hat{b}-b)$, so $(\widehat{\mathrm{EO}} - |a|)/r = \max(s_a Z_a^N, s_b Z_b^N - \vartheta_N)$, and $\mathrm{EO} = \max(|a|,|b|) = |a| + r\max(0, -\vartheta_N)$. Therefore

$$\frac{\widehat{\mathrm{EO}} - \mathrm{EO}}{r} = \max\left(s_a Z_a^N,\ s_b Z_b^N - \vartheta_N\right) - \max(0, -\vartheta_N) \Rightarrow \max(s_a Z_a,\ s_b Z_b - \vartheta) - \max(0, -\vartheta)$$

by the continuous mapping theorem. With $U_1 = s_a Z_a$, $U_2 = s_b Z_b - \vartheta$ we have $\max(U_1, U_2) = (U_1+U_2)/2 + |U_1 - U_2|/2$, $EU_1 = 0$, $EU_2 = -\vartheta$, and $U_1 - U_2 = Y \sim N(\vartheta, \sigma^2)$, $\sigma^2 = \mathrm{Var}(s_a Z_a - s_b Z_b)$. Hence the mean of the limit is $-\vartheta/2 + E|Y|/2 - \max(0,-\vartheta) = \frac{1}{2}\{E|Y| - |\vartheta|\}$ (check $\vartheta \ge 0$ and $\vartheta < 0$ separately), and $E|Y| \ge |EY| = |\vartheta|$ with strict inequality when $\sigma > 0$. (A Monte Carlo check with correlated Gaussian errors and $\vartheta \in \{-2, -1, 0, 0.7, 1\}$ agrees with this mean to three decimals. The bound $\max(0,-\vartheta)$ of an earlier version referred to $(\widehat{\mathrm{EO}} - |a|)/r$, not to $(\widehat{\mathrm{EO}} - \mathrm{EO})/r$.) The limit is not normal in general: for $\vartheta = 0$ and independent standard normal $s_a Z_a, s_b Z_b$ the maximum has density $2\varphi(x)\Phi(x)$ and mean $\pi^{-1/2} > 0$. At $a = b = 0$, $\widehat{\mathrm{EO}} = \max(|\hat{a}|, |\hat{b}|) = r\max(|Z_a^N|, |Z_b^N|)$ exactly, which gives the stated limit. The regime with $|a|/r$ or $|b|/r$ bounded but not both zero needs the limit of $\max(|a + rZ_a|, |b + rZ_b|)$ with finite offsets and is not treated.

(c) $\{\mathrm{EO}_m \ge \delta\} = \{a \ge \delta\} \cup \{a \le -\delta\} \cup \{b \ge \delta\} \cup \{b \le -\delta\}$. The procedure rejects iff the four one-sided tests reject: $Z_a(-\delta) > z_{1-\alpha}$, $Z_a(\delta) < -z_{1-\alpha}$, and the same for b; equivalently the $(1-2\alpha)$ Wald intervals of a and b lie in $(-\delta, \delta)$. If $\mathrm{EO}_m \ge \delta$ at least one of the four null hypotheses holds; pass to a subsequence along which it is always the same one. By Theorem 6(c) (and its mirror image for the lower margin) the corresponding one-sided test has $\limsup P(\text{reject}) \le \alpha$, and the intersection test rejects only if this test does. The argument uses only the marginal assumptions for a and b and no condition on ties. □

S4 Additional testing simulations

This section gives the protocol and the full tables of the simulations summarised in Section 6.7. All results come from `experiments/testing_sim.py` (library: `tcs/labeling_tests.py`, unit tests: `tests/test_labeling_tests.py`) and are stored in `results/testing_sim.json`; a summary is in `results/testing_sim_summary.json`.

S4.1 Protocol

Population with known nulls.

Covariates $x \in \mathbb{R}^5$ are standard normal, the group has prevalence 0.3, and $Y \sim \mathrm{Bernoulli}\{\mathrm{expit}(\beta^\top x - 0.2 + 0.3G)\}$ with $\beta = (1, -0.8, 0.6, 0, 0)$, as in the non-smooth simulation. Model m predicts $1\{\mathrm{expit}(s_m(\beta_m^\top x + o_m G)) > 1/2\}$. Models 0 to 3 have the weight on x_4 of the form $\beta_{m4} = \pm c_m$, $c_m \in [0.6, 1]$ and $\beta_{m5} = 0$; model $m+4$ copies model m except that the weight is on x_5. Model 8 uses neither. Since (x_4, x_5) are exchangeable, independent of each other and of (x_1, x_2, x_3, G, Y), the law of $(\hat{Y}_m, Y, G)$ equals that of $(\hat{Y}_{m+4}, Y, G)$, so the three contrasts of the pairs $(m, m+4)$, $m = 0, \ldots, 3$, are exactly zero in the population: 12 true nulls out of 108. The Monte Carlo value of these 12 contrasts from 2×10^7 draws is at most 0.001 in absolute value, consistent with the Monte Carlo error; they are set to zero. The smallest $|\theta_k|$ among the other contrasts is 0.007, and 80 contrasts have

$|\theta_k| \geq 0.02$ ("material" false nulls). Because the contrasts are zero only in the population, the null family is tested only for the population target; for the pool target the family in which every hypothesis is true ($H_k : \theta_k = \theta_k^{\text{pool}}$) is used.

Estimators and variances.

SRS and PATTERN use the plug-in (expansion) estimator, PATTERN-DIF the difference estimator with the ensemble-mean proxy, strata (G, K), plain Neyman allocation with a floor of 10 labels per stratum, and no variance floor. The covariance matrix $\hat{\Sigma}$ is the stratified covariance of the residual influence values (difference estimator) or of the influence values (plug-in); its diagonal equals the variances of `tcs.labeling_dif` and `tcs.labeling`, which is checked in the unit tests. For the population target the pool covariance $\hat{\Sigma}_{\text{pool}}$ is added. Holm's step-down procedure [12] uses the Wald p-values. The single-step max-t critical value uses 20,000 standard normal draws with a fixed seed, transformed by the eigen-decomposition of $\hat{R}$ (negative eigenvalues set to zero and the margins renormalised); it is the exact normal quantile for $K = 1$ and the Sidak value if $\hat{R} = I$ (unit tests). Bonferroni and Holm have the same FWER when all hypotheses are true, which is why the tables show identical values for the all-true families.

Replications.

Synthetic study: $R = 1000$ new pools per cell (no cell was reduced), six (N, L) cells, $L \leq N/4$; seeds are a function of the cell. Undefined estimates (an empty stratum or a non-positive denominator total) occurred in no replication. Real pools: the eleven settings of the main study (contrasts that are identically zero in the pool are excluded: $K = 108$, except 57 in Law school), $L \in \{250, 500, 1000, 2000\}$ with $L \leq N/4$, $R = 400$ (reduced from 1000 because of the number of cells), pool contrasts from all labels. Power curves: seven budgets between 200 and $\min(4000, N/4)$, $R = 1000$; the synthetic pool for this study and the next is one fixed pool of $N = 20{,}000$ units (pool-conditional target). Power-optimal design: $R = 1000$ per allocation. The budget for 80% power is found by linear interpolation of power in $\log L$ between the grid points straddling 0.8; it is censored if the curve starts above or ends below 0.8.

Predictions.

The predicted power uses $\Phi(-z + |\Delta|/\sqrt{V}) + \Phi(-z - |\Delta|/\sqrt{V})$ with V the exact design variance of the estimator, $V_k(n) = \sum_h N_h^2(1/n_h - 1/N_h)S_{hk}^2$, where S_{hk}^2 is the finite-population variance within stratum h of the (residual) influence value of contrast k at the true totals of the pool, and n_h the allocation used at that budget. The predicted budget is found on a grid of 40 budgets with the same interpolation as for the simulated curves. Contrasts were chosen per setting among those whose predicted budgets for SRS, PATTERN and PATTERN-DIF are all inside the grid, six at evenly spaced ranks of the SRS budget. Law school, Spambase and Churn have no such contrasts (budget range too small or effects too small or too large) and are not in the power study.

Power-optimal allocation.

The maximin allocation minimises $\max_k V_k(n)/\delta_k^2$ over $\sum_h n_h = L$, $\min(N_h, 10) \leq n_h \leq N_h$, with V_k from the proxy variance model of the residual influence values (label free); it is solved by mirror descent on the dual weights, with the duality gap reported ($\leq 0.13\%$ in all eight problems) and the integer allocation obtained by rounding. Performance is evaluated by simulating the labelling with each allocation and testing, for every targeted contrast, the null value θ_k^0 whose distance from the truth is the target effect (for E1, $\theta_k^0 = 0$; for E2, $\theta_k^0 = \theta_k - \text{sign}(\theta_k) \times 0.05$), using the true contrasts to define the targets. The minimum over contrasts of a simulated power is biased downwards by Monte Carlo noise (about 0.02 to 0.04 for $R = 1000$); the predicted minimum, computed from the exact variances of the allocation used, is free of it and agrees with the simulated one within this bias.

S4.2 Size

Size is correct to within Monte Carlo error for the pool-conditional tests of the pool contrast and the two-phase tests of the population contrast in all designs and cells (Table S1); the lowest rejection rates of single contrasts are 0.03 at $\alpha = 0.05$, so no contrast is systematically over-covered either. SRS has the largest worst-contrast rates at $L = 250$ (0.081 at $\alpha = 0.05$). The one-sided tests are as accurate as the two-sided ones. The TOST at the boundary is conservative on average, as expected

Table S1 Size of the two-sided Wald test of the true value, mean and maximum over 108 contrasts, by design, budget and level ($R = 1000$). "Pool, cond.": pool-conditional test of the pool contrast. "Pop., two-phase": two-phase test of the population contrast. The maximum of a calibrated test has mean 0.019, 0.068, 0.125 at $\alpha = 0.01$, 0.05, 0.10.

			$\alpha = 0.01$				$\alpha = 0.05$				$\alpha = 0.1$			
			Pool, cond.		Pop., two-phase		Pool, cond.		Pop., two-phase		Pool, cond.		Pop., two-phase	
N	L	Design	mean	max	mean	max	mean	max	mean	max	mean	max	mean	max
2,000	250	SRS	0.012	0.024	0.013	0.023	0.055	0.075	0.056	0.078	0.106	0.131	0.107	0.133
		PATTERN	0.012	0.019	0.012	0.021	0.054	0.073	0.054	0.072	0.105	0.126	0.105	0.134
		PATTERN-DIF	0.011	0.024	0.011	0.019	0.051	0.066	0.051	0.067	0.100	0.124	0.101	0.133
2,000	500	SRS	0.011	0.018	0.010	0.016	0.052	0.072	0.052	0.069	0.103	0.124	0.102	0.128
		PATTERN	0.011	0.019	0.011	0.020	0.053	0.070	0.052	0.069	0.104	0.130	0.101	0.122
		PATTERN-DIF	0.012	0.019	0.011	0.022	0.053	0.069	0.051	0.069	0.103	0.122	0.102	0.125
8,000	250	SRS	0.012	0.028	0.012	0.029	0.056	0.081	0.056	0.078	0.107	0.134	0.108	0.133
		PATTERN	0.012	0.021	0.012	0.021	0.053	0.073	0.053	0.070	0.103	0.131	0.102	0.123
		PATTERN-DIF	0.011	0.018	0.010	0.020	0.052	0.069	0.051	0.066	0.101	0.121	0.101	0.121
8,000	500	SRS	0.012	0.026	0.012	0.022	0.055	0.070	0.054	0.070	0.106	0.134	0.106	0.123
		PATTERN	0.010	0.017	0.010	0.019	0.051	0.066	0.050	0.063	0.100	0.124	0.100	0.122
		PATTERN-DIF	0.009	0.017	0.010	0.021	0.048	0.064	0.049	0.067	0.098	0.116	0.099	0.123
8,000	1000	SRS	0.010	0.021	0.010	0.018	0.051	0.069	0.053	0.066	0.101	0.121	0.103	0.124
		PATTERN	0.010	0.021	0.009	0.019	0.050	0.063	0.048	0.067	0.101	0.122	0.097	0.123
		PATTERN-DIF	0.011	0.020	0.009	0.019	0.050	0.069	0.048	0.063	0.098	0.124	0.096	0.122
8,000	2000	SRS	0.010	0.020	0.011	0.022	0.049	0.063	0.050	0.068	0.097	0.117	0.100	0.120
		PATTERN	0.010	0.017	0.011	0.024	0.051	0.071	0.052	0.065	0.101	0.124	0.102	0.122
		PATTERN-DIF	0.011	0.021	0.011	0.019	0.052	0.074	0.053	0.071	0.103	0.125	0.104	0.131

Table S2 Rejection rate of the pool-conditional two-sided Wald test of the *population* value (PATTERN-DIF, $R = 1000$). The pool term is ignored, so the size exceeds the level when ϖ is large.

			$\alpha = 0.01$		$\alpha = 0.05$		$\alpha = 0.10$	
N	L	ϖ	mean	max	mean	max	mean	max
2000	250	0.39	0.028	0.045	0.092	0.133	0.158	0.210
2000	500	1.11	0.069	0.112	0.165	0.233	0.244	0.309
8000	250	0.08	0.013	0.023	0.060	0.081	0.114	0.145
8000	500	0.20	0.017	0.031	0.070	0.097	0.127	0.156
8000	1000	0.46	0.029	0.051	0.094	0.140	0.159	0.215
8000	2000	1.19	0.077	0.140	0.175	0.257	0.252	0.342

when the margin is small relative to the standard error.

S4.3 Familywise error and power

When all hypotheses are true the simultaneous coverage of the max-t bands for PATTERN-DIF is 0.929 to 0.951 (pool target) and 0.935 to 0.951 (population target), below the nominal 0.95 in most cells, while Bonferroni and Holm are at 0.95 to 0.97. The shortfall of max-t is largest for SRS and PATTERN at small budgets (coverage 0.91 to 0.93 at $L = 250$), where the estimated covariance of 108 contrasts from 250 labels is noisy, and for the pool target at small pools. We therefore describe max-t as approximately, not exactly, calibrated at these budgets. With the 12 true nulls the FWER of every adjusted procedure is at most 0.012. The unadjusted tests have FWER 0.35 to 0.48 in this mixed family and a simultaneous coverage of 0.08 to 0.24 in the all-true family.

S4.4 Real pools

The per-cell values (Table S5) show that the shortfalls of the simultaneous procedures occur where the individual intervals already undercover at small budgets: Law school at $L = 250$ (Bonferroni 0.72, max-t 0.65) and $L = 500$ (max-t 0.89), MAGIC at $L = 250$ (max-t 0.885, Bonferroni 0.92) and Spambase at $L = 250$ (max-t 0.897). The zero-variance floor repairs most of Law school at $L = 250$ (Bonferroni minimum over cells 0.92, max-t 0.89). PATTERN-DIF max-t is at or above 0.93 in 51% of the cells; in the summary (Table S6) its mean coverage rises with the budget from 0.90 to 0.945. Undefined estimates occurred in at most 0.5% of the replications. Because $\hat{\Sigma}$ is estimated from the same strata as the standard errors, the simultaneous procedures inherit the small-budget deficiencies of the single intervals and add the noise of the correlation matrix.

Table S3 Rejection rates at the boundary of the null for pool-conditional tests of the pool contrast: one-sided tests ($H_0 : \theta \geq \theta_0$ and $H_0 : \theta \leq \theta_0$, the first being a non-inferiority test with margin θ_0) and TOST of $H_0 : |\theta| \geq \delta$ with $\delta = |\theta_0|$ so that the truth is on the boundary (contrasts with $|\theta| \geq 0.02$ only). Mean and maximum over contrasts.

			$\alpha = 0.05$						$\alpha = 0.1$					
			$H_1 : \theta < \theta_0$		$H_1 : \theta > \theta_0$		TOST		$H_1 : \theta < \theta_0$		$H_1 : \theta > \theta_0$		TOST	
N	L	Design	mean	max	mean	max	mean	max	mean	max	mean	max	mean	max
2,000	250	SRS	0.054	0.083	0.052	0.067	0.022	0.073	0.105	0.132	0.104	0.126	0.051	0.131
		PATTERN	0.052	0.072	0.053	0.070	0.023	0.061	0.101	0.125	0.103	0.127	0.054	0.117
		PATTERN-DIF	0.049	0.062	0.051	0.071	0.026	0.060	0.099	0.122	0.101	0.122	0.063	0.115
2,000	500	SRS	0.053	0.079	0.050	0.067	0.030	0.076	0.103	0.131	0.101	0.121	0.068	0.128
		PATTERN	0.052	0.070	0.052	0.071	0.035	0.064	0.102	0.120	0.103	0.130	0.077	0.114
		PATTERN-DIF	0.051	0.072	0.051	0.071	0.038	0.063	0.101	0.125	0.103	0.124	0.081	0.116
8,000	250	SRS	0.055	0.080	0.052	0.069	0.021	0.075	0.106	0.145	0.103	0.130	0.050	0.139
		PATTERN	0.053	0.072	0.051	0.066	0.019	0.070	0.102	0.128	0.100	0.121	0.048	0.116
		PATTERN-DIF	0.051	0.069	0.051	0.062	0.023	0.060	0.101	0.122	0.101	0.118	0.057	0.114
8,000	500	SRS	0.054	0.073	0.052	0.072	0.028	0.070	0.104	0.133	0.103	0.129	0.062	0.132
		PATTERN	0.049	0.060	0.051	0.069	0.029	0.062	0.097	0.119	0.102	0.124	0.068	0.119
		PATTERN-DIF	0.049	0.061	0.049	0.066	0.034	0.062	0.098	0.118	0.099	0.116	0.074	0.112
8,000	1000	SRS	0.051	0.067	0.050	0.070	0.035	0.066	0.100	0.127	0.100	0.121	0.076	0.127
		PATTERN	0.051	0.069	0.050	0.063	0.040	0.067	0.100	0.120	0.098	0.124	0.084	0.118
		PATTERN-DIF	0.049	0.064	0.049	0.063	0.041	0.063	0.099	0.118	0.099	0.124	0.086	0.123
8,000	2000	SRS	0.049	0.064	0.048	0.061	0.040	0.064	0.099	0.122	0.097	0.117	0.084	0.117
		PATTERN	0.050	0.073	0.051	0.072	0.043	0.073	0.100	0.121	0.103	0.128	0.089	0.121
		PATTERN-DIF	0.051	0.070	0.052	0.073	0.046	0.070	0.100	0.124	0.102	0.130	0.092	0.122

Table S4 Synthetic population, $R = 1000$, familywise level 0.05. FWER: at least one rejection among the 12 true nulls of $H_k : \theta_k = 0$ (population target, two-phase covariance). Power: mean rejection rate over the 80 false nulls with $|\theta_k| \geq 0.02$. Covered: share of replications in which no hypothesis of the family $H_k : \theta_k = \theta_k^{\text{true}}$ (all 108 true) is rejected, for the pool target (pool-conditional covariance) and the population target (two-phase covariance); one minus this is the FWER of the all-true family.

			FWER (12 true nulls)				Power, 80 material false nulls				Pool family covered				Population family covered			
N	L	Design	Un.	Bonf.	Holm	Max-t	Un.	Bonf.	Holm	Max-t	Un.	Bonf.	Holm	Max-t	Un.	Bonf.	Holm	Max-t
2,000	250	SRS	0.433	0.005	0.005	0.008	0.348	0.068	0.069	0.077	0.082	0.933	0.933	0.912	0.088	0.942	0.942	0.928
		PATTERN	0.480	0.008	0.010	0.011	0.398	0.113	0.116	0.124	0.083	0.934	0.934	0.920	0.094	0.934	0.934	0.916
		PATTERN-DIF	0.398	0.006	0.008	0.009	0.467	0.183	0.189	0.198	0.195	0.964	0.964	0.943	0.133	0.953	0.953	0.936
2,000	500	SRS	0.425	0.002	0.002	0.002	0.497	0.192	0.199	0.206	0.096	0.954	0.954	0.934	0.095	0.960	0.960	0.946
		PATTERN	0.426	0.007	0.008	0.008	0.557	0.267	0.276	0.280	0.103	0.958	0.958	0.937	0.098	0.945	0.945	0.928
		PATTERN-DIF	0.406	0.007	0.008	0.008	0.609	0.324	0.335	0.338	0.193	0.960	0.960	0.929	0.118	0.952	0.952	0.936
8,000	250	SRS	0.471	0.009	0.009	0.011	0.349	0.068	0.070	0.078	0.075	0.931	0.931	0.910	0.070	0.936	0.936	0.925
		PATTERN	0.430	0.007	0.008	0.012	0.393	0.112	0.115	0.124	0.106	0.945	0.945	0.921	0.101	0.945	0.945	0.923
		PATTERN-DIF	0.353	0.005	0.006	0.009	0.473	0.200	0.205	0.217	0.201	0.967	0.967	0.947	0.183	0.964	0.964	0.945
8,000	500	SRS	0.430	0.006	0.006	0.008	0.497	0.194	0.201	0.208	0.104	0.936	0.936	0.922	0.108	0.938	0.938	0.918
		PATTERN	0.428	0.010	0.010	0.010	0.555	0.268	0.276	0.281	0.098	0.954	0.954	0.937	0.094	0.954	0.954	0.939
		PATTERN-DIF	0.375	0.002	0.003	0.004	0.635	0.352	0.365	0.370	0.205	0.967	0.967	0.947	0.146	0.966	0.966	0.950
8,000	1000	SRS	0.424	0.002	0.007	0.006	0.634	0.364	0.378	0.378	0.116	0.965	0.965	0.951	0.093	0.965	0.965	0.946
		PATTERN	0.418	0.003	0.006	0.004	0.697	0.439	0.455	0.451	0.122	0.969	0.969	0.960	0.126	0.963	0.963	0.956
		PATTERN-DIF	0.386	0.007	0.011	0.009	0.759	0.518	0.540	0.534	0.236	0.969	0.969	0.951	0.156	0.967	0.967	0.951
8,000	2000	SRS	0.417	0.005	0.009	0.007	0.754	0.525	0.544	0.536	0.113	0.959	0.959	0.944	0.103	0.947	0.947	0.933
		PATTERN	0.445	0.007	0.008	0.007	0.810	0.602	0.624	0.613	0.112	0.963	0.963	0.946	0.104	0.952	0.952	0.940
		PATTERN-DIF	0.450	0.006	0.008	0.008	0.842	0.658	0.683	0.670	0.197	0.957	0.957	0.934	0.100	0.950	0.950	0.935

S4.5 Power curves and the sample-size check

The predicted and observed budget ratios agree in every setting (Table S7); the ratio of the budgets is 0.96 to 1.03 times the prediction by setting. Two contrasts (one in EEG, one in Phoneme) were censored in the simulation. The ratio of variances at the budget of PATTERN-DIF overstates the budget ratio by 3% to 20% in the median by setting, because of the finite-pool factor. The comparison with the main-study efficiency shows that the latter depends on the budget and on the contrast set: it is the median MSE ratio over all live contrasts at $L = 500$, whereas the budget ratio applies to contrasts that SRS can detect within the grid.

S4.6 Power-optimal allocation

The maximin allocation lowers the largest variance-to-effect ratio of the targeted contrasts (by 1.5 to 2.0 in the real pools, 1.05 on the synthetic pool) and the simulated minimum power follows the prediction. It moves labels to the strata that matter for the hardest contrasts and away from the others, which lowers the mean power

Table S5 Simultaneous coverage of the K live pool contrasts at familywise level 0.05 ($R = 400$ per row; Monte Carlo standard error about 0.011 at 0.95). PATTERN-DIF without floor; Max-t (floor) uses the zero-variance floor of `tcs.labeling_dif`. Bonferroni and Holm coincide when every hypothesis is true. $c_{\max t}/c_{\mathrm{Bonf}}$: mean max-$t$ critical value over the Bonferroni value.

Setting	L	K	Unadj.	Bonf./Holm	Max-t	Max-t (floor)	$c_{\max t}/c_{\mathrm{Bonf}}$
Adult (sex)	250	108	0.185	0.955	0.925	0.938	0.945
	500	108	0.185	0.968	0.935	0.943	0.949
	1000	108	0.215	0.963	0.922	0.930	0.950
	2000	108	0.190	0.983	0.958	0.958	0.947
Adult (race)	250	108	0.198	0.973	0.953	0.965	0.946
	500	108	0.188	0.960	0.932	0.935	0.947
	1000	108	0.220	0.985	0.953	0.953	0.946
	2000	108	0.265	0.975	0.945	0.945	0.930
Bank marketing	250	108	0.130	0.935	0.905	0.910	0.954
	500	108	0.130	0.948	0.912	0.915	0.955
	1000	108	0.208	0.950	0.925	0.925	0.952
	2000	108	0.240	0.970	0.948	0.948	0.940
Credit card	250	108	0.245	0.958	0.920	0.943	0.937
	500	108	0.243	0.993	0.968	0.980	0.937
	1000	108	0.208	0.973	0.950	0.968	0.937
	2000	108	0.213	0.968	0.930	0.938	0.934
Law school	250	57	0.427	0.722	0.653	0.910	0.825
	500	57	0.720	0.935	0.890	0.892	0.818
	1000	57	0.688	0.980	0.905	0.905	0.811
	2000	57	0.917	0.995	0.950	0.950	0.678
MAGIC	250	108	0.208	0.920	0.885	0.887	0.949
	500	108	0.213	0.930	0.910	0.905	0.952
	1000	108	0.200	0.968	0.945	0.945	0.954
EEG eye state	250	108	0.165	0.958	0.935	0.943	0.961
	500	108	0.162	0.975	0.953	0.955	0.963
	1000	108	0.157	0.960	0.945	0.945	0.964
Phoneme	250	108	0.172	0.945	0.927	0.930	0.953
	500	108	0.162	0.958	0.932	0.935	0.954
Spambase	250	108	0.155	0.938	0.897	0.900	0.954
Churn	250	108	0.132	0.955	0.917	0.917	0.953
	500	108	0.220	0.943	0.912	0.912	0.940
Electricity	250	108	0.182	0.945	0.925	0.930	0.955
	500	108	0.160	0.968	0.953	0.955	0.957
	1000	108	0.218	0.978	0.948	0.948	0.957
	2000	108	0.160	0.960	0.943	0.943	0.957

and increases the number of contrasts below 80% in several pools. The allocation for the residual variance with the default weights ("resid") is not better than the default.

S4.7 Limitations

- The exact nulls exist only in the population and there are only 12 of them among 108 hypotheses, so the FWER of the mixed family is far below the nominal level for every adjusted procedure; tightness is assessed through the all-true family.

Table S6 Simultaneous coverage of the family of pool contrasts on the eleven real pools, mean and minimum over setting–budget cells at the budget L.

		Mean over cells				Minimum over cells			
L	Design	Un.	Bonf.	Holm	Max-t	Un.	Bonf.	Holm	Max-t
250 (11 cells)	SRS	0.049	0.473	0.473	0.453	0.00	0.08	0.08	0.08
	PATTERN	0.123	0.824	0.824	0.801	0.06	0.45	0.45	0.44
	PATTERN-DIF	0.200	0.928	0.928	0.895	0.13	0.72	0.72	0.65
	PATTERN-DIF + floor	0.238	0.952	0.952	0.925	0.13	0.92	0.92	0.89
500 (10 cells)	SRS	0.072	0.632	0.632	0.609	0.02	0.23	0.23	0.22
	PATTERN	0.181	0.913	0.913	0.891	0.09	0.72	0.72	0.71
	PATTERN-DIF	0.238	0.957	0.957	0.930	0.13	0.93	0.93	0.89
	PATTERN-DIF + floor	0.248	0.958	0.958	0.933	0.14	0.93	0.93	0.89
1000 (8 cells)	SRS	0.114	0.802	0.802	0.786	0.01	0.57	0.57	0.54
	PATTERN	0.206	0.961	0.961	0.932	0.09	0.94	0.94	0.88
	PATTERN-DIF	0.264	0.969	0.969	0.937	0.16	0.95	0.95	0.91
	PATTERN-DIF + floor	0.270	0.970	0.970	0.940	0.16	0.95	0.95	0.91
2000 (6 cells)	SRS	0.127	0.911	0.911	0.890	0.07	0.81	0.81	0.80
	PATTERN	0.276	0.969	0.969	0.949	0.10	0.94	0.94	0.92
	PATTERN-DIF	0.331	0.975	0.975	0.945	0.16	0.96	0.96	0.93
	PATTERN-DIF + floor	0.335	0.976	0.976	0.947	0.16	0.96	0.96	0.94

Table S7 Label budget for 80% power of the two-sided Wald test of $H_0 : \theta_k = 0$ by setting. Contr.: number of contrasts with both budgets observed / selected. $V_{\mathrm{SRS}}/V_{\mathrm{DIF}}$: exact variance ratio at the PATTERN-DIF budget; MSE ratio ($L = 500$): median efficiency of PATTERN-DIF against SRS in the main study (`results/label_efficiency_dif.json`); pred. and obs.: median over contrasts of the predicted and observed ratio of budgets; Within ±20%: share of contrasts with observed within 20% of predicted. The last block gives SRS against PATTERN (plug-in).

Setting	Contr.	$V_{\mathrm{SRS}}/V_{\mathrm{DIF}}$ (oracle, at L_{DIF})	MSE ratio, $L = 500$ (main study)	$L_{\mathrm{SRS}}/L_{\mathrm{DIF}}$ pred.	obs.	Obs./pred. median	$L_{\mathrm{SRS}}/L_{\mathrm{PATTERN}}$ pred.	obs.	Within ±20%
Synthetic pool	6/6	2.88	–	2.79	2.60	0.97	1.53	1.47	1.00
Adult (sex)	6/6	5.97	6.80	5.40	5.35	1.02	3.31	3.28	1.00
Adult (race)	6/6	3.57	12.44	3.29	3.37	0.97	2.93	2.98	1.00
Bank marketing	6/6	5.88	12.07	4.92	4.88	1.00	5.18	5.02	1.00
Credit card	6/6	4.37	5.26	3.66	3.59	0.96	3.42	3.36	1.00
MAGIC	6/6	3.52	4.82	3.11	3.07	1.00	2.51	2.44	1.00
EEG eye state	5/6	4.35	3.73	3.76	3.93	1.03	1.71	1.77	1.00
Phoneme	2/3	2.94	5.79	2.49	2.28	0.97	2.21	2.16	1.00
Electricity	6/6	3.01	4.54	2.83	2.87	1.01	2.66	2.67	1.00

- There is one synthetic design and eleven real pools; the real-pool study has only the all-true family and $R = 400$, so coverages are resolved to about 0.01.
- The max-t procedure uses an estimated correlation matrix and showed mild anticonservatism (up to 0.09 FWER) at small budgets, which the theory does not quantify.
- Power was studied for the pool-conditional target and for contrasts whose budgets fall inside the grid; contrasts that need many more labels, or fewer, are not represented, and the target effects are the true contrasts.
- The power-optimal allocation uses the true effect sizes and a proxy variance model; with a poorly calibrated proxy the predicted gain need not materialise, and the gains in minimum power are small in absolute terms.
- Equivalence and non-inferiority were examined at the boundary of the null only, not their power; the Bonferroni, Holm and max-t procedures were examined for two-sided tests only.

S5 Checking the theory by simulation

Section 5 gives three predictions of the efficiency $\mathrm{Eff}_t = \mathrm{MSE}_{\mathrm{SRS}}/\mathrm{MSE}_{\mathrm{PATTERN}}$ of a comparison estimand t, in decreasing order of generality and increasing order of simplicity:

LIN the exact linearised design effect of the allocation actually used, $N^2(1/L - 1/N)S_t^2/V_t(n)$ with V_t from (12); it is exact up to the linearisation of Proposition 1 and includes the compromise between targets, the proxy-based

Table S8 Default and maximin allocations for PATTERN-DIF ($R = 1000$). E1: the K_1 contrasts with $|\theta_k| \geq 0.02$ at effect $|\theta_k|$; E2: all K contrasts at a common effect 0.05. min and mean: minimum and mean over the targeted contrasts of the simulated power; pred. min: minimum predicted power; $< 80\%$: number of targeted contrasts with simulated power below 0.8. "Default (plain)" is the allocation of the main study, "Default (resid)" the Neyman allocation for the residual variance with weights $1/S_k^2$, "Maximin, E1/E2" the solution for the targets E1 and E2.

			E1: $\lvert\theta_k\rvert \geq 0.02$, effect $\lvert\theta_k\rvert$					E2: all live contrasts, effect 0.05				
Pool	L	Allocation	min	mean	pred. min	$< 80\%$	K_1	min	mean	pred. min	$< 80\%$	K
Synthetic pool	500	Default (plain)	0.05	0.70	0.07	36	75	0.14	0.38	0.17	100	108
		Default (resid)	0.07	0.70	0.07	36		0.16	0.38	0.16	100	
		Maximin, E1	0.07	0.70	0.07	38		0.15	0.37	0.17	102	
		Maximin, E2	0.07	0.70	0.07	36		0.16	0.37	0.17	103	
Synthetic pool	1000	Default (plain)	0.10	0.84	0.09	18	75	0.29	0.60	0.30	72	108
		Default (resid)	0.09	0.84	0.09	19		0.28	0.60	0.30	72	
		Maximin, E1	0.11	0.83	0.10	20		0.30	0.59	0.30	78	
		Maximin, E2	0.10	0.83	0.10	19		0.28	0.60	0.32	75	
Adult (sex)	500	Default (plain)	0.08	0.67	0.08	24	55	0.09	0.75	0.08	51	108
		Default (resid)	0.11	0.67	0.08	25		0.10	0.75	0.08	51	
		Maximin, E1	0.09	0.57	0.10	35		0.09	0.71	0.11	52	
		Maximin, E2	0.12	0.57	0.10	37		0.13	0.71	0.11	52	
Adult (sex)	1000	Default (plain)	0.12	0.78	0.11	20	55	0.13	0.87	0.12	21	108
		Default (resid)	0.12	0.77	0.11	20		0.12	0.87	0.13	24	
		Maximin, E1	0.14	0.67	0.14	32		0.15	0.79	0.18	35	
		Maximin, E2	0.15	0.69	0.15	28		0.18	0.79	0.18	35	
Bank marketing	500	Default (plain)	0.08	0.52	0.06	24	40	0.10	0.76	0.09	36	108
		Default (resid)	0.08	0.52	0.06	24		0.12	0.75	0.09	36	
		Maximin, E1	0.07	0.52	0.07	24		0.13	0.75	0.12	36	
		Maximin, E2	0.06	0.50	0.07	24		0.12	0.74	0.13	36	
Bank marketing	1000	Default (plain)	0.10	0.61	0.08	21	40	0.17	0.83	0.14	29	108
		Default (resid)	0.10	0.61	0.08	21		0.16	0.82	0.15	29	
		Maximin, E1	0.10	0.61	0.10	23		0.23	0.83	0.22	32	
		Maximin, E2	0.08	0.58	0.10	24		0.23	0.81	0.23	35	
Electricity	500	Default (plain)	0.10	0.85	0.10	21	78	0.23	0.83	0.22	32	108
		Default (resid)	0.11	0.85	0.10	21		0.24	0.83	0.24	32	
		Maximin, E1	0.15	0.80	0.12	28		0.38	0.80	0.35	44	
		Maximin, E2	0.12	0.80	0.12	28		0.35	0.80	0.35	44	
Electricity	1000	Default (plain)	0.15	0.95	0.15	8	78	0.37	0.95	0.38	11	108
		Default (resid)	0.16	0.95	0.16	8		0.41	0.95	0.42	10	
		Maximin, E1	0.20	0.93	0.21	8		0.64	0.95	0.63	8	
		Maximin, E2	0.21	0.93	0.21	9		0.65	0.95	0.64	11	

allocation, the minimum of 10 labels per stratum and the finite-population corrections;

OWN the own-target Neyman efficiency (24) on the pattern strata, which removes the compromise and the proxy but keeps the actual strata;

FORM the closed form of Corollary S8, (S8) and (S9), which assumes strata that separate the pair's agreements from its disagreements, and its small-target version $1/\rho$.

All three are computed from the full pool, using the labels only to evaluate the formulas, and are compared with Monte Carlo efficiencies. Everything in this section is produced by `experiments/theory_check.py` and stored in `results/theory_check.json`.

S5.1 Real pools

We re-ran the SRS and PATTERN arms of the main study (Section 7) on the 11 cached pools, at the same budgets, with $R = 200$ replications and the same random seeds; the GROUP and ORACLE samples are drawn as well, so that the random stream is identical. The resulting median efficiencies coincide with those in `results/label_efficiency.json` for all 35 pool–budget combinations (the medians in the main study exclude degenerate estimands, so in law school they differ from the medians over all targets reported here), so the per-target efficiencies below are those underlying the main results. A target is compared when its influence values are not identically zero and its design variance is positive. This removes the 45 targets of the 15 pairs of models with identical predictions in Law school, a further six targets in that pool, and all targets

of Law school at $L = 2000$, where the strata that carry the comparisons are fully enumerated and both the observed and the predicted efficiency are infinite (the json reports 4×10^{23}). That leaves 3519 target–budget pairs.

The linearised design effect is accurate.

Over the 3519 pairs the ratio of observed to predicted efficiency has median 0.998 and 10%–90% range 0.83–1.22; 77% of the ratios are within 20% and 99% within a factor 1.5, and the correlation of the logarithms is 0.988 (Figure S1, left). The spread is what Monte Carlo error alone produces: each MSE is an average of 200 squared errors, with a relative standard error of about $\sqrt{2/200} = 0.10$ for approximately normal errors, so the ratio of two independent MSEs has a relative standard error of about 0.14 and a 10%–90% range of roughly 0.84–1.20. The medians per pool and budget agree as well: the ratio of the observed median efficiency to the median LIN prediction lies between 0.88 and 1.20 for the 34 pool–budget combinations, and for each type of estimand separately the median of these ratios is 0.99 (TPR), 0.98 (FPR) and 1.00 (error). The agreement includes the cases with very large gains: for Churn at $L = 500$ the observed efficiency of 32.4 (predicted 32.2) arises because the disagreement strata are fully enumerated, which only the finite-population corrections in LIN capture. The lower correlations for Electricity, Phoneme and MAGIC reflect the narrow range of efficiencies across targets in those pools (most lie between 2 and 4), not a larger error. We conclude that, for the purpose of predicting efficiency, the linearised variance of Section 5.2 is adequate already at $L = 250$.

The closed form overstates the gain on real pools.

FORM is a much poorer predictor: the median ratio of observed efficiency to FORM is 0.47 (10%–90%: 0.17–1.17), and the rule of thumb $1/\rho$ is more optimistic still (Table S9). The reason is the one anticipated in Section 5.6: with nine models the strata (G, K) do not isolate the disagreements of a particular pair. The share of the pool on which some model disagrees, ρ_{any}, is between 0.18 and 0.73, while the median pairwise rate is between 0.001 and 0.33, and by (S13) the attainable gain lies between $1/\rho_{\text{any}}$ and $1/\rho$. OWN, which uses the actual strata, is much closer (median ratio of observed to OWN 0.87): the remaining 13% is the price of serving all 108 estimands with one allocation, which Proposition 5(d) makes explicit. Law school is a degenerate case for FORM, since the median pair disagrees on 0.1% of the pool and many targets have nearly zero variance; FORM and the observed efficiencies are negatively correlated there. FORM is therefore useful as an explanation of where the gain comes from and as an upper guide for a pair-specific design, but not as a numerical prediction for the nine-model design; for prediction, LIN should be used, and it requires nothing beyond the pool and a pilot or proxy estimate of the stratum variances.

S5.2 Synthetic pools with a controlled disagreement rate

To test the closed form where its assumptions hold, and to vary ρ over a wide range, we generated pools of $N = 200\,000$ with two groups ($\zeta_1 = 0.3$, $p_0 = 0.30$, $p_1 = 0.20$). A share $\rho \in \{1, 2, 5, 10, 20, 40\}\%$ of instances is placed in a disagreement region. Outside it, all models predict the same label, equal to Y with probability 0.85. Inside it, with two models, model 1 is right with probability q_g and model 2 predicts the opposite label; we use $q = (0.65, 0.35)$ (moderate group-wise gaps) and $q = (0.95, 0.05)$ (large gaps, one-sided disagreements). With five models, each model is right independently with a model- and group-specific probability between 0.40 and 0.70, conditionally on not all models agreeing. Each model's probability is $\text{logit}^{-1}\{1.73(2\hat{y} - 1) + \varepsilon\}$ with $\varepsilon \sim N(0, 1)$, and the proxy is their average, as in the real pools. The budget is $L = 1000$ and $R = 1000$, so the Monte Carlo relative standard error of an efficiency is about 0.06. With two models, the pattern strata separate agreement from disagreement exactly, and we also ran, for each target, its own Neyman allocation computed from the true stratum variances, which is the design FORM describes.

Efficiency scales as $1/\rho$***.***

In all three families the efficiency falls in proportion to $1/\rho$ over a forty-fold range of ρ (Figure S2). With two models and moderate gaps, ρ times the median observed efficiency (over all targets of

Table S9 Real pools: predicted and observed efficiency of PATTERN relative to SRS (medians over the 108 comparison estimands; $L = 500$, $L = 250$ for Spambase). ρ_{any}: share of the pool on which not all nine models agree; ρ: median pairwise disagreement rate. Obs: Monte Carlo ($R = 200$; identical to `results/label_efficiency.json`). LIN, OWN, FORM: predictions defined in the text. Last two columns: per-target ratio Obs/LIN, median and 10%–90% range, and the correlation of log Obs and log LIN across targets. For Law school the four medians (Obs, LIN, OWN, FORM) include the 51 estimands whose influence values vanish, whereas the last two columns use only the 57 compared targets; over those 57 the observed median is 37.6, as in Table 5. Source: `results/theory_check.json`.

Pool	N	ρ_{any}	ρ	Obs	LIN	OWN	FORM	$1/\rho$	Obs/LIN	corr
Adult-sex	19537	0.51	0.069	4.08	3.97	5.43	13.22	14.5	1.03 (0.83, 1.27)	0.95
Adult-race	19537	0.51	0.070	6.02	6.68	7.47	18.71	14.2	0.98 (0.77, 1.20)	0.98
Bank	18085	0.18	0.053	7.87	7.17	9.08	22.47	18.8	1.02 (0.83, 1.25)	0.96
Credit card	12000	0.56	0.055	3.79	3.88	4.48	12.96	18.3	0.99 (0.84, 1.18)	0.97
Law school	8320	0.19	0.001	15.33	13.79	25.74	3.54	924	1.06 (0.96, 1.22)	1.00
MAGIC	7608	0.40	0.166	2.76	2.83	3.31	5.68	6.0	0.96 (0.78, 1.16)	0.82
EEG	5992	0.73	0.329	1.83	1.75	2.24	3.19	3.0	1.03 (0.88, 1.27)	0.85
Phoneme	2162	0.39	0.162	3.60	3.61	2.91	5.53	6.2	1.00 (0.84, 1.19)	0.81
Spambase	1841	0.32	0.086	5.55	5.63	5.13	12.02	11.6	1.00 (0.85, 1.24)	0.92
Churn	2000	0.19	0.084	32.36	32.15	5.95	11.29	11.9	1.02 (0.89, 1.21)	1.00
Electricity	12000	0.34	0.128	2.84	2.93	3.34	6.69	7.8	0.96 (0.82, 1.18)	0.61

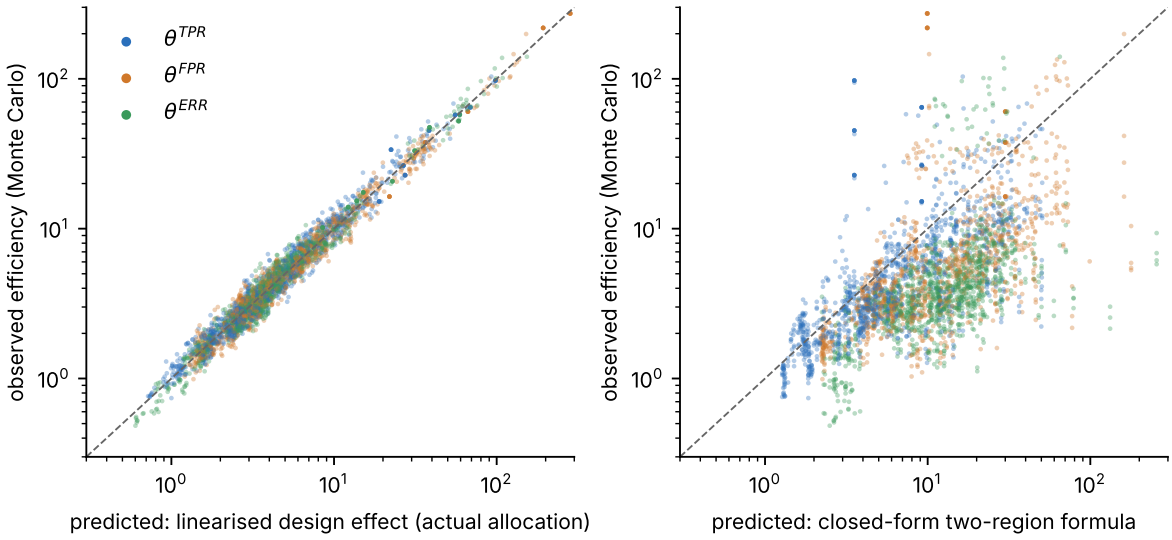


Fig. S1 Real pools, all budgets: observed efficiency of PATTERN relative to SRS for each of the 3519 compared target–budget pairs against the linearised prediction LIN (left) and the closed-form two-region formula FORM (right). Dashed: equality. Source: `results/theory_check.json`.

the scenario, the three types of estimand pooled; Table S10 shows them separately) is between 0.86 and 1.13 for $\rho \geq 2\%$, so the rule of thumb is accurate to about 15%; at $\rho = 1\%$ it is 1.9, because the budget of 1000 labels is then about half the size of the disagreement region and the finite-population correction roughly doubles the gain. LIN agrees with the observed efficiencies throughout (median ratio per scenario between 0.88 and 1.04), with a tendency to be optimistic by a few per cent that we attribute to the second-order terms neglected by the linearisation.

Table S10 Synthetic pools: observed (Obs), linearised (LIN) and closed-form (FORM) efficiency of PATTERN relative to SRS for θ^{TPR} and θ^{ERR} (median over pairs for five models), as a function of the disagreement rate ρ ($N = 200\,000$, $L = 1000$, $R = 1000$). Source: `results/theory_check.json`.

	Two models, moderate gaps						Two models, large gaps						Five models					
	θ^{TPR}			θ^{ERR}			θ^{TPR}			θ^{ERR}			θ^{TPR}			θ^{ERR}		
ρ	Obs	LIN	FORM	Obs	LIN	FORM	Obs	LIN	FORM	Obs	LIN	FORM	Obs	LIN	FORM	Obs	LIN	FORM
1%	190.3	216.0	102.2	174.1	181.1	115.5	56.2	55.8	46.2	665.4	608.0	517.0	172.9	198.3	170.7	158.1	162.9	203.2
2%	54.3	57.4	43.7	56.7	58.2	55.7	22.2	26.0	23.4	251.5	240.7	261.2	44.9	64.2	82.9	49.8	55.3	99.7
5%	14.9	18.6	16.6	23.2	20.2	22.1	10.0	9.7	9.5	88.1	83.1	103.7	17.4	21.0	32.9	19.5	19.6	40.1
10%	7.2	8.9	8.4	8.6	9.6	10.9	4.4	5.1	5.1	45.6	42.5	53.5	9.2	11.0	17.5	8.9	9.4	20.1
20%	4.4	4.7	4.6	5.0	4.8	5.5	2.4	2.8	2.8	16.7	20.3	25.6	4.9	5.4	9.0	4.7	4.6	10.0
40%	2.3	2.5	2.5	2.4	2.4	2.7	1.7	1.6	1.8	9.5	10.1	12.4	2.5	2.8	4.7	2.3	2.3	5.0

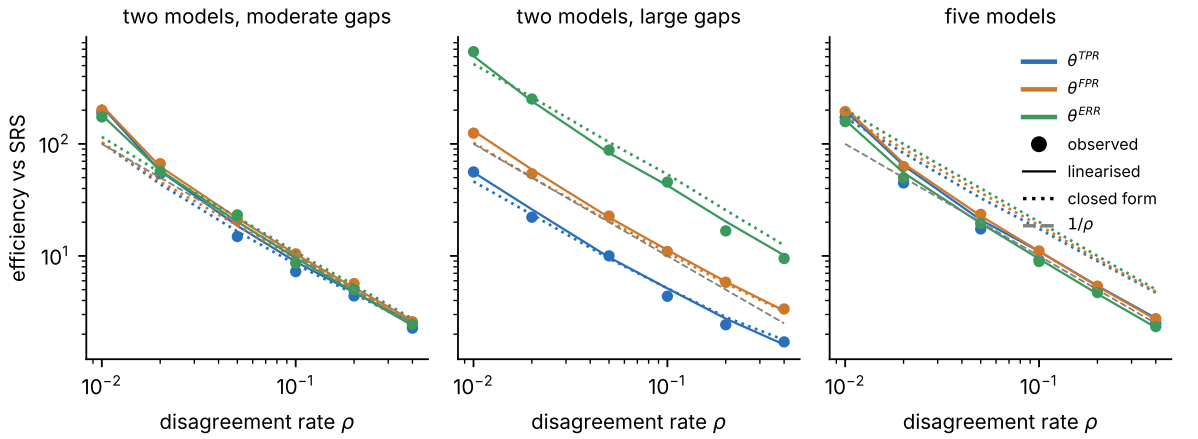


Fig. S2 Synthetic pools: efficiency of PATTERN relative to SRS against the disagreement rate ρ (share of the pool in the disagreement region), median over pairs for each type of estimand. Points: Monte Carlo ($R = 1000$); solid: linearised prediction LIN; dotted: closed form FORM; grey dashed: $1/\rho$. Source: `results/theory_check.json`.

The closed form is accurate, and conservative, when its assumptions hold.

With two models, the observed efficiency of the own-target Neyman design divided by FORM is between 0.90 and 1.25 for moderate gaps and $\rho \geq 5\%$, and between 0.95 and 1.47 for large gaps; the LIN predictions for the same designs give the same ratios, so the discrepancy is in the closed form and not in the simulation. FORM is below the truth for two reasons stated in Corollary S8(a): it treats the agreement region as one stratum, whereas the pattern design splits it by the common prediction ($K = 0$ versus $K = M$, which have very different positive rates), and it ignores the finite-population correction, which matters when the disagreement region is small (at $\rho = 1\%$ the ratio is 1.5 to 2.5). The large-gap family also shows the role of the target size in Corollary S8(c): the TPR gaps are large and the disagreements one-sided ($|a_g| = 0.9$), and the gain for θ^{TPR} is only 0.45 to 0.7 times $1/\rho$, in line with the factor $(\beta + \lambda)^{-2}$, which is about 0.4 for group 0 ($\pi_{\mathcal{D}} \approx \pi_{\mathcal{A}} \approx 0.3$) before the group factor Γ is applied, while for the error difference, whose disagreements are nearly deterministic within each group ($|\mu_g| = 0.9$), the gain is several times $1/\rho$, as (S9) predicts.

With more models, the relevant rate is between ρ_{any} and the pairwise rate.

In the five-model family the pairwise disagreement rate is about half of ρ, and FORM, which assumes that each pair's disagreements are isolated, overstates the gain by a factor of about 1.9 (median ratio of observed to FORM between 0.53 and 0.56 for $\rho \geq 2\%$). The observed efficiency is instead close to $1/\rho_{\text{any}}$ (ρ_{any} times the median efficiency

is between 0.94 and 1.06 for $\rho \geq 2\%$), the lower end of (S13). This is expected: in this family the models err independently inside the disagreement region, so a pair's disagreements are spread evenly over the values of K. In the real pools, where models make correlated errors, the median OWN efficiency lies inside the interval $[1/\rho_{\text{any}}, 1/\rho]$ for every pool (Table S9), mostly in its lower half.

S5.3 Misspecified proxy

Proposition S9 bounds the loss from a wrong proxy by the ratio (S14). We took the five-model pool with $\rho = 5\%$ and compared allocations computed from the default proxy, from the true labels (oracle), from a constant proxy equal to the overall positive rate (uninformative), from one minus the default proxy (inverted), and from an adversarial proxy that sets $\pi_i = 0.5$ where all models agree and $\pi_i = 0.001$ where they do not, with proportional allocation on the same strata as a reference.

None of the four wrong proxies costs a measurable amount of efficiency (Table S11): the predicted loss is at most 0.2%, the observed differences (between -2.4% and $+1.2\%$) are within Monte Carlo error, and every one of the 30 estimands is estimated at least 15 times more efficiently than under SRS. Proportional allocation on the same strata, by contrast, has eight times the weighted variance. The explanation is Proposition 2: the proxy enters the anticipated stratum variances only through label rates, while the dominant feature, namely that the comparison estimands receive almost no variance from the strata where all models agree, is fixed by the predictions. With a constant or adversarial proxy the agreement strata are still anticipated to have variance of order δ_g^2, which is small, and the disagreement strata still have variance of order one because D_i varies within them; so all these allocations put almost the whole budget into the disagreement strata and differ only in how it is divided among them. The Kantorovich bounds (1.21–1.24) are valid but loose here. This robustness is specific to the structure of the problem and should not be read as a general guarantee: Proposition S9(d) shows that a proxy that misjudges a high-variance stratum can do arbitrarily badly in other stratified problems, and the empirical comparison with the oracle allocation in Section 9 is the relevant check for real data.

S5.4 The difference estimator (PATTERN-DIF)

Section 5.4 makes four predictions about the difference estimator (14) with the proxy mean (13): it is exactly unbiased for the totals for any proxy and its plug-in variance is unbiased (Proposition 1); its linearised variance is $V_\psi = \sum_h c_h S_{e,h}^2$ with residual influence values $e_i = z_i - \tilde{z}_i$; its gain over the plain estimator on the same sample is V_0/V_ψ, which a label-free calculation forecasts under calibration (Proposition 4); and a miscalibrated proxy can lose to the plain estimator (Proposition S5). We check them with `experiments/theory_check_dif.py` (results in `results/theory_check_dif.json`). Every comparison is between the plain and the difference estimator on the *same* PATTERN sample, so the ratio MSE(plain)/MSE(DIF) isolates the estimator. Three predictions are used: **LIN**, $\sum_h c_h S_{z,h}^2 / \sum_h c_h S_{e,h}^2$, computed with the true labels; **ANT**, the label-free forecast $\sum_h c_h(\bar{\sigma}_h^2 + S_{\tilde{z},h}^2)/\sum_h c_h \bar{\sigma}_h^2$ of (21), which uses the proxy only; and **TUNE**, $V_\psi/V(\lambda)$ with $\lambda_h = \min\{1, \max(0, \beta_h)\}$, the most that tuning the coefficient could add to $\lambda = 1$.

Exactness and the variance formulae.

On a pool of 14 units in two strata (6 and 8 units, 2 and 3 labels) we enumerated all 840 samples for three proxies (a random mixture of u_i, v_i, an arbitrary bounded vector, and the anti-correlated vector $1 - t_i$) and three coefficients ($\lambda = 0, \frac{1}{2}, 1$). The mean of $\hat{T}_{\lambda\psi}$, its covariance matrix and the mean of the plug-in variance estimator agree with T, the formula of Proposition 1(a) and each other to within 3×10^{-13} in absolute terms (relative to the variances, 10^{-14}), that is machine precision, for every proxy. By Monte Carlo ($R = 2000$) the totals are also unbiased on the four real pools and six synthetic pools: the largest standardised bias over the 29–49 coordinates is $|z| = 2.7$ for the difference estimator and 2.6 for the plain estimator, which share their samples and have strongly correlated coordinates, and the largest bias is 0.06 standard deviations (the Monte Carlo resolution is 0.022). The variance formula holds: the

Table S11 Proxy misspecification (five models, $\rho = 5\%$, $L = 1000$, $R = 1000$). Loss: weighted variance $\sum_t w_t V_t$ relative to the oracle allocation, predicted by the linearisation (pred.) and observed with MSEs (obs.). Eff: harmonic mean over the 30 estimands of the efficiency relative to SRS, predicted and observed, and the smallest observed efficiency. Kant.: Kantorovich bound of Proposition S9(c) evaluated with the actual allocation. Source: `results/theory_check.json`.

Allocation	Loss pred.	Loss obs.	Eff pred.	Eff obs.	min Eff obs.	Kant.
Default proxy	1.00	1.01	22.0	20.5	15.4	1.24
Oracle	1.00	1.00	21.9	20.6	17.1	1.07
Uninformative	1.00	1.01	21.9	20.5	16.6	1.23
Inverted	1.00	0.98	22.0	21.2	17.2	1.24
Adversarial	1.00	0.98	21.9	21.0	15.9	1.21
Proportional	8.48	7.85	2.6	2.6	2.1	56.21

median over estimands of the Monte Carlo variance divided by the linearised V_ψ is between 0.99 and 1.02 on the real pools (Table S12) and 1.01 to 1.07 on the synthetic pools (up to 1.10 for θ^{TPR}, a second-order effect of the random denominators at $L = 600$), the mean of $\hat{V}_\psi$ divided by the Monte Carlo variance is between 0.98 and 1.02, and the Wald coverage of the difference estimator (0.943 to 0.951 on the real pools, 0.947 to 0.953 on the synthetic pools) is that of the plain estimator.

Predicted against observed gain.

On the four real pools the observed per-estimand gain divided by LIN has median 1.00 and 10%–90% range 0.97–1.04 (432 estimands, correlation of the logarithms 0.997; Figure S3, left), which is Monte Carlo error. We then compared LIN with the gains stored in `results/label_baselines.json`, which were produced independently by `experiments/label_baselines.py` with $R = 200$. In the 34 cells with a defined gain (all 35 except Law school at $L = 2000$, where both estimators are exact) the observed median gain is 1.47 (10%–90% range 1.28–2.10, extremes 1.22 and 11.1), LIN has median 1.49, and the ratio of observed to LIN has median 0.98 (10%–90% range 0.93–1.01, 97% of cells within 20%, correlation of the logarithms 0.996, 0.978 without the three Law school cells); Figure S3 (right). LIN uses the labels, so it is not a forecast, but it confirms that the residual-variance formula V_ψ describes the estimator. The label-free forecast ANT has median ratio 1.01 (0.90–1.37) over cells, and on the four real pools ratio 1.12 (0.86–1.53) with correlation 0.78 across estimands; the observed gain is 1.30 and 1.37 times ANT in EEG and Spambase. It fails in Law school (forecast 1.3–1.4, observed 5.1–11.1), where the disagreement sets of most pairs contain a handful of units per stratum: the realised residual variance is then a single draw from the label distribution, and it is an order of magnitude below its expectation under the proxy (we checked the stratum variances for one estimand). Without Law school the correlation of ANT with the observed gain is 0.62. The gain is therefore not a doubling in general: it is 1.2 to 1.7 in most cells, about 1.8 to 2.1 in EEG and Spambase, and 5 to 11 only in Law school. By estimand type the order is that of (S4): the gain is largest for θ^{FPR}, next for θ^{TPR} and smallest for θ^{ERR} (Adult-sex: 2.25, 1.67 and 1.13; Table S13 for the synthetic pools).

Tuning the coefficient.

TUNE is small: the median over the 34 cells is 1.02 (10%–90% 1.01–1.04), and 1.01 to 1.04 on the four real pools, so a perfectly tuned coefficient would add at most a few per cent to $\lambda = 1$ except in Law school (2.3–3.5). The observed estimated-coefficient version (PATTERN-PPI) has median gain 1.03 over $\lambda = 1$ (10%–90% 0.99–1.10), and it costs coverage: mean coverage over the 35 cells is 0.946 for PATTERN-DIF and 0.929 for PATTERN-PPI, and 8 of 35 cells fall below 0.93 for PATTERN-PPI against 1 of 35 for PATTERN-DIF and for the plain estimator. This is the trade-off of Proposition S6.

Allocation.

The residual-variance allocation of Proposition S7 does not improve on the design's plain-variance allocation for the difference estimator. The linearised prediction of the ratio of V_ψ under the plain allocation to V_ψ under the residual allocation

Table S12 Four real pools, $R = 2000$: efficiency of the difference estimator relative to the plain estimator on the same PATTERN sample (median over the 108 estimands). Obs: Monte Carlo; LIN, ANT, TUNE: predictions defined in the text; Obs/LIN: per-target ratio, median and 10%–90% range, and correlation of the logarithms; Var: Monte Carlo variance of $\hat{\theta}_{\mathrm{DIF}}$ over V_ψ; $\hat{V}$: mean estimated over Monte Carlo variance; Cov: Wald coverage of plain and difference estimator; max $|z|$: largest standardised bias of the 46–49 totals. Source: `results/theory_check_dif.json`.

Pool	L	Obs	LIN	ANT	TUNE	Obs/LIN	corr	Var	$\hat{V}$	Cov plain	Cov DIF	max $\lvert z\rvert$
Adult-sex	500	1.50	1.46	1.47	1.01	1.01 (0.99, 1.04)	0.998	0.99	1.02	0.949	0.951	1.6
Bank	500	1.45	1.48	1.46	1.01	0.99 (0.96, 1.03)	0.994	1.02	0.99	0.942	0.943	2.0
EEG	500	1.90	1.91	1.43	1.04	1.00 (0.97, 1.03)	0.997	1.00	1.00	0.947	0.948	2.2
Spambase	250	2.30	2.34	1.97	1.02	1.00 (0.96, 1.03)	0.998	1.00	0.99	0.944	0.945	1.4

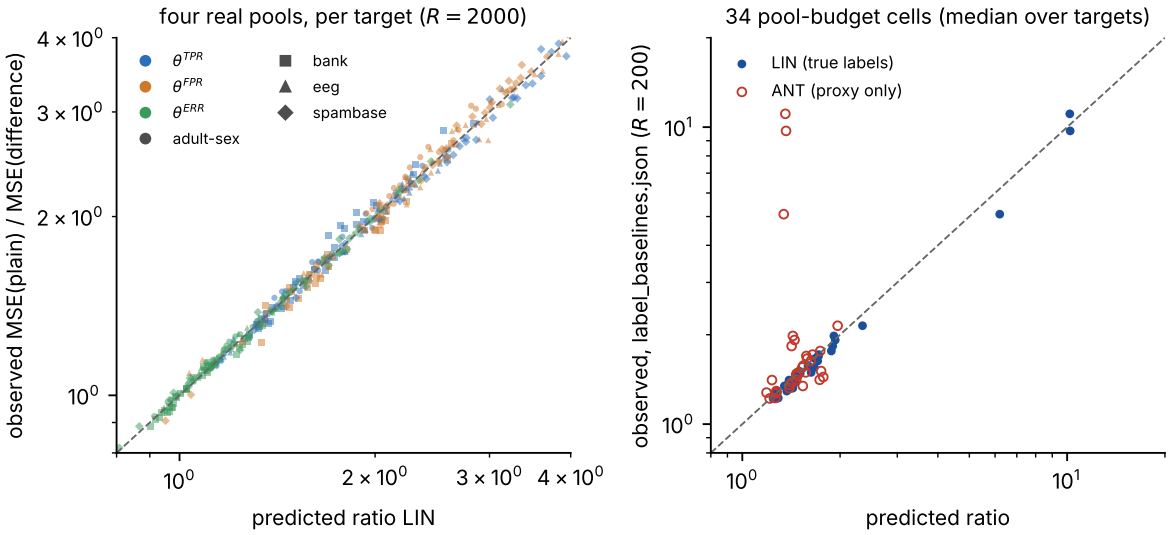


Fig. S3 Left: observed against linearised (LIN) gain of the difference estimator over the plain estimator, per estimand, on four real pools ($R = 2000$; 432 estimands). Right: median over estimands in each of 34 pool–budget cells, observed in `results/label_baselines.json` (PATTERN-DIF against PATTERN, $R = 200$) against LIN (filled) and ANT (open). Dashed: equality. Source: `results/theory_check_dif.json`.

has median 0.99 (10%–90% 0.95–1.00) over the 34 cells, and on four real pools the observed ratio of mean squared errors is 0.98, 0.98, 0.98 and 1.01, against linearised predictions 1.00, 1.00, 0.98 and 1.00, with the same coverage. Using the "wrong" variances therefore costs at most about 2% here, as the bound (S7) allows when the signal-to-noise ratio $\tilde{\nu}_h$ is similar across strata.

Calibrated and miscalibrated proxies.

In pools whose labels follow the proxy, ANT and LIN agree with each other and with the Monte Carlo gain within 2% (median observed/LIN between 0.99 and 1.00; Table S13, top), which verifies Proposition 4: the gain is 1.8 to 2.4 for the rate differences and 1.1 for the error difference, because in the disagreement strata the proxy is uncertain ($\overline{\pi(1-\pi)}$ is large) and the closed forms (S4) are bounded; a sharper latent score hardly changes them. For miscalibrated proxies (bottom), unbiasedness and coverage are unaffected (coverage 0.947–0.953), LIN keeps tracking the Monte Carlo gain (median ratio 0.99 to 1.01) and ANT, which assumes calibration, does not. The theory's two warnings appear. First, an inverted proxy ($\kappa = -1$) makes the difference estimator *worse* than the plain estimator for the error difference (gain 0.76, that is 1.3 times the mean squared error, as LIN predicts), while the rate differences still gain 1.4 to 1.5. The design-weighted slope $\bar{\beta} = \sum_h c_h \mathrm{Cov}_h(z, \tilde{z}) / \sum_h c_h S^2_{\tilde{z},h}$ of the error difference is -1.02, the clipped coefficient is 0 and

Table S13 Synthetic pools ($N = 60\,000$, five models, $Y_i \sim \text{Bernoulli}(\pi_i)$, $L = 600$, $R = 2000$): gain of the difference estimator over the plain estimator, observed (Obs), linearised (LIN) and label-free (ANT), median over the ten pairs. Top: calibrated proxy, sharpness a of the latent score. Bottom: $a = 2$, proxy $\pi_i^{\text{proxy}} = \text{logit}^{-1}(\kappa s_i)$ ($\kappa = 1$ calibrated, $\kappa = 0$ constant 0.5, $\kappa < 0$ inverted). Source: `results/theory_check_dif.json`.

	θ^{TPR}			θ^{FPR}			θ^{ERR}		
	Obs	LIN	ANT	Obs	LIN	ANT	Obs	LIN	ANT
$a = 0.5$	1.76	1.76	1.76	2.40	2.44	2.47	1.10	1.09	1.10
$a = 1$	2.03	1.97	1.98	2.33	2.33	2.36	1.09	1.09	1.10
$a = 2$	2.11	2.12	2.14	2.21	2.32	2.31	1.11	1.12	1.11
$a = 4$	2.23	2.23	2.24	2.26	2.27	2.26	1.11	1.12	1.12
$\kappa = -1$	1.43	1.48	2.29	1.52	1.56	2.15	0.76	0.76	1.12
$\kappa = 0$	1.88	1.94	2.00	2.04	2.05	2.00	1.00	1.00	1.00
$\kappa = 0.5$	2.11	2.11	2.02	2.23	2.22	2.09	1.10	1.09	1.03
$\kappa = 1$	2.21	2.16	2.15	2.28	2.27	2.29	1.12	1.12	1.12
$\kappa = 2$	2.09	2.06	2.59	2.22	2.17	2.89	1.08	1.07	1.39
$\kappa = 4$	1.81	1.80	3.81	1.89	1.90	4.45	0.96	0.94	2.11

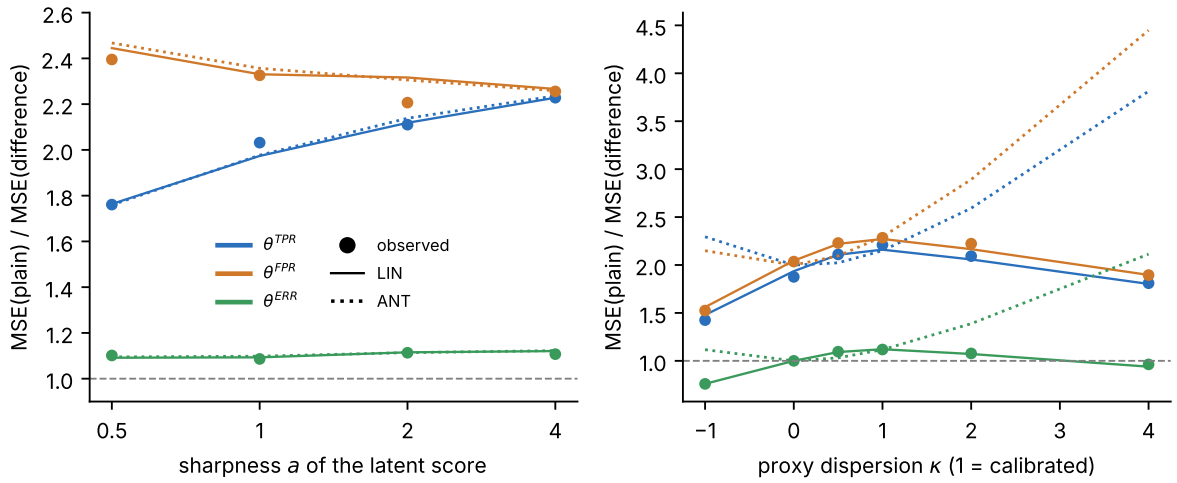


Fig. S4 Synthetic pools, gain of the difference estimator over the plain estimator (median over pairs, by estimand). Left: calibrated proxy, sharpness a of the latent score. Right: $a = 2$, proxy dispersion κ ($\kappa = 1$ calibrated). Points: Monte Carlo ($R = 2000$); solid: LIN; dotted: ANT, which assumes calibration. Dashed: no gain. Source: `results/theory_check_dif.json`.

TUNE, which here recovers the plain estimator, is $1.32 = 1/0.76$. Second, a constant proxy $\pi_i \equiv \frac{1}{2}$ ($\kappa = 0$) is uninformative for the error difference (gain exactly 1, since $\tilde{z} \equiv 0$) but not for the rate differences (1.9 and 2.0, TUNE 1.03 and 1.04), because $\tilde{z}_i$ still carries the known pattern of D_i. Third, the threshold $\beta < \frac{1}{2}$ of Proposition 3(a) is visible: for the error difference $\bar{\beta}$ is 1.02 at $\kappa = 1$, 0.62 at $\kappa = 2$ (gain 1.08) and 0.44 at $\kappa = 4$ (gain 0.96), so an over-sharp proxy gains little for the rate differences (1.8–1.9) and crosses to a loss for the error difference.

Zero-variance strata and the floor.

The residual floor (S6) was evaluated on four pools at $L = 250$ ($R = 1000$). At Churn the difference estimator alone raises the worst-estimand coverage from 0.52 (plain estimator, `results/label_baselines.json`) to 0.84 (0.87 in this run), because $\hat{e}_i = -\pi_i \hat{d}_i$ varies even when no positive is sampled; at Law school it does not

(0.64 for both). Where the problem persists the floor repairs it: for Law school mean coverage rises from 0.913 to 0.954 and the worst-estimand coverage from 0.65 to 0.89; for Adult-sex and Credit card default the mean rises from 0.942 to 0.946 and from 0.949 to 0.955 with the worst estimand unchanged (0.85 and 0.92); at Churn the floor is active in 0.2% of (replication, estimand) pairs and changes nothing (0.938). The floor widens the intervals by a median factor of 1.00 (90th percentile 1.08, 1.15, 1.33 for Adult-sex, Credit card and Law school). It does not alter the point estimate. As in Section 9 it is a safeguard chosen after seeing the failure, and these numbers are not an out-of-sample test.

S5.5 Summary

The linearised design effect predicts the efficiency of the implemented design to within Monte Carlo error on all 11 real pools and in the synthetic families (per-target median ratio 1.00, 10%–90% range 0.83–1.22 on the real pools). The closed form of Corollary S8 is accurate and conservative when the strata separate a pair's agreements from its disagreements (the own-target efficiency is 0.90 to 1.47 times the formula for $\rho \geq 5\%$), and it captures the $1/\rho$ scaling and the effect of the target size. With nine models it overstates the gain on real pools by a median factor of about two, because the counts K locate any disagreement but not that of a specific pair; the bounds $1/\rho_{\text{any}} \leq \text{Eff} \leq 1/\rho$ of (S13) describe this gap. The gain vanishes only as the disagreement rate approaches one, as in EEG ($\rho_{\text{any}} = 0.73$, efficiency 1.6–1.9). For the difference estimator the linearised residual variance predicts the observed gain over the plain estimator within Monte Carlo error (median ratio 0.98 over 34 pool–budget cells; Section S5.4), the label-free forecast does so except in Law school, and the gain is moderate (median 1.47) rather than a doubling.

S6 Baseline designs and detailed results

This section gives the full definitions of the baselines compared in Section 8, the per-setting results and the detailed coverage discussion.

Common ground.

To make the comparison about the choice of labels and estimators and nothing else, every method estimates the same vector of totals T (Section 3), plugs it into the same map $\hat{\theta} = f(\hat{T})$ for all $3\binom{M}{2} = 108$ comparison estimands $\theta^{\text{TPR}}_{mj}, \theta^{\text{FPR}}_{mj}, \theta^{\text{ERR}}_{mj}$, and reports 95% Wald intervals from a linearisation variance. Every method that uses a proxy uses the same $\pi_i = M^{-1}\sum_m \hat{p}_{mi}$, and every allocation over several estimands uses the same weights $w_k = 1/S_k^2$ as (9). The budget is counted in distinct labelled instances. Pool labels are used only to answer label queries, to compute the truth and, for reporting, to pick the three most accurate models. Estimands whose numerator is identically zero in the pool (the two models never disagree on the relevant instances) are estimated exactly by every design and are left out of all summaries. This removes 51 of the 108 estimands in the law-school setting, including all those among its three most accurate models, and none elsewhere.

S6.1 Baseline definitions

B1: importance sampling for active comparison.

Sawade et al. [13] compare the risks of two fixed models by drawing n instances with replacement from an instrumental distribution q and estimating each risk by the self-normalised weighted mean $\sum_k \ell_k/q_k \big/ \sum_k 1/q_k$ (with $p(x) = 1/N$ on a pool). Their Theorem 1 shows that the asymptotic variance of the estimated risk difference is minimised by $q^*(x) \propto p(x)\{\mathbb{E}[(\ell_1 - \ell_2 - \Delta)^2 \mid x]\}^{1/2}$. They replace $p(y \mid x)$ by an equal mixture of the models' predictive distributions, and for k models they use the equal mixture of the pairwise-optimal distributions. We keep all of this and change only the quantity being compared. With replacement draws $i_1, \ldots, i_n \sim q$, our totals are the Hájek (self-normalised) estimators

$$\hat{T}^{\text{IS}} = N\,\frac{\sum_{k=1}^n t_{i_k}(Y_{i_k})/q_{i_k}}{\sum_{k=1}^n 1/q_{i_k}},$$

$$\hat{\theta} - \theta \approx \frac{1}{n}\sum_{k=1}^n \frac{a_{i_k}}{q_{i_k}},$$

$$a_i = \nabla f(T)^\top \{t_i(Y_i) - T/N\}.$$

For $\theta^{\mathrm{ERR}}_{mj}$, $a_i = \{|\hat{y}_{mi} - Y_i| - |\hat{y}_{ji} - Y_i| - \theta^{\mathrm{ERR}}_{mj}\}/N$, which is exactly Sawade et al.'s centred loss difference. For $\theta^{\mathrm{TPR}}_{mj}$, the self-normalisation is automatic because f is a ratio, and $a_i = \sum_g s_g \, \mathbf{1}\{G_i = g\} \, Y_i\{(\hat{y}_{mi} - \hat{y}_{ji}) - (\mathrm{TPR}_{mg} - \mathrm{TPR}_{jg})\}/P_g$. The asymptotic variance is $n^{-1} \sum_i \mathbb{E}[a_i^2]/q_i$ [14], so the single-estimand optimum is $q_i \propto \{\mathbb{E}_\pi a_i^2\}^{1/2}$, with the expectation taken under $Y_i \sim \mathrm{Bernoulli}(\pi_i)$ and a evaluated at the proxy totals. This is our adaptation to TPR and FPR gaps. For a pair (m, j) we combine its three estimands as $q_i^{(mj)} \propto \{\sum_{k \in (m,j)} w_k \mathbb{E}_\pi a_{ik}^2\}^{1/2}$. The design distribution is the equal mixture over the 36 pairs, as in Sawade et al., mixed with the uniform distribution with weight 0.05 so that every q_i is bounded away from zero. Draws continue until L distinct instances are labelled; repeated draws reuse the label. The median number of draws was $1.12L$ (range $1.02L$–$1.48L$), so B1 effectively gets more than L draws for L labels. The variance is estimated by n^{-1} times the sample variance of $\hat{a}_{i_k}/q_{i_k}$.

B2: proxy-score stratification.

The classic approach to estimating a classifier's metrics with few labels stratifies on the classifier's score and allocates optimally [15, 16]. We form strata as group $\times$ within-group quantile bins of π_i (10 bins, Score-10; and 5 bins, Score-5), merge strata smaller than 50 as for PATTERN, and use the same estimator (6), variance (8) and proxy-Neyman allocation (9). B2 therefore differs from PATTERN only in the strata. It uses the proxy strongly but ignores which models disagree.

B3: disagreement-focused two-phase design.

Labelling only where models disagree is the "active $\neq$" baseline of Sawade et al. [13] and the discordant-pair approach of Musgrove et al. [17] and of the screening-test literature [18]. On its own it identifies only ratios, because the denominators P_g, Q_g need labels in the agreement set. Let $D = \{i : 0 < K_i < M\}$ and $A = U \setminus D$. Phase 1 draws a simple random sample s_0 of size n_0 from U. Phase 2 draws a simple random sample of $L - n_0$ from $D \setminus s_0$. Conditionally on $n_A = |s_0 \cap A|$, the labelled units in A are a simple random sample from A. Conditionally on $|s_0 \cap D|$, the union of $s_0 \cap D$ and the phase-2 sample is a simple random sample of size $n_D = |s_0 \cap D| + L - n_0$ from D. The combined estimator is therefore the two-stratum estimator $\hat{T} = N_A \bar{t}_A + N_D \bar{t}_D$ with variance (8) at the realised (n_A, n_D), which is conditionally design-unbiased. We take $n_0 = L/4$ (Dis-25) and $n_0 = L/2$ (Dis-50).

B4: prediction-powered inference.

PPI [19] corrects the predictions on the whole pool with the labelled residuals. With the prediction of t_i taken as its proxy mean $\tilde{t}_i = \pi_i u_i + (1 - \pi_i) v_i$, the PPI estimator of a total under SRS is the difference estimator

$$\hat{T}^{\mathrm{PPI}}_\lambda = \lambda \sum_{i \in U} \tilde{t}_i + \frac{N}{n} \sum_{i \in s} \{t_i(Y_i) - \lambda \tilde{t}_i\},$$

with $\lambda = 1$ (PPI). PPI++ [3] tunes λ for each estimand. In our finite-pool setting the predictions are known for every unit, so the $(1 + n/N)$ factor of PPI++ drops out and the variance-minimising value is the regression coefficient of $\nabla f^\top t_i$ on $\nabla f^\top \tilde{t}_i$ in the sample. We clip it to $[0, 1]$. StratPPI [4] applies PPI++ within strata with a stratum-specific λ_{hk} and allocates the budget by a Neyman rule on the residual standard deviation, estimated from the predictor's own confidence scores. We use the B2 strata (group $\times$ 10 proxy bins), per-stratum and per-estimand λ_{hk} clipped to $[0, 1]$, and the allocation $n_h \propto N_h \{\sum_k w_k \overline{\pi(1-\pi)(A_k - B_k)^2}_h\}^{1/2}$. Here $\pi_i(1 - \pi_i)(A_{ik} - B_{ik})^2$ is the proxy variance of the residual $z_{ik} - \mathbb{E}_\pi z_{ik}$, so this allocation is Fisch et al.'s confidence-based rule for our estimands. In all three versions the variance is the stratified formula (8) applied to the residuals $z_{ik} - \lambda \nabla f^\top \tilde{t}_i$.

B5: PATTERN with a model-assisted estimator (the proposed method).

The PPI correction is a difference or GREG-type estimator [6, Chapter 6] and can be applied to the PATTERN sample without changing the design. PATTERN-DIF uses the stratified difference estimator with $\lambda = 1$. PATTERN-PPI tunes λ_{hk} per stratum as in StratPPI. B5 is the proposed method (PATTERN-DIF, $\lambda = 1$) and a tuned-weight variant (PATTERN-PPI), and is reported in the last block of the results.

Not run.

Active testing with LURE [20], adaptive variance-minimising selection [21], and active inference [22, 23] select labels sequentially and target one estimand of one model. Their fixed-budget, one-shot analogue for our vector of estimands is B1, which uses the same proxy-optimal inclusion principle. Model-based estimators that need a fitted joint model of labels and scores (for example SSME) are outside the design-based comparison made here.

S6.2 Detailed results

Where a baseline beats PATTERN.

Table S14 and Figure S5 show where the ranking changes. First, StratPPI beats PATTERN at $L = 500$ in seven of the ten settings of the table (ratios 1.01–1.57 in Bank, Credit-card, Law school, MAGIC, EEG, Phoneme and Electricity, and 1.13 in Spambase at $L = 250$), and is below it in Adult-sex (0.97), Adult-race (0.93) and Churn (0.24). Over all cells its ratio exceeds 1 in 19 of 35 (median 1.12 at $L = 500$). The margin is small in most settings, and it is not a failure of the stratification: StratPPI uses the same sample size but a model-assisted estimator, and B5 shows that the same correction applied to the PATTERN sample improves PATTERN itself by a median factor of 1.3–1.7. Second, Dis-25 beats PATTERN on Churn at $L = 500$ (ratio 1.52); there the pool has a 19% disagreement set, PATTERN strata are mostly small, and Dis-25 labels most of its sample inside the disagreement set. Its intervals are not reliable in that cell: mean coverage 0.93 but worst-estimand coverage 0.66, so the apparent gain should be read with caution. Third, PPI++ beats PATTERN on EEG (ratio 1.19 at $L = 500$), the setting with the largest disagreement set (73%), where agreement cells hold little and the proxy is informative. Importance sampling never beats PATTERN in the all-estimand median (ratio below 1 in all 35 cells), but on the pairs among the three most accurate models it is competitive: its median efficiency against SRS on those pairs is 4.2, 4.1, 4.2 and 6.3 at $L = 250, \ldots, 2000$, against 3.4, 3.6, 4.0 and 6.8 for PATTERN, and it beats PATTERN on those pairs in 10 of 35 cells (Adult-sex: 1.49 at $L = 250$ and 1.10 at $L = 500$, while its all-estimand ratio there is 0.89). Where the question is only about the best few models, importance sampling is a serious alternative.

Where PATTERN wins clearly.

In the two settings with the lowest disagreement rate, Law school and Churn, PATTERN is 6–84 times more efficient than SRS at $L \leq 1000$ and PATTERN labels essentially every disagreement instance at $L = 2000$ in Law school (all non-degenerate estimands then have error close to zero, ratio above 10^{20}, so these cells are shown as >100 and excluded from the figure range). IS, Dis-50, score strata and PPI++ reach only 0.04–0.7 of PATTERN's efficiency there (Dis-25 on Churn is the exception discussed above), because none of them isolates the agreement region exactly: PATTERN gives it a stratum of its own that can be labelled sparsely, while the other designs either draw it at random or leave the disagreement set under-sampled.

Coverage and failures.

Over all 35 cells the mean 95% coverage is 0.925 (SRS), 0.947 (PATTERN), 0.950 (IS), 0.941 (Score-10), 0.942 (Dis-25), 0.929 (PPI++), 0.924 (StratPPI), 0.946 (PATTERN-DIF) and 0.929 (PATTERN-PPI). The shortfalls concentrate in the smallest cells. The lowest worst-estimand coverages over all cells are 0.14 for StratPPI (Churn, $L = 250$), 0.16 for SRS and for PPI++, 0.17 for Score-10, 0.20 for PATTERN-PPI, 0.45 for Dis-25 and 0.52 for PATTERN (Churn, $L = 250$). The tuned estimators (PPI++, StratPPI, PATTERN-PPI) undercover when per-stratum tuning parameters are estimated from a few labels, notably in Law school where PATTERN-PPI covers 0.70 at $L = 250$ and StratPPI 0.70–0.87 below $L = 2000$. PATTERN-DIF, which fixes $\lambda = 1$, keeps the coverage of PATTERN (0.946). Undefined estimates (a group with no sampled positive) occurred in at most 1% of replications for any method (Dis-25, Adult-race, $L = 250$: 1%; SRS 0.5%; PATTERN and the others none). Runtimes per replicate (design computation excluded) are 8–11 ms on average for the stratified and importance-sampling designs and 34–51 ms for the PPI-type estimators (Table 9, last column), which tune per estimand and stratum. Design computation is at most 0.6 s per cell for every method.

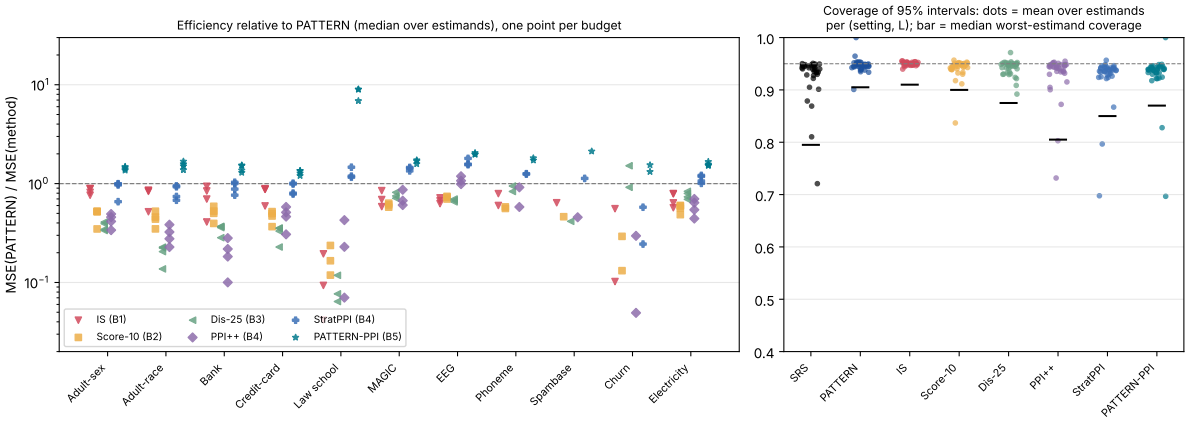


Fig. S5 Left: MSE(PATTERN)/MSE(method) by setting, one point per budget, median over comparison estimands (values above 1 mean that the method beats PATTERN; the axis is clipped to $[0.02, 30]$ and law school at $L = 2000$ is not shown in range because PATTERN labels the whole disagreement set there). Right: mean coverage of the nominal 95% interval over estimands for each (setting, L), with the median over cells of the worst-estimand coverage as a bar. Source: `results/label_baselines.json`.

Table S14 Efficiency relative to PATTERN, MSE(PATTERN)/MSE(method), median over estimands, by setting at $L = 500$ ($L = 250$ for Spambase, whose pool is too small for $L = 500$). Values above 1 mean that the method beats PATTERN. Second column: efficiency of PATTERN relative to SRS. Last two columns: mean coverage of the 95% intervals of StratPPI and of PATTERN-PPI. Source: `results/label_baselines.json`.

	PAT.	Baselines (B1–B4) vs PATTERN						B5 vs PATTERN		Coverage	
Setting	vs SRS	IS	Score-10	Dis-25	Dis-50	PPI++	StratPPI	PATTERN-DIF	PATTERN-PPI	StratPPI	PAT-PPI
Adult-sex	4.19	0.89	0.52	0.39	0.37	0.46	0.97	1.45	1.49	0.935	0.939
Adult-race	7.27	0.86	0.46	0.23	0.22	0.32	0.93	1.44	1.58	0.939	0.939
Bank	8.26	0.84	0.50	0.36	0.38	0.22	**1.03**	1.47	1.52	0.941	0.933
Credit-card	3.71	0.89	0.52	0.36	0.35	0.51	**1.01**	1.28	1.35	0.937	0.933
Law school	39.0	0.09	0.24	0.08	0.08	0.23	**1.47**	11.1	8.97	0.797	0.828
MAGIC	2.93	0.69	0.63	0.75	0.63	0.67	**1.43**	1.70	1.73	0.942	0.943
EEG	1.70	0.67	0.73	0.70	0.66	**1.19**	**1.57**	1.98	2.05	0.926	0.939
Phoneme	3.53	0.60	0.58	0.83	0.67	0.58	**1.26**	1.71	1.72	0.946	0.941
Spambase*	5.77	0.64	0.46	0.42	0.37	0.46	**1.13**	2.14	2.12	0.925	0.933
Churn	34.6	0.10	0.13	**1.52**	0.28	0.05	0.24	1.28	1.32	0.930	0.932
Electricity	2.86	0.79	0.59	0.80	0.73	0.64	**1.21**	1.49	1.53	0.942	0.933

The model-assisted extension.

PATTERN-DIF and PATTERN-PPI (bottom block of Table 9) improve on PATTERN in every cell except Law school at $L = 2000$, where PATTERN is already exact, by a median factor of 1.3–1.6 (by budget) and up to 11 in Law school at $L \leq 500$; they are the best methods in this study, with efficiency against SRS of 6.6, 6.2, 6.6 and 15.5 for PATTERN-DIF. The estimated-λ version adds little over $\lambda = 1$ (PATTERN-PPI is within 3% of PATTERN-DIF in the median) and costs coverage, so we recommend the fixed-λ version if the extension is used. Because the extension changes the estimator rather than the design, its variance expressions are those of the difference estimator under stratified sampling [6]. For $\lambda = 1$ (PATTERN-DIF) unbiasedness, the variance formula and the asymptotic coverage are covered by Proposition 1 with the proxy mean as ψ; for the tuned weights of PATTERN-PPI only the asymptotic statement of Proposition S6 holds, and the finite-sample undercoverage reported above is not covered by the theory.

Limitations (detail).

The comparison is restricted to fixed-budget, one-shot designs on cached pools of nine models with logistic-type probabilities; the proxy is the same for every method, so a better proxy for any of them (for example a calibrated or learned one)

could change the ranking, especially for StratPPI and PATTERN-PPI which rely on it directly. The importance-sampling baseline draws with replacement and so spends more than L draws for L labels; counting draws instead of labels would lower its efficiency. $R = 200$ replications give Monte Carlo errors of about 10% on a single mean squared error, so differences between 0.9 and 1.1 in Table S14 should not be read as ranking. Medians over estimands hide estimand-level differences: on the three types, PATTERN's median efficiency against SRS is 6.2 (TPR), 4.8 (FPR) and 3.8 (ERR), against 3.0, 3.5 and 3.0 for IS and 5.3, 5.2 and 2.3 for StratPPI.

S7 Ablation studies: details

This section gives the full protocol and discussion of the ablation studies summarised in Section 9. Tables for the stratifier (A1) and for small-sample inference (A4) are in the main text (Tables 10 and 11); the others are below.

This section varies one ingredient of PATTERN-DIF at a time and asks two questions: does the estimator remain valid, and how much efficiency is gained or lost relative to the default? The default is the method of the main study: strata (G_i, K_i) with strata smaller than 50 merged within group, weighted generalised Neyman allocation [24] computed with the ensemble-mean proxy π_i, a floor of 10 labels per stratum, the stratified difference estimator with $\lambda = 1$ and the same proxy, and 95% Wald intervals from the linearisation variance. The estimator that uses the proxy only to allocate labels, the plain (Horvitz–Thompson) estimator, is called HT or $\lambda = 0$. Because the proxy now enters the estimator as well as the allocation, the proxy ablations report the two roles separately, and every ablation reports the difference estimator and, where it is informative, HT on the same samples. All results come from `experiments/label_ablations_dif.py` (helpers in `tcs/labeling_variants_dif.py`) and are stored in `results/label_ablations_dif.json`. An earlier version of this study, in which the plain estimator was the default, is not reported; where its conclusions differ they are noted below.

Protocol.

We use the eleven cached pools and budgets $L \in \{250, 500, 1000\}$ with $L \leq N/4$ (29 setting–budget cells) and $R = 150$ replications per cell and design. The first seven designs (SRS, GROUP, PATTERN and four alternative stratifiers) draw the same samples as an earlier version of this ablation study that used the plain estimator throughout (`results/label_ablations.json` in the repository); we call it the earlier plug-in study and refer to its numbers where they are compared, so that differences between the two studies are due to the estimator alone. There are 108 comparison targets for the 36 pairs of nine models; targets whose numerator vanishes identically in the pool are exact for every design and are excluded (only Law school, where 57 remain). Efficiency is the per-target ratio $\mathrm{MSE}_{\mathrm{SRS}}/\mathrm{MSE}_{\mathrm{design}}$ with SRS and the plain estimator as reference, clipped to $[10^{-4}, 10^{4}]$, summarised by the median over targets and then over cells. "Rel." is the median over cells of $\mathrm{MSE}_{\text{PATTERN-DIF}}/\mathrm{MSE}_{\mathrm{variant}}$ (above one favours the variant) and "Win" the share of cells with Rel. above one. Coverage is the mean over targets and the worst target. With $R = 150$ the Monte Carlo standard error of one coverage is about 0.018 and the minimum over 108 targets that all cover exactly 95% has mean 0.90, so worst-target coverage near 0.90 is consistent with nominal behaviour and values well below 0.88 are not. A design with more strata than the budget can feed at 10 labels each uses the floor $\max(2, \lfloor 0.8L/H \rfloor)$. Undefined estimates occurred in at most 0.7% of replications.

S7.1 Stratifier (A1)

Table 10 varies the stratifier; per-cell values are in Table S18 of Section S8. On the same samples the difference estimator is more efficient than HT in all 29 cells (median ratio 1.67, range 1.36 to 14.5 in Law school), and its efficiency against SRS is 6.3 at every budget, compared with 3.6 to 4.3 for HT. The two gains are not additive: SRS with the difference estimator (SRS-DIF, i.e. PPI of Angelopoulos et al. 19 with the ensemble proxy) gains a factor 1.9, the PATTERN design with HT a factor 4.0 over all budgets, and the product of these medians (7.5) exceeds the combined gain.

The stratifier conclusions of the earlier study carry over. Group × agree/disagree keeps 74% of the default's precision (earlier 71%) and the joint

pattern of the three models agreeing most with the ensemble keeps 38% (earlier 37%): the count K_i is what matters, and three models cannot isolate the disagreement among 36 pairs. Splitting each (G, K) cell at the median of π_i is a tie (Rel. 1.01, better in 15 of 29 cells; by budget 0.99, 0.97, 1.04), as are tertile bins (0.99), whose extra strata eat the budget through the floor (3 to 9 labels where it binds). We keep K_i. The allocation tailored to the variance of the difference estimator (`'resid'` in `tcs.labeling_dif`) is not better than the plain allocation (Rel. 0.96, better in 13 cells). The references that drop an ingredient lose clearly: GROUP-DIF and SRS-DIF keep 28%, proportional allocation 57% (earlier 52%) and the gap-targeted allocation of A5 74% (earlier 74%). Coverage hardly depends on the stratifier (mean 0.938 to 0.946).

S7.2 Proxy and the role of λ (A2)

The proxy now has two jobs, and the ablation separates them (Table S15; Figure S8 in Section S8 shows the per-cell spread). The first block uses a proxy $\tilde{\pi}$ only to allocate labels and analyses the sample with HT. The second uses the same $\tilde{\pi}$ in the allocation and in the estimator, through the proxy mean $\psi_i = \tilde{\pi}_i u_i + (1 - \tilde{\pi}_i) v_i$. The third allocates with $\tilde{\pi}$ but keeps the ensemble proxy in the estimator.

Allocation only. As in the earlier study, the allocation is insensitive to the proxy. With HT, every proxy gives a median efficiency of 3.95 to 4.05 against 3.97 for the ensemble mean, including the reversed proxy $1 - \pi$, the shuffled proxy and the constant 0.5; allocation by the true labels gives 3.71. With the ensemble proxy kept in the estimator, the alternative allocations keep 99% to 101% of the default's precision (Rel. in Table S15, last block), with per-cell ratios between 0.86 and 1.4 except in Churn at $L = 500$, where the constant, naive Bayes and shuffled proxies label the small disagreement strata exhaustively and gain factors of 19, 13 and 6 (1.8 for the logistic proxy). The reason is the same as before: the influence values vary within a stratum mainly through the differences $\hat{y}_{mi} - \hat{y}_{ji}$, known without labels. What matters in the allocation is that it is Neyman-type rather than proportional (Rel. 0.57).

Proxy in the estimator. Here the proxy matters. Replacing the ensemble mean by a single model costs 13% (logistic regression) or 42% (naive Bayes, whose probabilities are extreme); the constant 0.5 costs 16%, a shuffled proxy 47%, a noisy proxy 39% and the reversed proxy 34% (Table S15, middle block). The difference estimator remains unbiased and its coverage is unaffected (mean 0.940 to 0.949), as the theory predicts, so a poor proxy costs efficiency only. A corrupted proxy also removes the gain over HT on the same sample: the median ratio of DIF to HT is 0.96 for the shuffled and noisy proxies and 1.00 for naive Bayes, against 1.67 for the ensemble, and DIF is less efficient than HT in 19, 18 and 15 of 29 cells for these three. The constant keeps most of the gain (ratio 1.40) because ψ_i still carries the known structure of u_i, v_i.

The weight λ. Table S26 (Section S8) compares $\lambda = 0$ (HT), $\lambda = 1$ (default) and a λ tuned per stratum and target on the labelled units and clipped to $[0, 1]$ [as in PPI++ and StratPPI; 3, 4]. With the ensemble proxy the tuned version is slightly more efficient than $\lambda = 1$ (median Rel. 1.016, better in 24 of 29 cells; 1.026 at $L = 250$) but it undercovers: mean 0.923, worst target 0.799 on average and 0.227 in Law school at $L = 250$, against 0.945, 0.877 and 0.633 for $\lambda = 1$. The tuned weights are estimated from a few labels per stratum and the variance estimate ignores that. With a degraded proxy the tuned version is the more efficient of the three (shuffled 5.2, noisy 5.1, reversed 5.4, against 3.7, 3.5 and 4.4 at $\lambda = 1$ and about 4.0 for HT), because it shrinks the correction, but again at coverage 0.92 to 0.93. Neither alternative is better than $\lambda = 1$ with the ensemble proxy in both efficiency and validity, so we keep $\lambda = 1$. Tuning is worth considering only when the proxy is known to be poor and some undercoverage is acceptable.

S7.3 Number of compared models (A3)

For $M \in \{2, 3, 5\}$ we drew three random subsets of the nine models per size (the subsets of the earlier study), rebuilt strata, proxy and estimands from the subset only, and compared PATTERN-DIF and PATTERN-HT with SRS on the $3\binom{M}{2}$ targets. Subsets in which all models agree on every instance are dropped. Table S16 and Figure S6 show, as before, that the gain is governed by the share of the pool on which the compared models

Table S15 Proxy ablation (A2). Eff.: median over all 29 cells of the median-over-targets efficiency vs SRS (plain estimator); Rel.: relative to PATTERN-DIF with the ensemble proxy (as in Table 10); worst: mean over cells of the worst-target coverage; mean: mean coverage. In the first block the proxy enters only the allocation and the sample is analysed with HT; in the second it enters allocation and estimator; in the third only the allocation, the estimator keeps the ensemble proxy. The true-label proxy is not defined as an estimator proxy (the estimator would be exact).

	Proxy in allocation only (HT)			Proxy in both (DIF)				DIF, ensemble in estimator	
Proxy	Eff.	Rel.	worst	Eff.	Rel.	mean	worst	Eff.	Rel.
Ensemble mean π (default)	3.97	0.65	0.858	6.27	1.00	0.945	0.877	6.27	1.00
Single model (logistic)	3.96	0.64	0.861	5.56	0.87	0.943	0.886	6.01	0.99
Single model (naive Bayes)	4.05	0.63	0.858	3.88	0.58	0.942	0.873	5.99	1.01
Constant 0.5	3.97	0.62	0.850	5.71	0.84	0.940	0.858	6.01	0.99
π shuffled within group	3.95	0.62	0.854	3.70	0.53	0.949	0.889	6.12	0.99
π with logit noise (sd 2)	4.05	0.64	0.854	3.46	0.61	0.949	0.877	5.91	0.99
Reversed, $1 - \pi$	3.99	0.63	0.859	4.38	0.66	0.941	0.872	6.23	0.99
True labels (allocation only)	3.71	0.64	0.865	–	–	–	–	6.01	1.00
Proportional allocation (no proxy in allocation)	2.05	0.38	0.819	3.75	0.57	0.943	0.877	–	–

disagree rather than by M: the Spearman correlation between log efficiency and the disagreement share across the 284 configurations is -0.72, -0.74 and -0.76 at $L = 250$, 500 and 1000 (HT: -0.81, -0.78, -0.79), and within settings at $L = 250$ it is negative in all eleven (median -0.43). At $L = 250$ the median efficiency of PATTERN-DIF is 17.2 when the compared models disagree on less than 10% of the pool, 12.5 for 10 to 20%, 4.8 for 20 to 40% and 3.8 above 40%. What the difference estimator changes is the size of the gain over HT, which grows with M from 1.15 at $M = 2$ to 1.67 at $M = 9$. The log ratio of DIF to HT is positively correlated with the disagreement share (0.09, 0.19, 0.34 by budget). The correction therefore helps most where the design alone helps least, but it does not remove the dependence on disagreement. Coverage is not affected by M (mean 0.945 to 0.950); the worst-target figures cannot be compared across rows, because the number of targets grows from 3 to 108.

S7.4 Small-sample inference (A4)

With the plain estimator the main study found worst-target coverage near 0.5 at $L = 250$ in Churn and Law school, traced to strata whose sampled units show no variation in the influence values. The difference estimator removes most of this: the residual $\xi_i = t_i - \psi_i = (Y_i - \pi_i)(u_i - v_i)$ varies even when no positive is sampled, no (replication, target) pair had a standard error of exactly zero (plain estimator: up to 1.0% in Law school), and the worst-target Wald coverage in Churn at $L = 250$ rises from 0.47 to 0.91 (Table 11, Figure S7). The shortfalls that remain are of a different kind. Over the 29 cells the worst-target Wald coverage is below 0.85 in five cells (Law school 0.63, MAGIC 0.78 and Adult-sex 0.82 at $L = 250$, Bank 0.83 and Phoneme 0.83 at $L = 500$), against five cells for HT and six for SRS-DIF. The worst target is a TPR contrast in 17 of 29 cells (FPR in 4, error difference in 8). On average over cells its mean standard error is 0.90 of the Monte Carlo standard deviation (0.99 for the median target) and its bias is below 0.16 standard deviations, so the intervals are too short rather than off-centre, which is consistent with a linearisation error in a ratio with few sampled positives in the smaller group (we did not test this further).

None of the standard corrections closes this gap. A t quantile with Satterthwaite [25] degrees of freedom adds 1% to the width and moves the mean worst target at $L = 250$ from 0.852 to 0.865. The stratified bootstrap percentile interval [26], which resamples labelled residuals within strata ($B = 999$, no finite-population correction; Rao and Wu 27 discuss a rescaled version), is 11% wider and a little conservative (mean coverage 0.961 over all cells), with worst target 0.878 and minimum 0.767. The proxy variance floor of `tcs.labeling_dif` replaces the sample variance of a stratum by its proxy residual variance when the former is exactly zero ('zero') or by the larger of the two in every stratum ('max'). The 'zero' floor leaves the width unchanged (1.002) and lifts the minimum over cells from 0.633 to 0.780, entirely through Law school (0.90 at $L = 250$); combined with the t quantile the mean worst target is 0.885 and four cells stay below 0.85. Only the 'max' floor reaches 0.92 in every cell (mean worst target 0.948), and it is conservative: mean coverage 0.985 at $L = 250$, width 1.67 times Wald on average, 1.25 in the median and 4.9 in Law school. For SRS-DIF Wald

Table S16 Number of compared models (A3). Configs: (setting, budget, subset) combinations; Disagr.: median share of the pool on which not all M models agree. Efficiency: median over configurations of the median-over-targets $\mathrm{MSE}_{\mathrm{SRS}}/\mathrm{MSE}$. DIF/HT: median ratio of the two efficiencies on the same sample. Coverage: PATTERN-DIF, Wald.

			PATTERN-DIF vs SRS			PATTERN-HT vs SRS		Coverage (DIF)	
M	Configs	Disagr.	$L = 250$	500	1000	all L	DIF/HT	mean	worst
2	87	0.133	9.48	12.58	9.28	8.37	1.15	0.950	0.933
3	84	0.193	8.29	10.70	8.57	6.44	1.27	0.949	0.919
5	84	0.294	7.62	6.64	7.08	4.78	1.52	0.946	0.904
9	29	0.396	6.27	6.18	6.37	3.97	1.67	0.945	0.877

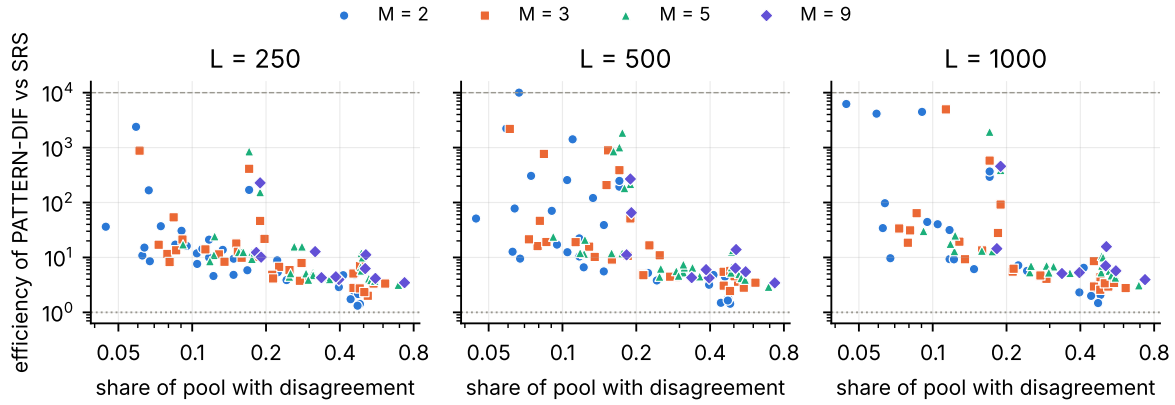


Fig. S6 Efficiency of PATTERN-DIF relative to SRS (median over targets) against the share of the pool on which the compared models do not all agree, for random subsets of M models in every setting (A3). Both axes are logarithmic; the dotted line marks parity and the dashed line the clipping value 10^4.

coverage is poor (minimum 0.27) and the bootstrap worse (0.10); 'max' repairs it (minimum 0.88) at 23% extra width. The floors were designed after seeing the failure of the plain estimator on these pools, so the 'zero' result for Law school is in-sample (the residual floor is discussed further in Section S5). An independent check on synthetic pools (`results/floor_check.json`) found that the floor leaves the interval width unchanged (median ratio 1.00).

S7.5 Ranking models by the equalised-odds gap (A5)

Practitioners also look at each model's equalised-odds gap $EO_m = \max(|\Delta_m^{TPR}|, |\Delta_m^{FPR}|)$ [28] and rank models by it. We computed the plug-in $\widehat{EO}_m = f_{EO}(\hat{T})$ from the totals of the same samples, with the difference estimator (DIF totals) and the plain estimator, and recorded the RMSE, the probability P that the model with the smallest estimated gap has the smallest true gap, Kendall's τ [29] between estimated and true gaps, and the bias. We also include a variant of PATTERN whose allocation targets the $2M$ per-model gaps instead of the comparison contrasts (gap-targeted). Table S17 summarises the results; per-cell values are in Table S24.

The plug-in $\widehat{EO}_m$ is biased upwards, as a maximum of absolute values must be (mean bias 0.042 for SRS, 0.026 for PATTERN with HT, 0.024 for PATTERN-DIF, 0.019 for the gap-targeted variant with DIF). PATTERN-DIF is more accurate than SRS in the median (efficiency 1.94 to 2.21) and identifies the best model more often (mean P 0.50 against 0.35 at $L = 250$; 0.46 against 0.21 without Law school, where six models tie, and EEG, where the best model leads by 0.039). PATTERN with HT reaches efficiency 1.80 to 1.94 and P 0.46 to 0.62, so the difference estimator adds little for this purpose. The gain is not uniform: PATTERN-DIF is less accurate than SRS in all Electricity cells (0.57 to 0.73) and in MAGIC at

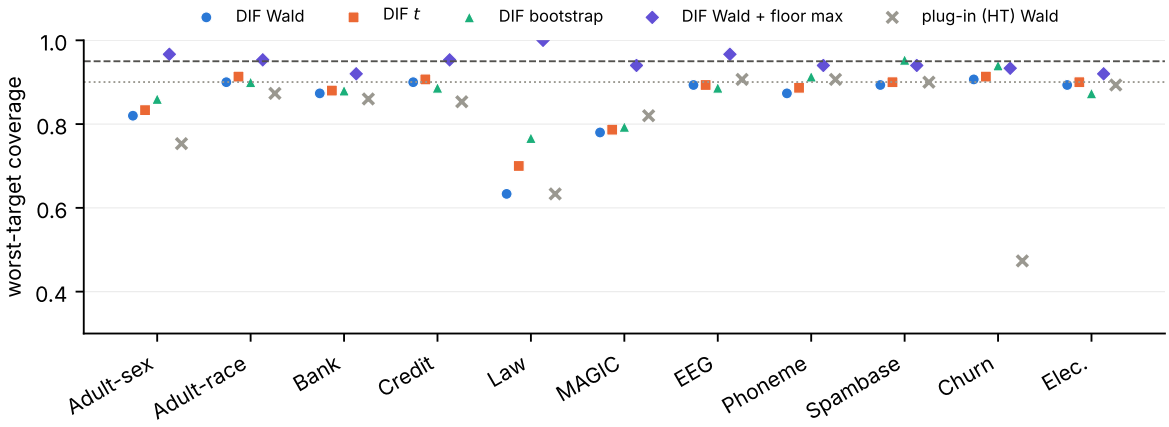


Fig. S7 Worst-target coverage at $L = 250$ in each setting for PATTERN-DIF under four interval methods and for the plain estimator (A4). Dashed line: nominal 0.95; dotted line: expected minimum over 108 targets of 150-replicate coverage if every target covered exactly 95% (0.90; Law school has 57 targets).

$L = 1000$, because the allocation spends labels on disagreement whereas the gap of one model depends on all of its positive predictions. SRS-DIF has about the same RMSE efficiency (2.08 to 2.28) but a lower P (0.42 at $L = 250$) and τ (0.41). The gap-targeted allocation with the difference estimator is the best design here (efficiency 2.30 to 2.61, $P = 0.49$, 0.56, 0.65 and $\tau = 0.48$, 0.54, 0.64 by budget; never worse than SRS except in Law school) but it keeps only 74% of the precision for the comparison contrasts (Table 10). In Law school the models are near-perfect and six have $EO_m = 0$ exactly; the plain estimator then returns exactly zero for them and wins every tie, whereas the difference estimator does not (MSE ratio clipped at 10^{-4}). When the true margin to the next model is 0.002 or less (Adult-sex, Bank, Credit card default, Churn) identification is hard for every design: P for PATTERN-DIF is at most 0.55 at the largest budget and 0.11 in Credit card default (margin 0.0003).

Summary.

With the difference estimator the ablation conclusions about the design are those of the earlier plug-in study: the count K_i is the stratifier worth having, Neyman rather than proportional allocation matters, the allocation is insensitive to the proxy, and the gain depends on how often the compared models disagree. Three things differ. First, the proxy now matters in the estimator, where a poor proxy costs up to half of the efficiency and removes the advantage over HT, although coverage is unaffected; the ensemble mean was never beaten there. Second, the difference estimator removes the zero-variance failure of the plain estimator, so the proxy variance floor has become a cheap safeguard ('zero', no loss of width) rather than a 19% widening, and the remaining under-coverage of rate contrasts at $L = 250$ is not fixed by t quantiles, the bootstrap or the floor; only the conservative 'max' floor reaches nominal worst-target coverage. Third, for ranking models by EO_m itself the gap-targeted allocation with the difference estimator is better than the default. Neither tuned λ nor proxy-bin strata beat the default consistently, and we keep it; we recommend reporting the Wald interval with the 'zero' floor, which costs nothing in width.

S8 Ablations: full tables

This appendix gives the per-cell results behind Section 9: 29 setting–budget cells, $R = 150$ replications and $B = 999$ bootstrap resamples. All values are stored in `results/label_ablations_dif.json`, which is produced by `experiments/label_ablations_dif.py`; its `report` mode (`experiments/label_ablations_dif_report.py`) writes the tables of this appendix and of Section 9 to `manuscript_v2/tables/` and draws the figures. Abbreviations: Credit = Credit

Table S17 Equalised-odds gap (A5). Eff.: median over cells of the median over models of $\mathrm{MSE}_{\mathrm{SRS}}/\mathrm{MSE}$ for $\widehat{EO}_m$ (cells in which both are exactly zero are skipped); P: mean probability that the estimated smallest-gap model has the smallest true gap; τ: mean Kendall τ between estimated and true EO_m. DIF: totals from the difference estimator; plug-in: plain estimator.

	$L = 250$			$L = 500$			$L = 1000$		
Design	Eff.	P	τ	Eff.	P	τ	Eff.	P	τ
SRS (plug-in)	1.00	0.35	0.32	1.00	0.42	0.39	1.00	0.49	0.48
SRS-DIF (PPI)	2.08	0.42	0.41	2.22	0.47	0.45	2.28	0.53	0.55
GROUP, plug-in	1.07	0.37	0.34	1.10	0.43	0.42	1.63	0.51	0.52
PATTERN, plug-in (HT)	1.83	0.46	0.45	1.80	0.51	0.49	1.94	0.62	0.62
PATTERN-DIF (default)	1.94	0.50	0.48	1.94	0.53	0.51	2.21	0.61	0.62
Gap-targeted, plug-in	2.03	0.44	0.43	2.29	0.52	0.51	2.54	0.59	0.61
Gap-targeted, DIF	2.30	0.49	0.48	2.61	0.56	0.54	2.51	0.65	0.64
ORACLE-ALLOC, DIF	2.02	0.50	0.47	1.77	0.54	0.52	1.93	0.57	0.60

card default, Law = Law school, Elec. = Electricity; DIF = stratified difference estimator with $\lambda = 1$; HT = plain (expansion) estimator.

Implementation details.

Stratifiers: the binary stratifier crosses group with the indicator $0 < K_i < M$. The three-model pattern uses the models with the highest agreement between $\hat{y}_{mi}$ and $\mathbf{1}\{\pi_i \geq 0.5\}$ (chosen without labels), with the eight patterns ordered by their number of ones so that merging of small strata goes to a neighbouring pattern. Proxy bins split each (G, K) cell at the quantiles of π_i within the cell. All stratifiers merge strata below 50 units within group, as in the default; the default design has $H = 6$ (Churn) to 20 (Electricity) strata. When H strata at 10 labels would use more than 80% of the budget, the floor is lowered to $\max(2, \lfloor 0.8L/H \rfloor)$; this affects only the proxy-bin stratifiers at $L \leq 500$ (floors 3 to 9, in 14 cells for tertile bins). Proxies: "single model" uses the predicted probabilities of the first logistic regression (LR) or of Gaussian naive Bayes (NB); the constant proxy is 0.5; "noise" adds $N(0, 2^2)$ to $\mathrm{logit}(\pi_i)$; "shuffle" permutes π within each group (fixed seed); "reversed" is $1 - \pi_i$; "true labels" uses Y_i in the allocation only. A proxy $\tilde{\pi}$ in the estimator means $\psi_i = \tilde{\pi}_i u_i + (1 - \tilde{\pi}_i) v_i$ in the difference estimator and, for the tuned weight, in λ_{hk} of the stratified PPI estimator of `tcs/labeling_baselines.py` (clipped to $[0, 1]$, estimated per stratum and target). Satterthwaite degrees of freedom are $\nu = (\sum_h a_h s^2_{e,h})^2 / \sum_h (a_h s^2_{e,h})^2/(n_h - 1)$ with $a_h = N_h^2(1 - n_h/N_h)/n_h$ and $s^2_{e,h}$ the sample variance of the residual influence values, computed per target; for "t + floor" the floored variances replace $s^2_{e,h}$ in both numerator and denominator. The bootstrap resamples the n_h labelled units within each stratum with replacement, recomputes $T = \sum_U f_i + \sum_h N_h \bar{e}^*_h$ with the pool-wide proxy total held fixed, and takes the 2.5% and 97.5% quantiles of the plug-in estimate. The two proxy variance floors are those of `tcs/labeling_dif.py`. A3 subsets were drawn once per size with seed 20261104 (the seed of the earlier study, so the subsets are identical). A5 computes $\widehat{EO}_m$ from the estimated totals (DIF totals for the difference estimator, expansion totals otherwise); the margin is the true EO difference between the best model and the next model with a larger gap.

S9 Data sets and models

All data sets were obtained from OpenML through scikit-learn's `fetch_openml` interface. Adult (version 2, 48,842 records; target income above 50K; groups: sex, female versus male, and race, non-White versus White); Bank Marketing (version 1, 45,211 records; target subscription; group: age below 30); default of credit card clients (version 1, 30,000 records; target default; group: sex); law school admission (binarised version, 20,800 records; group: race, non-White versus White); MAGIC gamma telescope (19,020 records); EEG eye state (14,980 records); Phoneme (5,404 records); Spambase (4,601 records); Churn (5,000 records); Electricity (45,312 records, subsampled to 30,000). For the last six the group is the indicator that the first numeric feature exceeds its median, and the feature is removed from the inputs. In all data sets the minority class is the positive class; for Adult, Bank Marketing and credit-card default this coincides with the

Table S18 Efficiency relative to SRS (median over live targets of $\mathrm{MSE}_{\mathrm{SRS}}/\mathrm{MSE}$, SRS with the plain estimator) for every cell and every stratifier (A1), difference estimator unless stated. Pat.: PATTERN-DIF (default); Pat. HT: the same sample with the plain estimator; Grp: GROUP; SRS: SRS-DIF; Bin.: group × agree/disagree; J3: three-model pattern; KP2, KP3: $K\times$ 2 or 3 proxy bins; Res.: residual allocation; Prop.: proportional allocation over the default strata.

Setting	L	Pat.	Pat. HT	Grp	SRS	Bin.	J3	KP2	KP3	Res.	Prop.
Adult-sex	250	6.27	3.63	2.10	1.83	3.24	2.60	6.75	6.79	5.64	5.96
Adult-sex	500	6.38	3.90	2.02	2.01	3.47	3.46	6.71	6.34	6.33	4.64
Adult-sex	1000	7.04	4.48	2.05	1.90	3.58	3.56	7.80	7.45	7.27	3.75
Adult-race	250	11.18	6.02	3.43	1.82	5.51	4.39	12.31	11.09	11.80	10.66
Adult-race	500	13.90	6.30	3.61	1.70	5.89	4.41	12.18	10.35	12.98	6.83
Adult-race	1000	15.73	8.30	3.66	2.03	6.55	6.35	16.29	14.01	12.31	4.77
Bank	250	12.33	7.58	3.72	1.52	11.66	4.03	10.52	11.28	10.13	8.67
Bank	500	11.14	8.00	3.90	1.60	10.75	4.58	11.37	11.39	11.59	5.98
Bank	1000	14.44	9.85	4.10	1.65	15.00	4.77	15.26	15.34	13.26	3.90
Credit	250	4.15	2.83	2.00	1.76	2.75	2.92	4.42	4.74	4.31	4.33
Credit	500	5.48	3.97	1.67	1.74	2.45	3.13	5.00	4.44	5.18	4.27
Credit	1000	5.69	4.20	2.02	1.77	2.51	3.31	6.56	5.78	6.09	3.26
Law	250	227.66	15.71	22.01	10.85	147.87	21.67	327.79	325.04	246.06	106.46
Law	500	267.97	36.85	20.83	11.87	134.54	17.68	512.00	379.54	323.57	85.76
Law	1000	456.17	78.85	22.25	11.21	299.51	20.73	>1000	>1000	410.76	76.27
MAGIC	250	3.96	2.56	2.01	1.85	3.83	2.07	4.43	4.42	4.34	3.15
MAGIC	500	4.24	2.53	2.10	1.96	4.02	2.81	4.44	4.21	4.66	3.06
MAGIC	1000	5.30	3.15	1.95	2.01	5.24	2.61	5.76	5.54	5.99	2.72
EEG	250	3.47	1.72	2.17	1.93	2.50	2.48	3.85	4.06	3.39	3.10
EEG	500	3.44	1.75	1.95	1.93	2.50	2.55	3.65	3.49	3.13	2.61
EEG	1000	3.94	1.95	2.10	1.95	2.85	2.83	4.45	4.64	4.23	3.05
Phoneme	250	4.44	2.48	2.09	2.03	4.20	2.45	3.59	3.67	4.16	3.28
Phoneme	500	5.99	3.59	2.06	2.03	5.12	2.83	6.13	5.46	5.71	2.92
Spambase	250	12.75	5.55	2.88	2.12	11.77	3.46	9.37	7.00	13.20	3.60
Churn	250	10.13	6.51	1.55	1.50	9.38	2.82	8.22	7.59	9.42	3.35
Churn	500	65.20	31.59	1.67	1.55	48.70	2.78	37.77	26.93	32.61	2.93
Elec.	250	4.28	2.73	1.81	1.72	4.33	1.47	4.26	4.31	4.17	3.85
Elec.	500	4.29	2.82	1.77	1.90	4.17	2.13	4.40	4.03	4.63	3.23
Elec.	1000	5.09	3.37	1.80	1.87	4.39	2.14	4.83	4.96	4.82	2.42

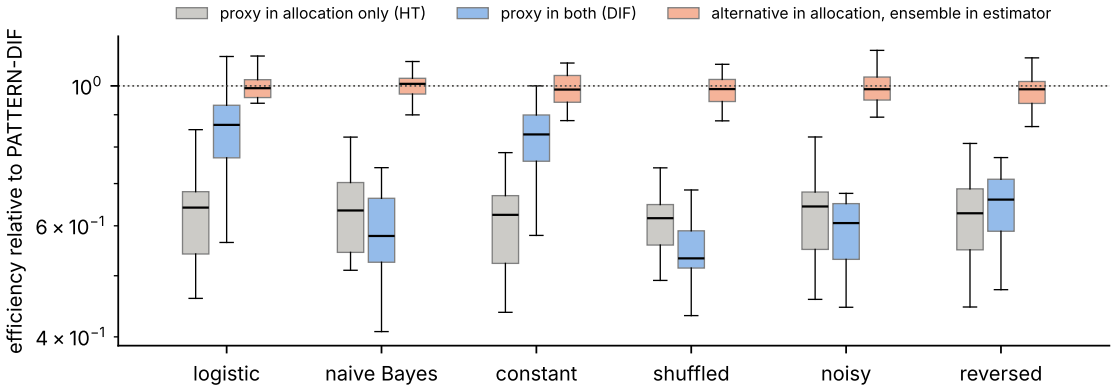


Fig. S8 Efficiency of alternative proxies relative to PATTERN-DIF with the ensemble proxy (Rel.), one value per cell (29 cells; boxes: quartiles, whiskers: 1.5 interquartile ranges). Grey: proxy in the allocation only, plain estimator; blue: proxy in allocation and estimator; orange: proxy in the allocation only, difference estimator with the ensemble proxy.

conventional positive class (income above 50K, subscription, default).

Each data set was split once, stratified by the label, into a training part (60%) and a pool (40%), with seed 0. Numeric features were standardised and categorical features one-hot encoded inside each model's pipeline. The nine models are: logistic regression (default, and with inverse regularisation strength 0.01), decision trees of depth

Table S19 Efficiency relative to SRS for every cell and allocation proxy (A2), with the same proxy in allocation and difference estimator (LR, NB, C = constant, Sh. = shuffled, Noi. = noise, Rev. = $1 - \pi$). Orc.: allocation by the true labels, estimator with the ensemble proxy. Pat.: default.

Setting	L	Pat.	LR	NB	C	Sh.	Noi.	Rev.	Orc.
Adult-sex	250	6.27	5.81	3.64	4.46	2.57	2.91	4.19	5.94
Adult-sex	500	6.38	5.38	4.19	5.76	3.69	3.47	4.38	7.51
Adult-sex	1000	7.04	7.08	3.63	7.01	3.88	3.97	4.72	6.91
Adult-race	250	11.18	11.28	6.21	9.37	5.72	6.52	7.22	12.30
Adult-race	500	13.90	11.60	7.34	10.05	5.96	7.94	7.63	11.25
Adult-race	1000	15.73	14.73	7.72	10.60	8.22	9.50	7.37	17.39
Bank	250	12.33	9.51	6.72	9.35	3.74	3.26	6.88	10.39
Bank	500	11.14	12.33	8.01	10.53	5.13	4.57	8.01	12.32
Bank	1000	14.44	14.40	9.60	12.26	6.74	7.42	9.81	15.13
Credit	250	4.15	4.51	2.90	3.85	3.70	3.10	3.01	4.86
Credit	500	5.48	5.23	3.32	4.81	3.78	3.36	3.62	5.34
Credit	1000	5.69	5.56	3.88	5.71	4.26	4.00	4.88	6.01
Law	250	227.66	672.95	28.53	31.71	3.17	18.29	9.02	228.93
Law	500	267.97	794.02	36.52	47.31	4.85	20.28	11.45	262.24
Law	1000	456.17	>1000	82.86	161.42	5.38	38.04	48.88	331.39
MAGIC	250	3.96	2.97	2.31	3.49	2.40	2.30	2.52	4.59
MAGIC	500	4.24	3.42	2.64	3.73	2.66	2.90	2.97	4.43
MAGIC	1000	5.30	4.43	3.28	4.72	2.93	3.25	3.23	5.51
EEG	250	3.47	2.42	1.55	2.82	1.94	1.55	1.74	3.32
EEG	500	3.44	2.29	1.49	2.64	1.77	1.73	1.69	3.46
EEG	1000	3.94	2.75	1.59	2.90	2.21	2.16	1.97	4.43
Phoneme	250	4.44	3.33	2.56	3.49	1.82	2.17	2.84	4.19
Phoneme	500	5.99	4.36	4.27	4.38	2.61	3.46	3.42	5.47
Spambase	250	12.75	10.07	8.62	9.20	4.60	6.92	5.19	14.59
Churn	250	10.13	6.07	5.94	9.07	4.71	6.38	6.39	10.21
Churn	500	65.20	70.28	>1000	>1000	255.08	68.66	37.93	60.60
Elec.	250	4.28	3.70	2.90	3.79	2.67	2.72	3.10	4.35
Elec.	500	4.29	3.59	3.18	4.15	2.82	2.96	3.17	4.19
Elec.	1000	5.09	4.70	3.46	4.57	3.25	3.46	3.83	5.03

3 and 12, a random forest with 200 trees and minimum leaf size 3, histogram gradient boosting with default settings, 25-nearest neighbours, Gaussian naive Bayes, and a multilayer perceptron with hidden layers of 64 and 32 units and early stopping. Pool predictions, predicted probabilities, group indicators and labels are cached by `experiments/preference_validation.py` and used by all experiments.

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Table S20 As Table S19 for the plain estimator (HT): the proxy enters the allocation only. Pat.: ensemble proxy.

Setting	L	Pat.	LR	NB	C	Sh.	Noi.	Rev.	Orc.
Adult-sex	250	3.63	3.12	3.55	3.36	3.53	3.69	3.57	3.20
Adult-sex	500	3.90	3.79	4.05	3.97	3.86	3.89	3.99	3.69
Adult-sex	1000	4.48	4.33	4.28	4.30	4.14	4.40	4.28	4.50
Adult-race	250	6.02	6.01	5.69	5.58	5.30	5.67	6.32	5.87
Adult-race	500	6.30	6.06	6.54	5.94	5.96	6.78	6.08	6.33
Adult-race	1000	8.30	7.02	7.14	7.37	7.04	6.97	7.70	7.58
Bank	250	7.58	6.88	7.26	6.82	7.29	7.37	6.81	7.13
Bank	500	8.00	8.68	8.41	7.70	8.52	8.21	7.71	8.03
Bank	1000	9.85	10.31	10.57	8.73	8.50	9.65	10.65	10.31
Credit	250	2.83	3.08	3.13	2.95	3.20	3.49	3.08	3.06
Credit	500	3.97	3.96	4.00	3.70	3.95	4.05	3.27	3.71
Credit	1000	4.20	4.30	4.42	4.24	4.25	4.05	4.12	3.78
Law	250	15.71	19.71	19.82	18.07	21.84	15.26	17.50	25.35
Law	500	36.85	36.00	29.92	37.65	36.07	39.21	34.64	43.36
Law	1000	78.85	113.76	84.30	79.61	75.75	66.62	78.81	150.09
MAGIC	250	2.56	2.38	2.23	2.53	2.46	2.48	2.43	2.61
MAGIC	500	2.53	2.75	2.79	2.70	2.78	2.69	2.95	2.85
MAGIC	1000	3.15	3.25	3.37	3.40	3.36	3.17	3.20	3.37
EEG	250	1.72	1.71	1.66	1.67	1.66	1.73	1.56	1.79
EEG	500	1.75	1.82	1.66	1.67	1.70	1.67	1.72	1.84
EEG	1000	1.95	1.92	1.95	2.02	2.00	2.03	1.96	2.15
Phoneme	250	2.48	2.49	2.37	2.29	2.47	2.43	2.53	2.33
Phoneme	500	3.59	3.55	3.39	3.45	3.42	3.55	3.41	3.20
Spambase	250	5.55	5.83	6.37	5.14	5.25	5.69	5.75	5.87
Churn	250	6.51	6.32	11.01	7.45	7.12	6.65	6.45	6.53
Churn	500	31.59	59.30	>1000	>1000	312.39	53.69	31.05	53.34
Elec.	250	2.73	2.65	2.81	2.81	2.75	2.83	2.67	2.75
Elec.	500	2.82	2.77	3.01	2.83	2.65	2.97	2.89	2.85
Elec.	1000	3.37	3.30	3.24	3.18	3.12	3.34	3.12	3.00

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Table S21 Tuned weight λ (A2), efficiency relative to SRS for every cell. $\lambda = 1$: default; tuned: λ chosen per stratum and target on the labelled units; C, Sh., Rev.: constant, shuffled and reversed proxy with tuned λ; SRS λ=1 and SRS tuned: one stratum (PPI and PPI++ with the ensemble proxy).

Setting	L	$\lambda = 1$	tuned	C tuned	Sh. tuned	Rev. tuned	SRS λ=1	SRS tuned
Adult-sex	250	6.27	7.46	5.83	5.20	5.41	1.83	2.05
Adult-sex	500	6.38	6.47	6.84	5.86	5.64	2.01	2.18
Adult-sex	1000	7.04	7.26	7.08	5.88	5.84	1.90	2.00
Adult-race	250	11.18	12.86	10.44	9.90	8.07	1.82	2.51
Adult-race	500	13.90	14.37	11.72	9.73	9.06	1.70	2.03
Adult-race	1000	15.73	16.60	11.27	10.12	8.54	2.03	2.16
Bank	250	12.33	13.14	10.41	8.27	10.01	1.52	2.15
Bank	500	11.14	11.51	10.93	10.04	10.79	1.60	1.95
Bank	1000	14.44	14.26	13.47	9.27	14.16	1.65	1.82
Credit	250	4.15	4.79	4.49	4.63	4.42	1.76	1.86
Credit	500	5.48	5.47	4.79	4.83	4.48	1.74	1.80
Credit	1000	5.69	5.85	5.84	5.09	5.40	1.77	1.78
Law	250	227.66	268.17	65.97	29.97	26.70	10.85	10.77
Law	500	267.97	240.23	106.40	43.99	35.00	11.87	11.90
Law	1000	456.17	429.49	313.84	98.94	102.98	11.21	11.21
MAGIC	250	3.96	4.00	3.81	3.26	3.36	1.85	1.98
MAGIC	500	4.24	4.33	3.96	3.39	4.03	1.96	2.00
MAGIC	1000	5.30	5.45	5.35	4.34	4.36	2.01	2.01
EEG	250	3.47	3.73	3.54	2.73	2.54	1.93	2.07
EEG	500	3.44	3.56	3.18	2.56	2.47	1.93	2.02
EEG	1000	3.94	4.14	3.31	3.28	2.99	1.95	2.04
Phoneme	250	4.44	4.23	3.96	2.99	3.48	2.03	2.01
Phoneme	500	5.99	6.06	4.61	4.25	4.30	2.03	2.03
Spambase	250	12.75	12.91	9.93	7.41	8.19	2.12	2.35
Churn	250	10.13	11.19	10.91	9.83	8.74	1.50	1.73
Churn	500	65.20	68.21	>1000	431.21	45.99	1.55	1.67
Elec.	250	4.28	4.19	4.15	3.48	3.56	1.72	1.83
Elec.	500	4.29	4.39	4.39	3.37	3.75	1.90	1.95
Elec.	1000	5.09	5.20	4.57	4.05	4.55	1.87	1.92

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Table S22 Coverage of PATTERN-DIF per cell (A4). H: number of strata after merging. Mean: Wald coverage averaged over targets; the remaining coverage columns give the worst target under Wald, with the 'zero' floor, with the 'max' floor, with t (Satterthwaite) and with the stratified bootstrap. Last two columns: worst-target coverage of PATTERN with the plain estimator and of SRS-DIF, both Wald. The Monte Carlo reference for the worst of 108 targets at exact 95% coverage is 0.90 (57 targets in Law school).

				Worst target, PATTERN-DIF					Worst, other	
Setting	L	H	Mean	Wald	Zero	Max	t	Boot.	HT	SRS-DIF
Adult-sex	250	19	0.946	0.820	0.820	0.967	0.833	0.860	0.753	0.913
Adult-sex	500	19	0.946	0.880	0.880	0.920	0.880	0.900	0.880	0.907
Adult-sex	1000	19	0.947	0.873	0.873	0.920	0.873	0.887	0.873	0.880
Adult-race	250	17	0.951	0.900	0.900	0.953	0.913	0.900	0.873	0.913
Adult-race	500	17	0.954	0.893	0.907	0.947	0.893	0.887	0.907	0.900
Adult-race	1000	17	0.949	0.913	0.913	0.927	0.920	0.933	0.913	0.907
Bank	250	16	0.942	0.873	0.873	0.920	0.880	0.880	0.860	0.900
Bank	500	16	0.939	0.833	0.833	0.927	0.833	0.860	0.860	0.900
Bank	1000	16	0.948	0.907	0.907	0.927	0.907	0.933	0.900	0.900
Credit	250	19	0.947	0.900	0.900	0.953	0.907	0.887	0.853	0.753
Credit	500	19	0.957	0.913	0.920	0.940	0.913	0.907	0.913	0.907
Credit	1000	19	0.946	0.907	0.907	0.933	0.907	0.933	0.913	0.907
Law	250	10	0.898	0.633	0.900	1.000	0.700	0.767	0.633	0.267
Law	500	10	0.951	0.933	0.933	1.000	0.933	0.953	0.920	0.400
Law	1000	10	0.945	0.907	0.907	0.987	0.913	0.953	0.900	0.640
MAGIC	250	19	0.935	0.780	0.780	0.940	0.787	0.793	0.820	0.833
MAGIC	500	19	0.933	0.887	0.887	0.940	0.887	0.900	0.880	0.913
MAGIC	1000	19	0.947	0.900	0.900	0.947	0.900	0.920	0.887	0.900
EEG	250	18	0.947	0.893	0.893	0.967	0.893	0.887	0.907	0.887
EEG	500	18	0.950	0.907	0.907	0.933	0.907	0.920	0.907	0.920
EEG	1000	18	0.949	0.887	0.887	0.927	0.887	0.927	0.900	0.893
Phoneme	250	14	0.947	0.873	0.873	0.940	0.887	0.913	0.907	0.913
Phoneme	500	14	0.945	0.833	0.833	0.940	0.833	0.927	0.900	0.893
Spambase	250	9	0.945	0.893	0.893	0.940	0.900	0.953	0.900	0.840
Churn	250	6	0.946	0.907	0.907	0.933	0.913	0.940	0.473	0.880
Churn	500	6	0.940	0.880	0.880	0.927	0.880	0.953	0.800	0.900
Elec.	250	20	0.948	0.893	0.893	0.920	0.900	0.873	0.893	0.893
Elec.	500	20	0.944	0.893	0.893	0.927	0.900	0.907	0.860	0.907
Elec.	1000	20	0.950	0.907	0.907	0.933	0.907	0.920	0.900	0.900

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Table S23 Average interval width relative to Wald for PATTERN-DIF per cell (A4), over replications and targets: 'zero' and 'max' floors, t with Satterthwaite degrees of freedom, and stratified bootstrap percentile.

Setting	L	Zero	Max	t	Boot.
Adult-sex	250	1.005	1.45	1.03	1.01
Adult-sex	500	1.003	1.33	1.01	1.03
Adult-sex	1000	1.001	1.27	1.00	1.07
Adult-race	250	1.005	1.32	1.02	1.02
Adult-race	500	1.002	1.25	1.01	1.06
Adult-race	1000	1.001	1.22	1.00	1.14
Bank	250	1.001	1.33	1.02	1.10
Bank	500	1.000	1.25	1.01	1.10
Bank	1000	1.000	1.18	1.00	1.20
Credit	250	1.008	1.24	1.02	1.02
Credit	500	1.003	1.13	1.01	1.04
Credit	1000	1.001	1.10	1.00	1.08
Law	250	1.018	4.90	1.02	1.05
Law	500	1.002	4.85	1.01	1.15
Law	1000	1.000	4.89	1.01	1.40
MAGIC	250	1.003	1.25	1.02	1.00
MAGIC	500	1.001	1.19	1.01	1.05
MAGIC	1000	1.000	1.17	1.00	1.14
EEG	250	1.004	1.53	1.01	0.99
EEG	500	1.001	1.53	1.00	1.04
EEG	1000	1.000	1.53	1.00	1.12
Phoneme	250	1.001	1.19	1.01	1.11
Phoneme	500	1.000	1.19	1.01	1.32
Spambase	250	1.002	1.37	1.01	1.31
Churn	250	1.000	1.78	1.01	1.14
Churn	500	1.000	1.64	1.01	1.42
Elec.	250	1.003	1.15	1.01	0.99
Elec.	500	1.001	1.12	1.01	1.03
Elec.	1000	1.000	1.10	1.00	1.10

Table S24 Equalised-odds gaps per cell (A5). Margin: true EO gap between the best and the next distinct model (in Law school six models tie at $EO_m = 0$, so the plain estimator, which returns exactly zero for them, wins every tie). P(best): probability that the estimated smallest-gap model has the smallest true gap for SRS, PATTERN with the plain estimator (HT), PATTERN-DIF (DIF) and the gap-targeted allocation with DIF (Gap-DIF). Kendall τ for SRS and PATTERN-DIF. Eff.: median over models of $\mathrm{MSE}_{\mathrm{SRS}}/\mathrm{MSE}$ of $\widehat{EO}_m$ for PATTERN-DIF.

				P(best)				Kendall τ		Eff.
Setting	L	Margin	Ties	SRS	HT	DIF	Gap-DIF	SRS	DIF	DIF
Adult-sex	250	0.002	1	0.19	0.29	0.43	0.42	0.06	0.34	2.29
Adult-sex	500	0.002	1	0.27	0.33	0.45	0.59	0.20	0.48	2.56
Adult-sex	1000	0.002	1	0.32	0.48	0.55	0.57	0.30	0.61	2.27
Adult-race	250	0.025	1	0.13	0.27	0.39	0.35	0.03	0.25	7.30
Adult-race	500	0.025	1	0.18	0.37	0.42	0.45	0.09	0.36	6.00
Adult-race	1000	0.025	1	0.22	0.57	0.63	0.72	0.14	0.52	3.41
Bank	250	0.002	1	0.11	0.20	0.29	0.23	0.01	0.18	3.62
Bank	500	0.002	1	0.17	0.27	0.32	0.30	0.10	0.16	2.95
Bank	1000	0.002	1	0.12	0.40	0.48	0.43	0.11	0.32	3.44
Credit	250	0.000	1	0.15	0.11	0.15	0.10	0.05	0.26	1.04
Credit	500	0.000	1	0.16	0.13	0.13	0.13	0.07	0.24	1.37
Credit	1000	0.000	1	0.10	0.13	0.11	0.17	0.09	0.21	2.15
Law	250	0.002	6	1.00	1.00	0.45	0.58	0.87	0.68	0.00
Law	500	0.002	6	1.00	1.00	0.49	0.49	0.90	0.71	0.00
Law	1000	0.002	6	1.00	1.00	0.59	0.63	0.94	0.76	0.00
MAGIC	250	0.008	1	0.17	0.39	0.55	0.55	0.40	0.53	1.03
MAGIC	500	0.008	1	0.29	0.56	0.73	0.79	0.49	0.63	1.19
MAGIC	1000	0.008	1	0.57	0.76	0.84	0.91	0.60	0.75	0.95
EEG	250	0.039	1	1.00	0.99	1.00	1.00	0.64	0.73	2.69
EEG	500	0.039	1	0.98	1.00	1.00	1.00	0.74	0.83	2.52
EEG	1000	0.039	1	1.00	1.00	1.00	1.00	0.80	0.90	2.63
Phoneme	250	0.027	1	0.28	0.46	0.59	0.59	0.45	0.57	1.94
Phoneme	500	0.027	1	0.47	0.51	0.65	0.68	0.53	0.65	1.32
Spambase	250	0.027	1	0.31	0.76	0.88	0.79	0.35	0.84	1.94
Churn	250	0.002	1	0.17	0.29	0.46	0.34	0.04	0.14	4.94
Churn	500	0.002	1	0.17	0.45	0.53	0.59	0.06	0.23	2.85
Elec.	250	0.007	1	0.35	0.35	0.37	0.48	0.62	0.74	0.73
Elec.	500	0.007	1	0.46	0.49	0.58	0.60	0.77	0.82	0.66
Elec.	1000	0.007	1	0.57	0.59	0.65	0.73	0.83	0.92	0.57

Table S25 Model subsets (A3) at $L = 250$: for each of the nine random subsets (three of each size M, in the order drawn) the share of the pool on which the subset's models do not all agree and the median-over-targets efficiency of PATTERN-DIF relative to SRS, written as share/efficiency. "exact": the subset's models agree on every instance, so every target is zero and the subset is dropped. The last column is the full set of nine models.

Setting	$M = 2$	$M = 3$	$M = 5$	$M = 9$
Adult-sex	0.06/10.8, 0.40/2.9, 0.46/2.4	0.45/2.8, 0.48/2.7, 0.08/11.6	0.49/6.7, 0.49/5.8, 0.12/11.1	0.51/6.3
Adult-race	0.06/15.1, 0.41/4.8, 0.46/4.9	0.45/5.1, 0.48/7.0, 0.08/8.2	0.49/12.0, 0.50/10.0, 0.12/24.3	0.51/11.2
Bank	0.07/8.5, 0.12/21.1, 0.12/10.0	0.13/11.4, 0.14/8.3, 0.07/17.0	0.16/12.6, 0.18/9.4, 0.09/17.3	0.18/12.3
Credit	0.04/36.2, 0.47/2.2, 0.48/1.4	0.52/2.0, 0.50/2.3, 0.09/13.5	0.53/4.1, 0.54/3.9, 0.12/8.5	0.56/4.2
Law	0.06/2386.8, 0.17/166.5, 0.17/169.5	0.17/410.7, 0.19/46.2, exact	0.17/854.4, 0.19/154.5, exact	0.19/227.7
MAGIC	0.10/7.6, 0.22/5.4, 0.24/3.9	0.29/4.1, 0.27/3.8, 0.11/14.0	0.36/4.0, 0.31/5.3, 0.29/5.1	0.40/4.0
EEG	0.09/30.6, 0.44/1.7, 0.47/1.3	0.55/3.3, 0.61/3.3, 0.18/10.3	0.69/3.2, 0.56/4.0, 0.56/3.8	0.73/3.5
Phoneme	0.07/37.0, 0.13/13.7, 0.15/9.5	0.25/5.9, 0.23/6.7, 0.15/17.9	0.31/4.7, 0.31/4.8, 0.34/4.4	0.39/4.4
Spambase	0.09/17.1, 0.17/5.8, 0.22/8.9	0.20/21.8, 0.28/7.9, 0.09/21.2	0.26/15.8, 0.28/15.8, 0.15/12.5	0.32/12.7
Churn	0.10/11.9, 0.11/14.3, 0.07/166.7	0.15/12.7, 0.08/53.8, 0.06/878.6	0.18/12.5, 0.18/12.3, 0.16/12.6	0.19/10.1
Elec.	0.10/16.2, 0.15/4.8, 0.12/4.6	0.21/4.8, 0.21/4.1, 0.16/9.9	0.25/4.5, 0.30/3.9, 0.25/5.2	0.34/4.3

Table S26 The weight λ (A2). Eff.: median over cells of efficiency vs SRS with the plain estimator; mean and worst: mean coverage and worst-target coverage, averaged over cells. The proxy named in the row is used in the allocation and, for $\lambda > 0$, in the estimator; λ tuned is chosen per stratum and target and clipped to $[0, 1]$.

	$\lambda = 0$ (HT)			$\lambda = 1$ (DIF)			λ tuned		
Proxy and design	Eff.	mean	worst	Eff.	mean	worst	Eff.	mean	worst
Ensemble π, PATTERN	3.97	0.944	0.858	6.27	0.945	0.877	6.47	0.923	0.799
Ensemble π, SRS	1.00	0.921	0.675	1.90	0.938	0.844	2.01	0.926	0.701
Constant 0.5, PATTERN	3.97	0.940	0.850	5.71	0.940	0.858	5.84	0.934	0.823
Shuffled π, PATTERN	3.95	0.940	0.854	3.70	0.949	0.889	5.20	0.924	0.783
Noisy π, PATTERN	4.05	0.943	0.854	3.46	0.949	0.877	5.07	0.923	0.806
Reversed $1 - \pi$, PATTERN	3.99	0.942	0.859	4.38	0.941	0.872	5.41	0.932	0.835